

Texture: statistical models of vision



Somewhere in Cinque Terre, May 2005

CS180: Intro to Computer Vision and Comp. Photo
Efros & Ng, UC Berkeley, Fall 2026

Object recognition

Is it really so hard?

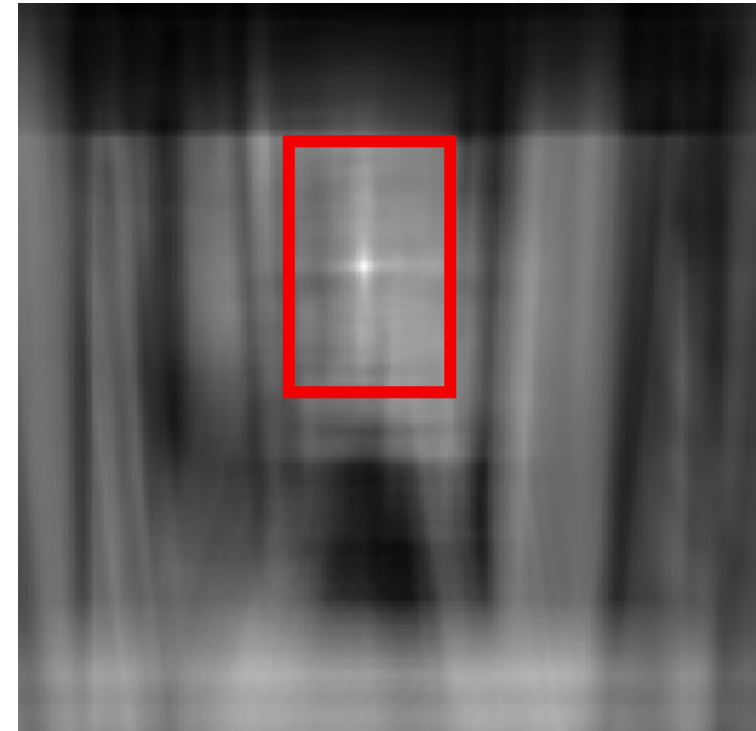
Find the chair in this image



This is a chair



Output of normalized correlation

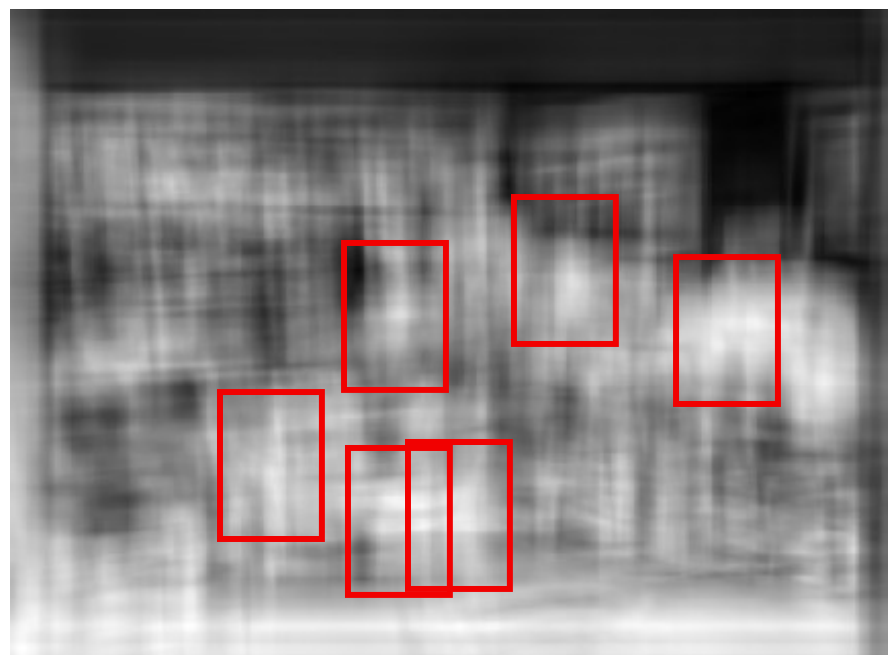
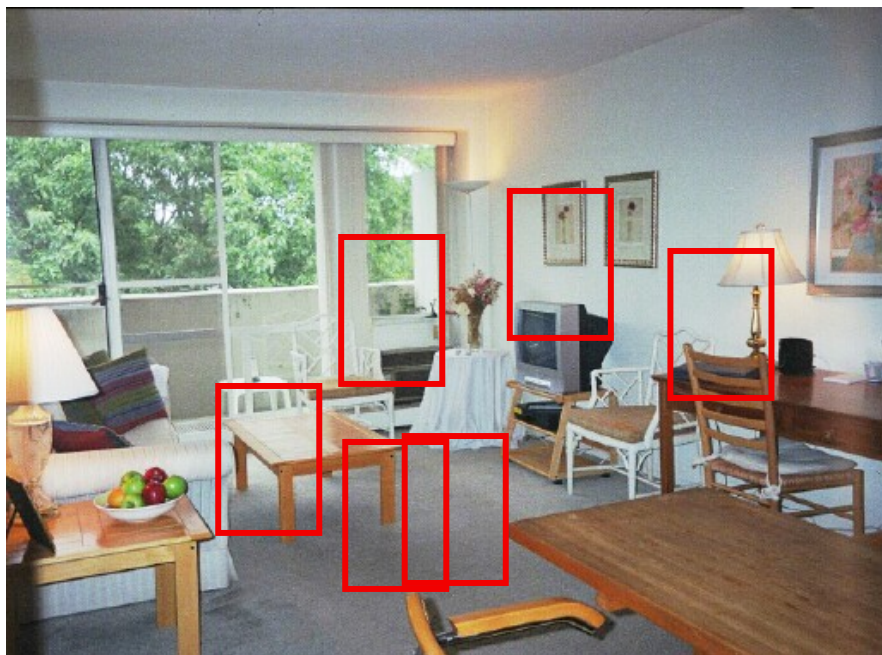




Object recognition

Is it really so hard?

Find the chair in this image



Recognition: Instance vs. Category

- Instance recognition:
 - “Find me this particular chair again”
 - Often simple template matching works OK
 - Even better with many small templates, a.k.a. feature descriptors
- Category recognition:
 - “find me a chair”
 - Templates don’t work. Why?
 - Focus on things that might be invariant across the category
 - A more statistical notion of appearance

What is Texture?

- Texture depicts spatially repeating patterns
- Many natural phenomena are textures



radishes



rocks



yogurt

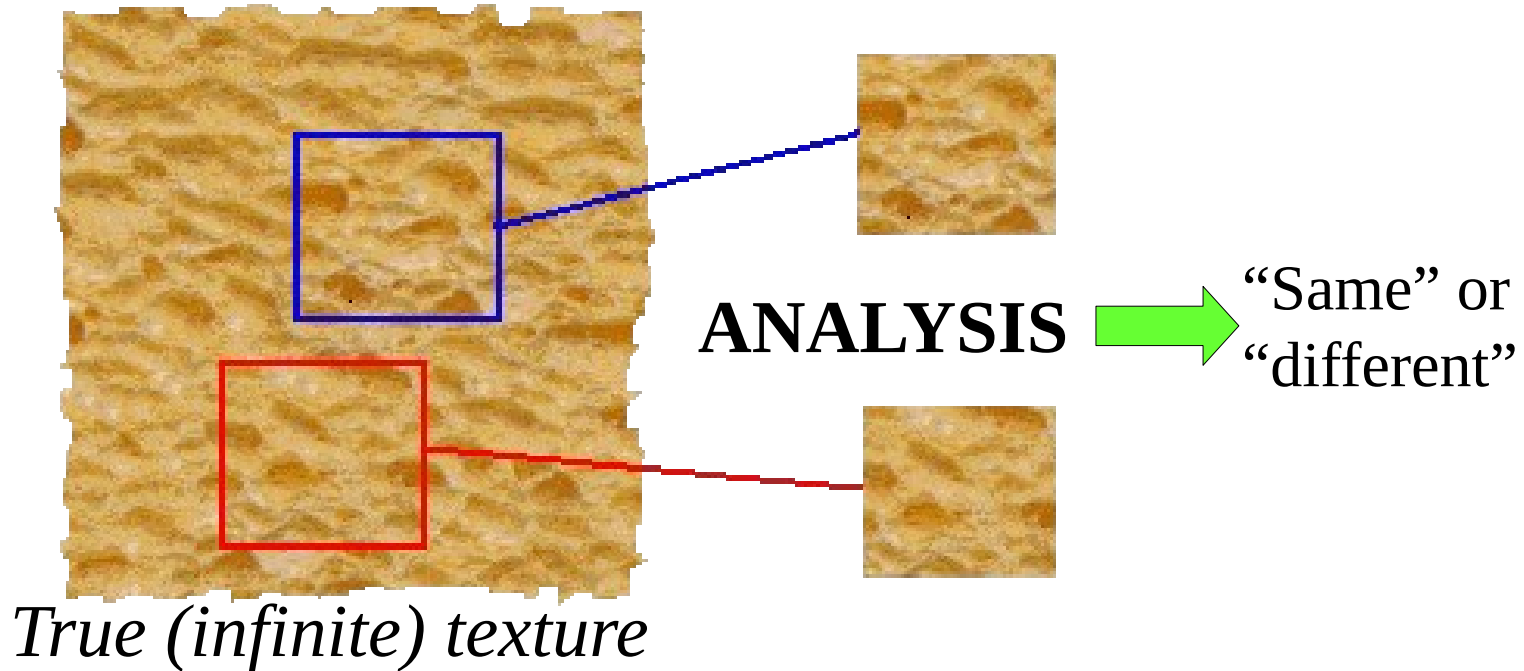
When are two textures similar?



Texture as “stuff”

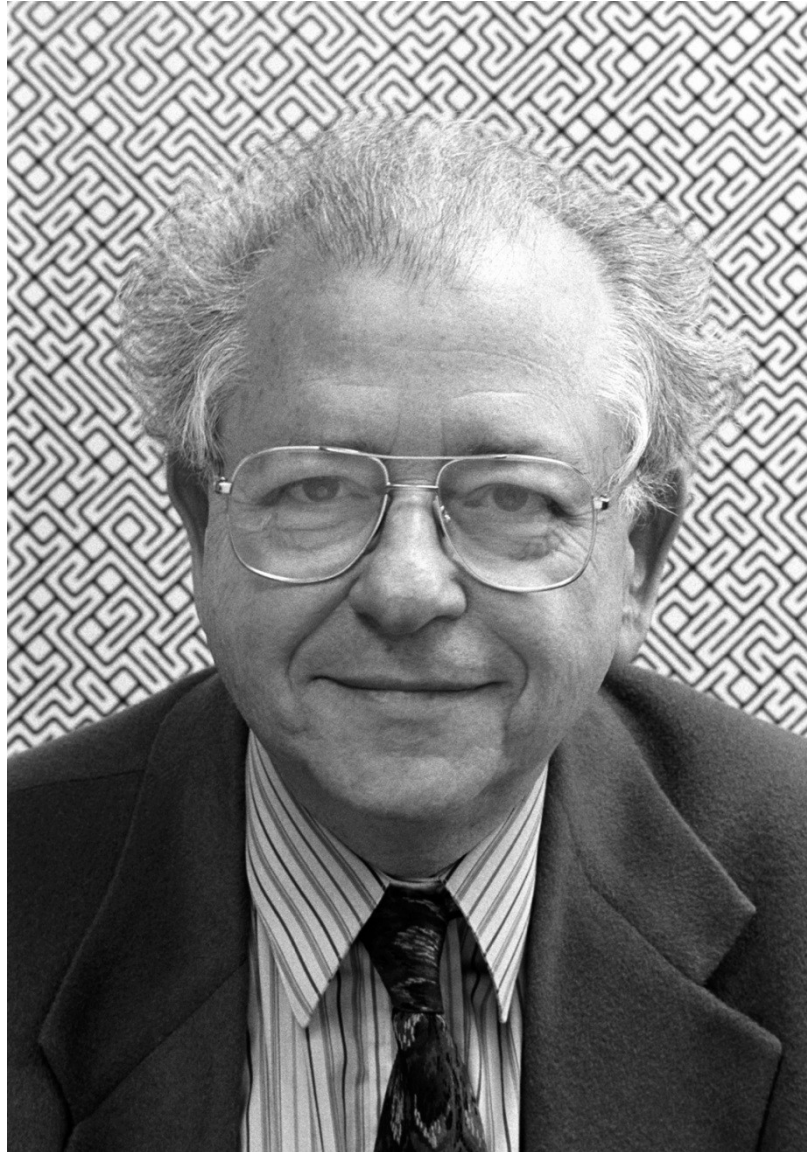


Texture Analysis



Compare textures and decide if they're made of the same “stuff”.

Béla Julesz, father of texture



REVIEW ARTICLES

Textons, the elements of texture perception, and their interactions

Bela Julesz

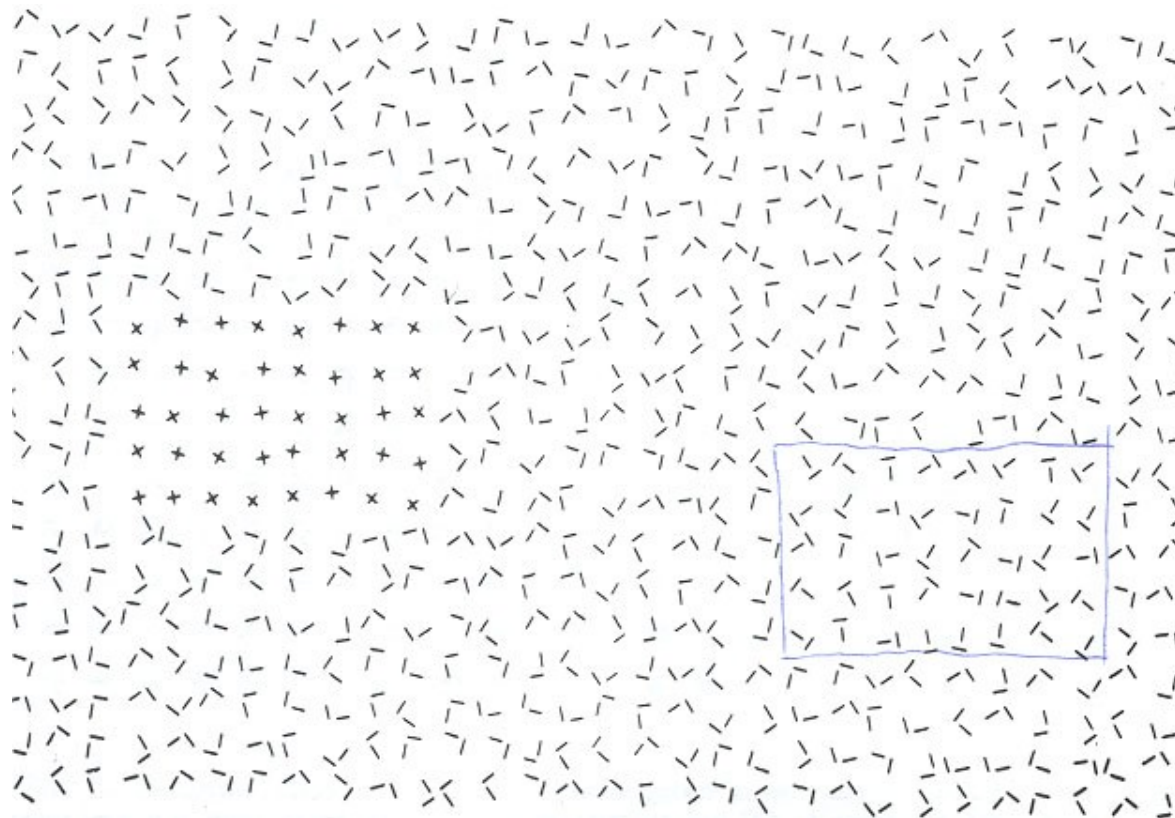
Bell Laboratories, Murray Hill, New Jersey 07974, USA

Research with texture pairs having identical second-order statistics has revealed that the pre-attentive texture discrimination system cannot globally process third- and higher-order statistics, and that discrimination is the result of a few local conspicuous features, called textons. It seems that only the first-order statistics of these textons have perceptual significance, and the relative phase between textons cannot be perceived without detailed scrutiny by focal attention.



Bela Julesz, "Textons, the Elements of Texture Perception, and their Interactions". Nature 290: 91-97. March, 1981.

Texton Discrimination (Julesz)

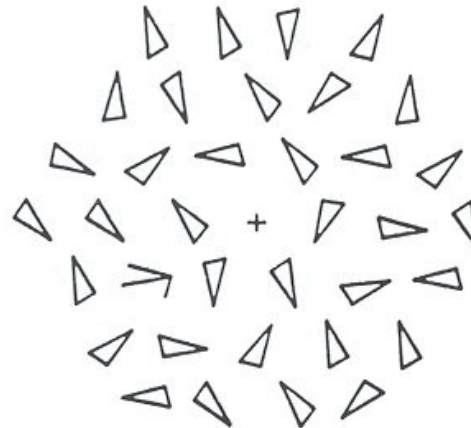
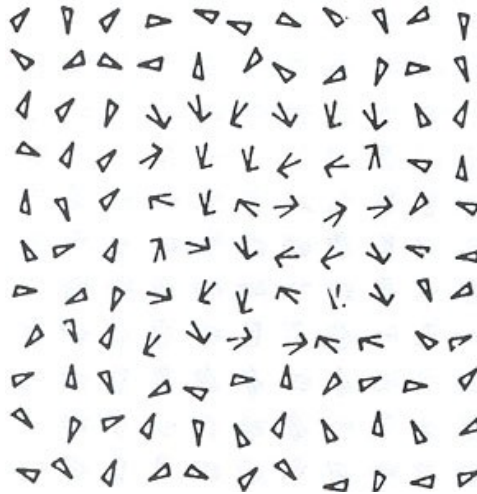


Human vision is sensitive to the difference of some types of elements and appears to be “numb” on other types of differences.

Search Experiment I



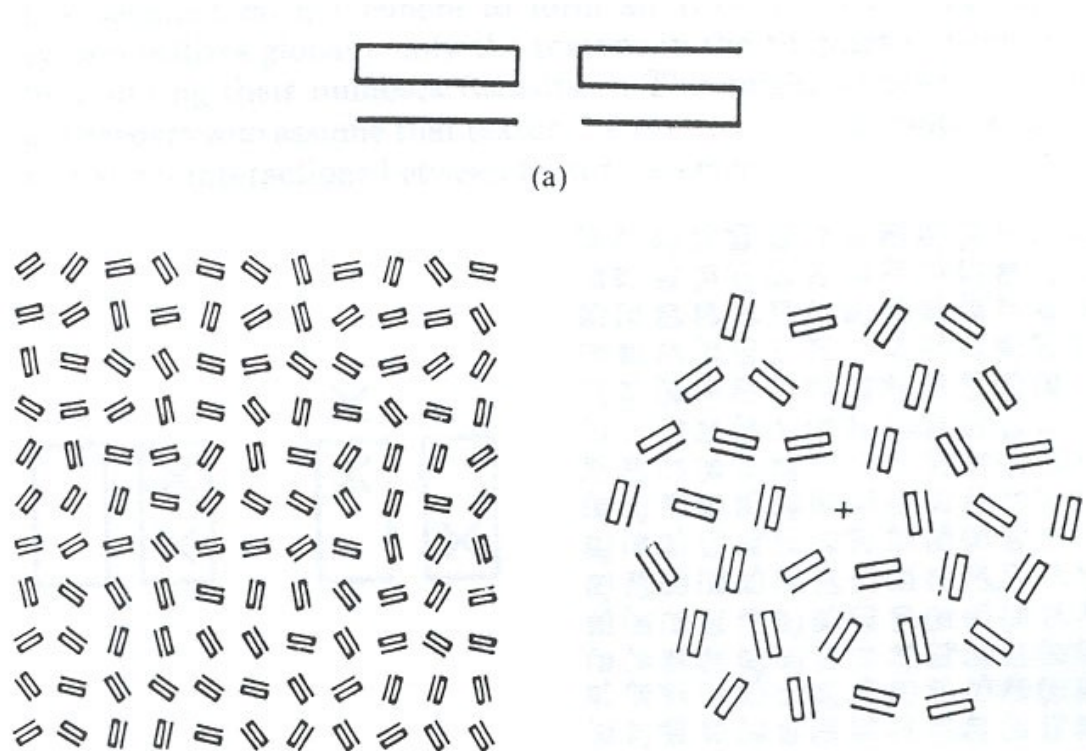
(a)



The subject is told to detect a target element in a number of background elements.

In this example, the detection time is independent of the number of background elements.

Search Experiment II



In this example, the detection time is proportional to the number of background elements,
And thus suggests that the subject is doing element-by-element scrutiny.

Preattentive vs Attentive Vision (Julesz)

Human vision operates in two distinct modes:

1. Preattentive vision

parallel, instantaneous ($\sim 100\text{--}200\text{ms}$),
without scrutiny,
independent of the number of patterns,
covering a large visual field.

2. Attentive vision

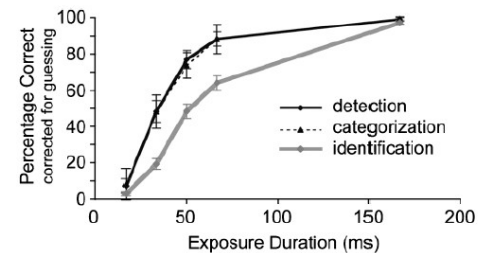
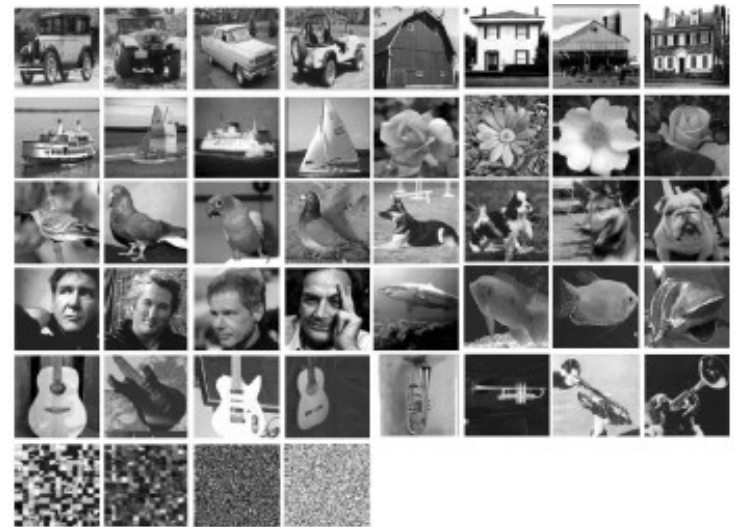
serial search by focal attention in 50ms steps

Evidence for Pre-attentive Recognition (Thorpe)

On a task of judging **animal** **vs no animal**, humans can make mostly correct saccades in 150 ms (Kirchner & Thorpe, 2006)

- Comparable to synaptic delay in the retina, LGN, V1, V2, V4, IT pathway.
- Doesn't rule out feed back but shows **feed forward only is very powerful**

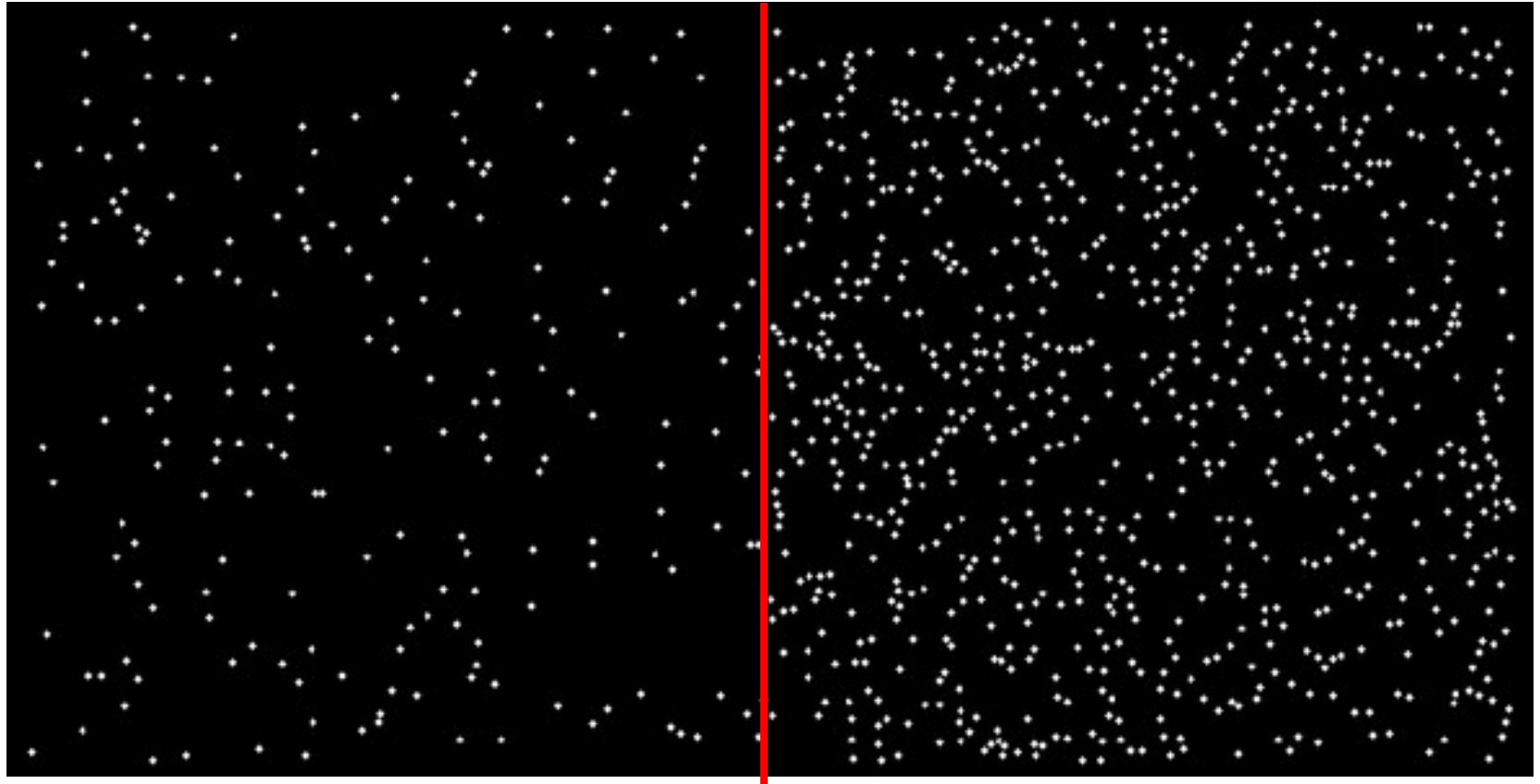
Detection and categorization are practically simultaneous (Grill-Spector & Kanwisher, 2005)



Julesz Conjecture

*Textures cannot be spontaneously discriminated if they have the **same first-order and second-order statistics** of texture features (textons) and differ only in their third-order or higher-order statistics.*

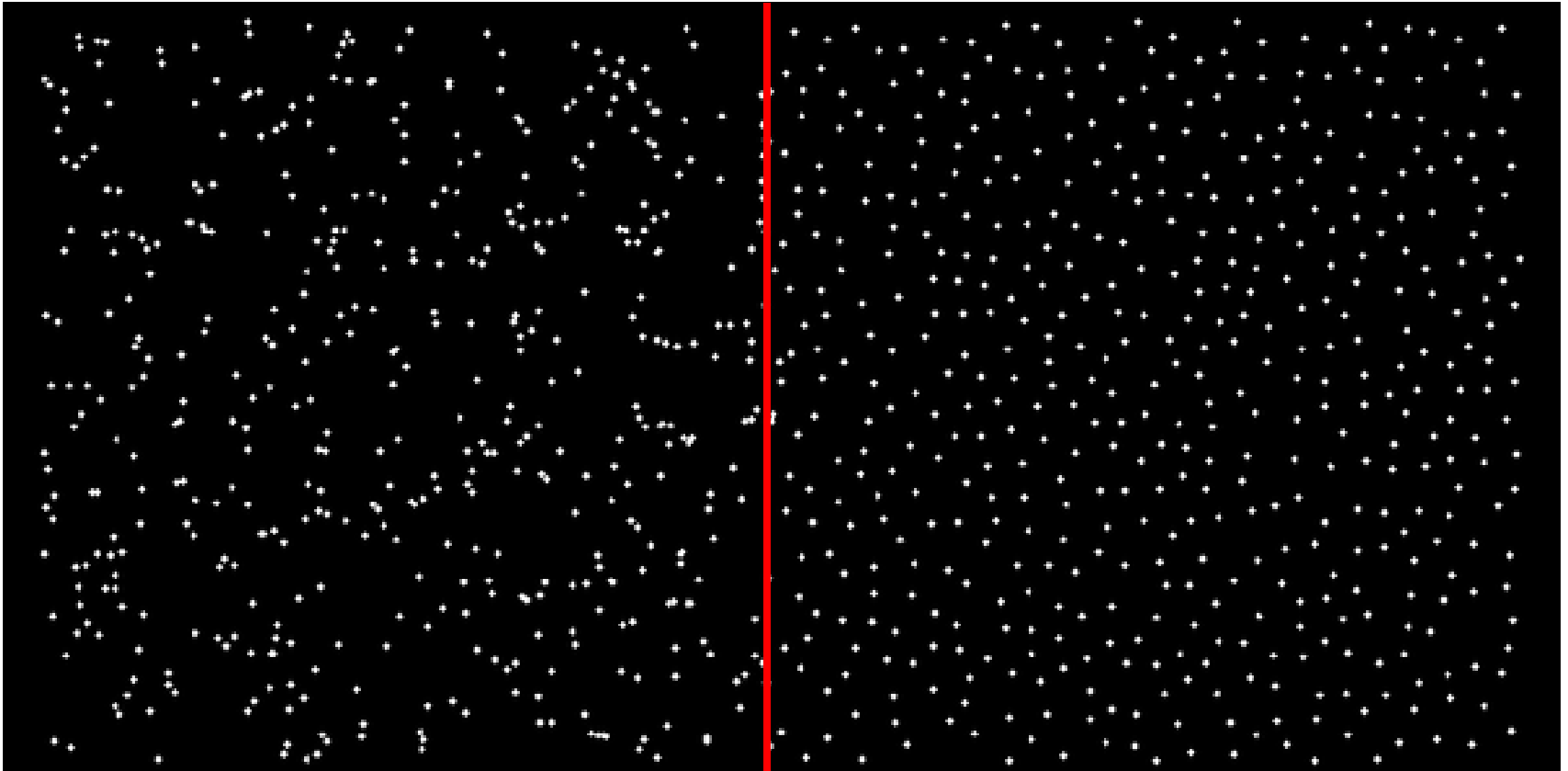
1st Order Statistics



5%
white

20%
white

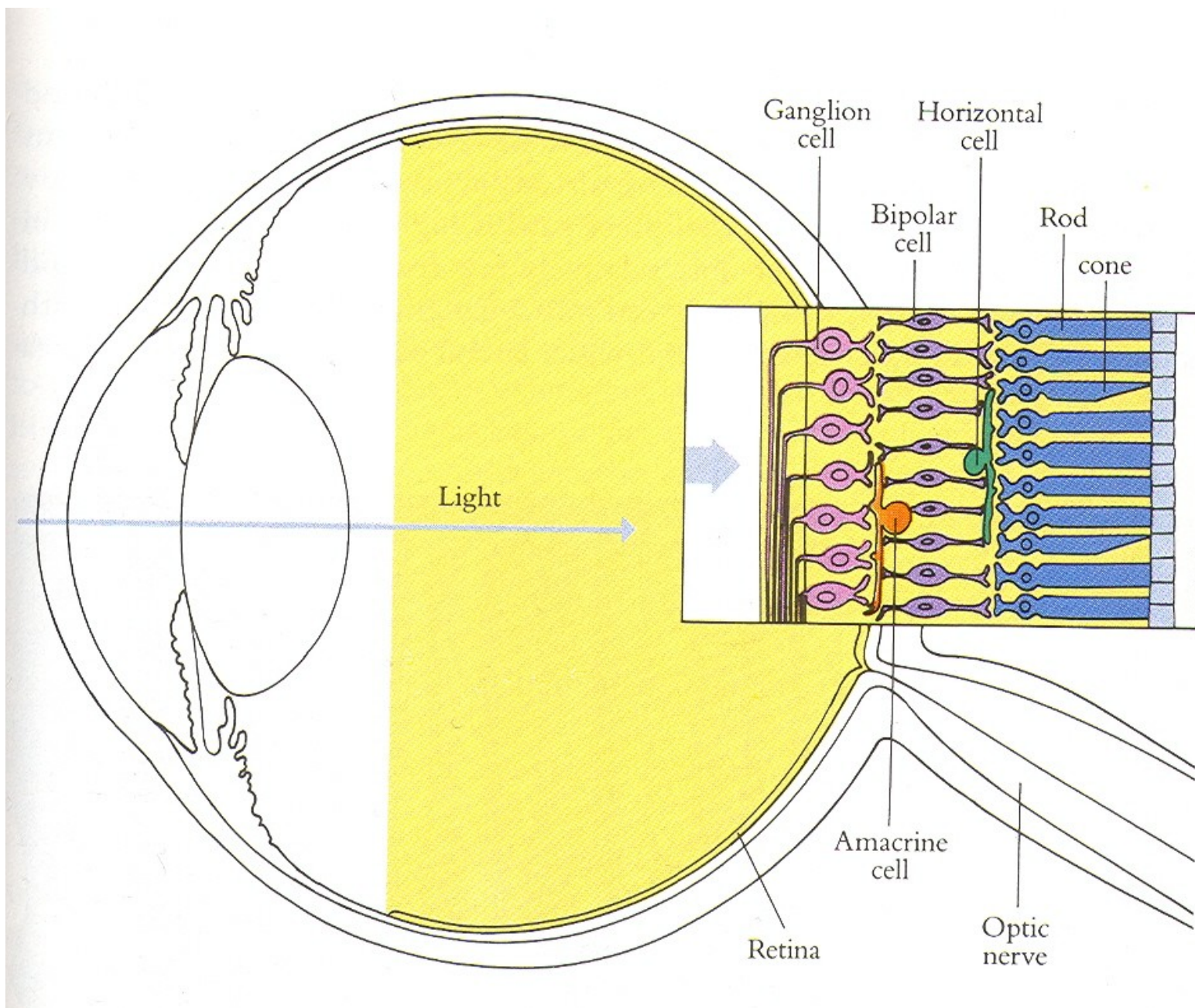
2nd Order Statistics

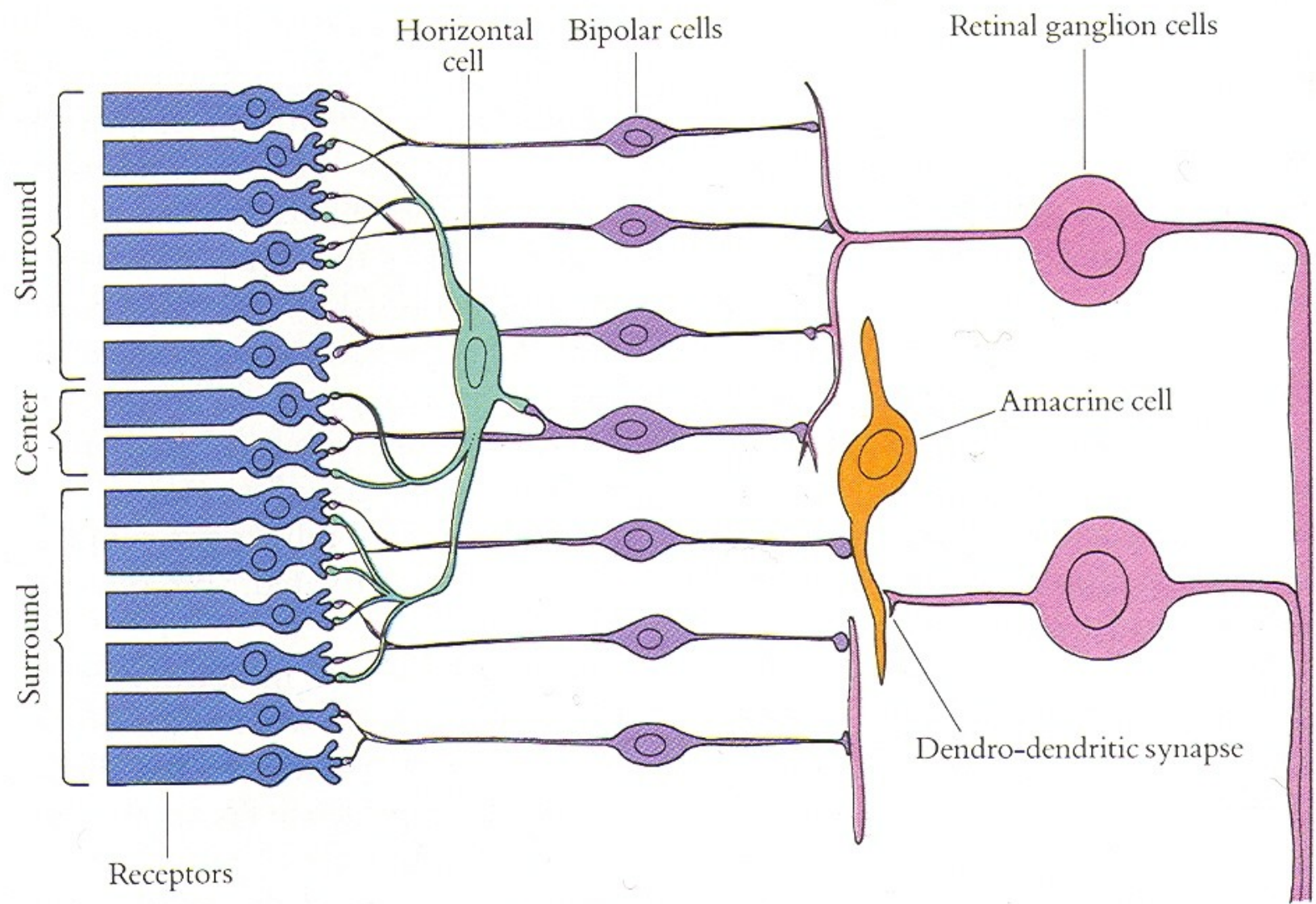


10%
white

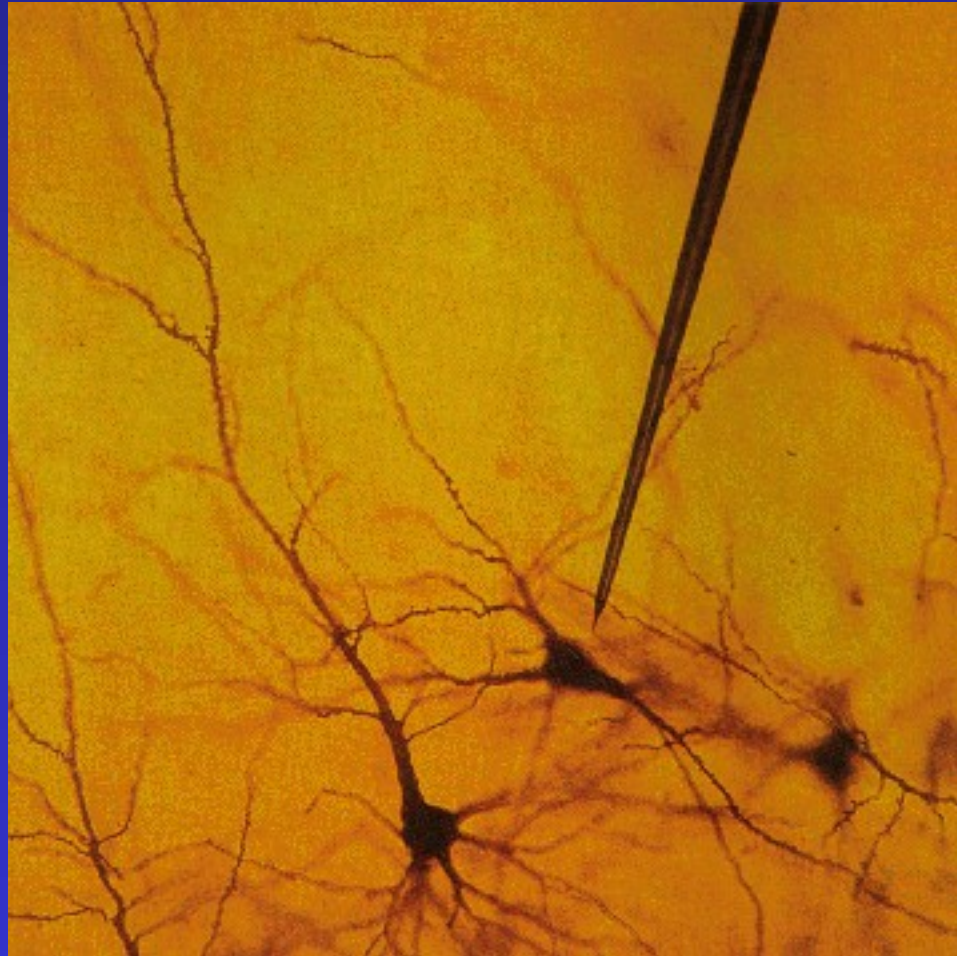
Big Question

What is the statistical unit (texton) of texture in real images?



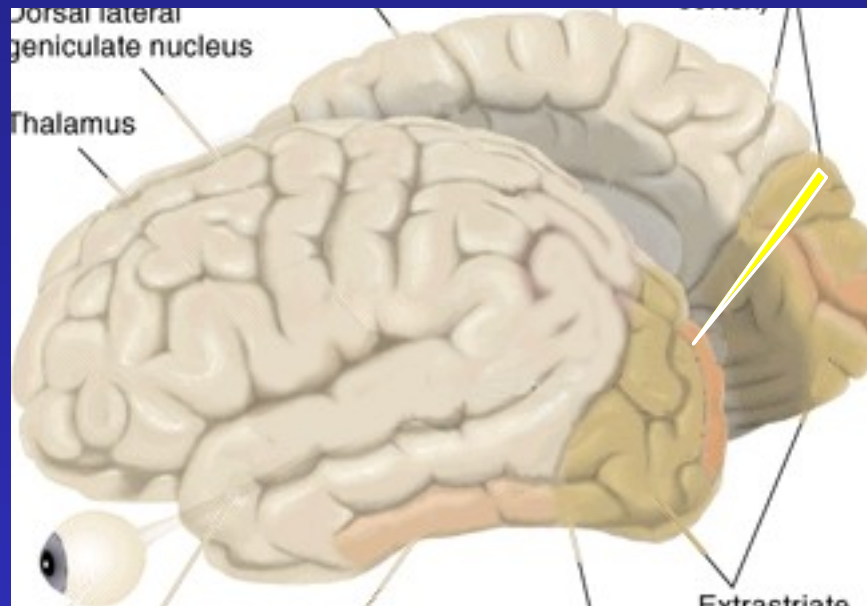


Single Cell Recording



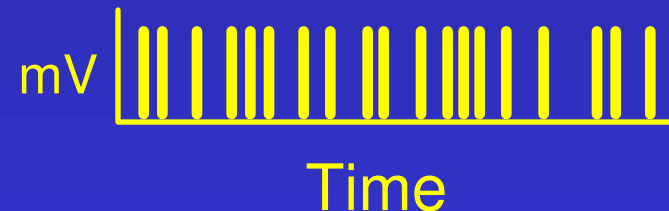
Single Cell Recording

Microelectrode



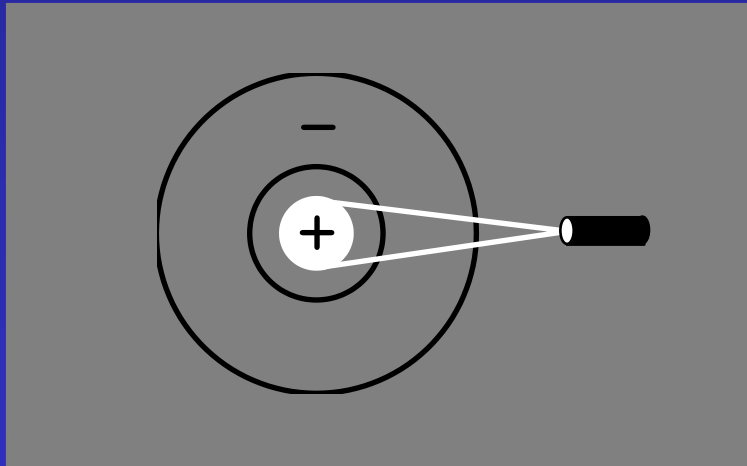
Amplifier

Electrical response
(action potentials)

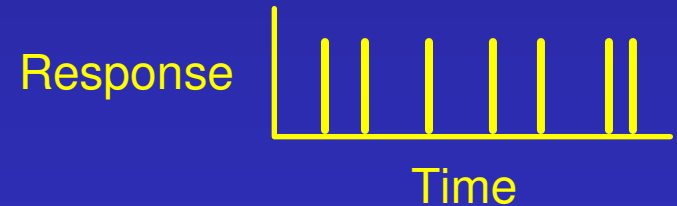


Retinal Receptive Fields

Receptive field structure in ganglion cells:
On-center Off-surround



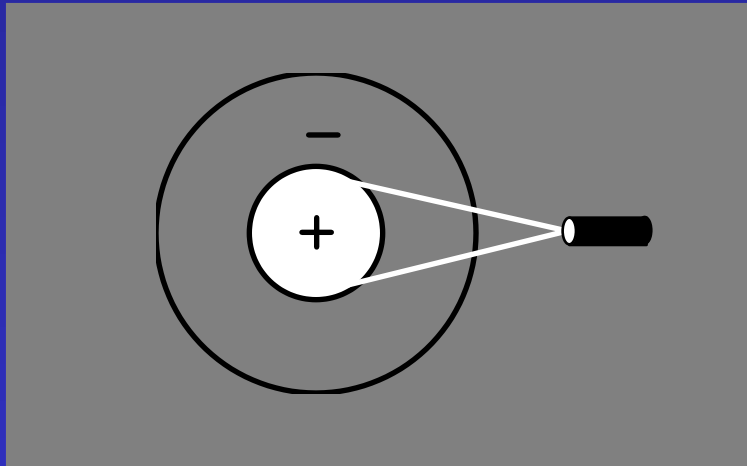
Stimulus condition



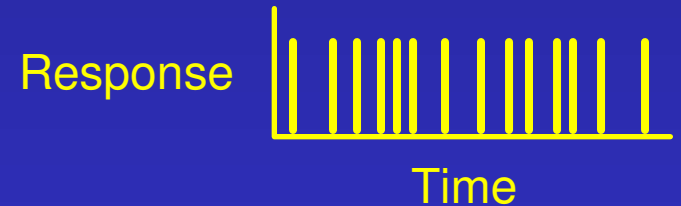
Electrical response

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On-center Off-surround



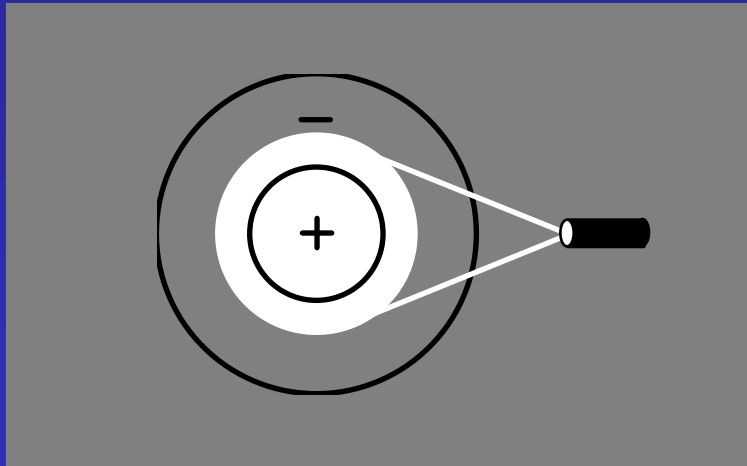
Stimulus condition



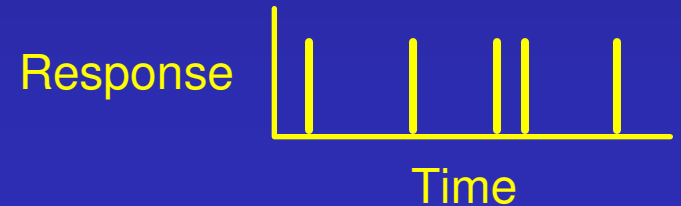
Electrical response

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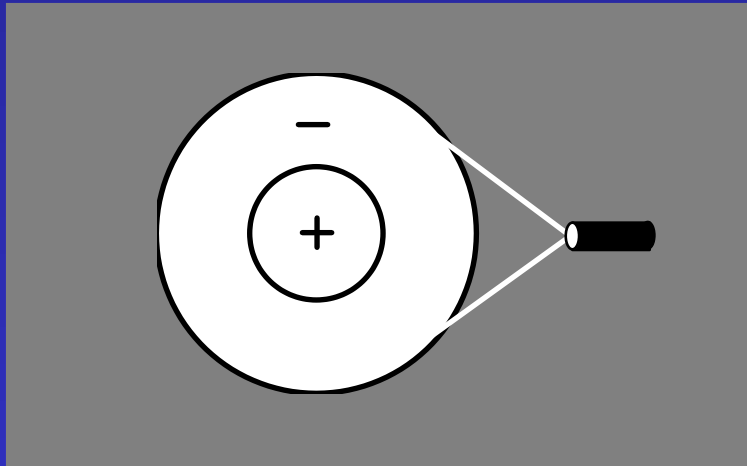
Stimulus condition



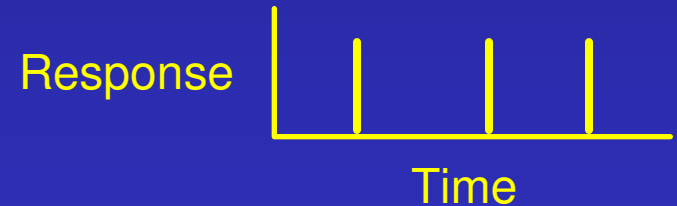
Electrical response

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On-center Off-surround



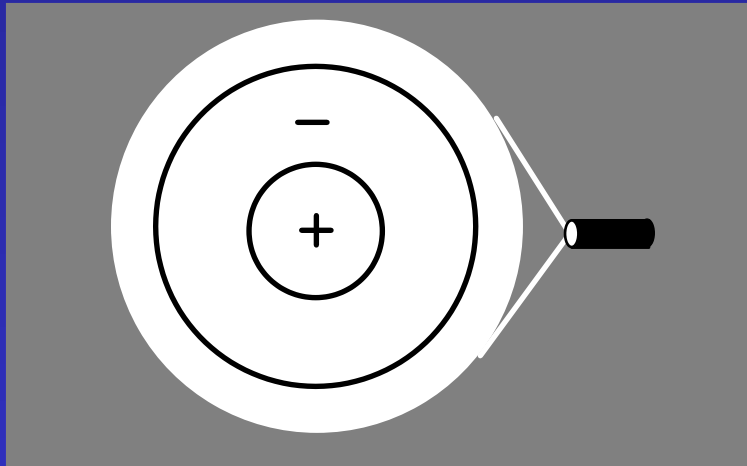
Stimulus condition



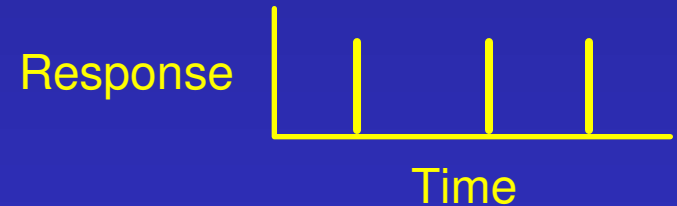
Electrical response

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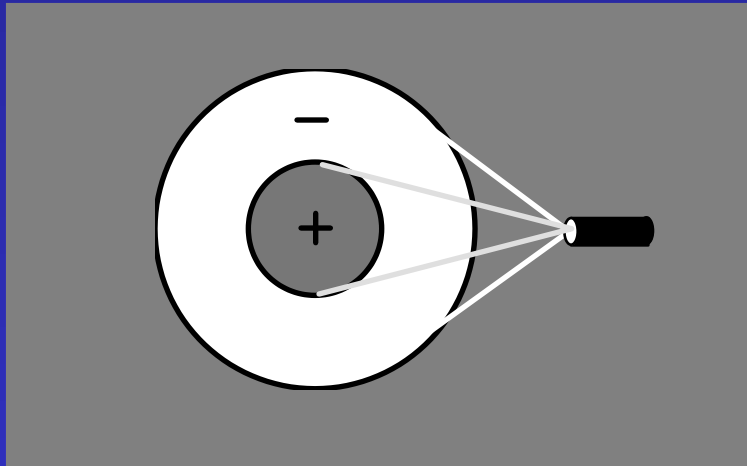
Stimulus condition



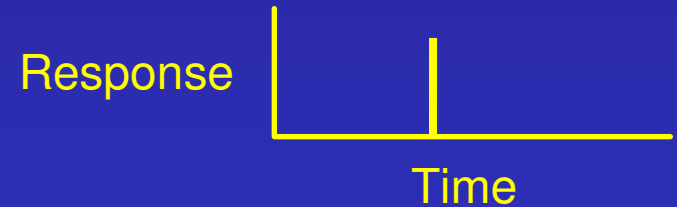
Electrical response

Retinal Receptive Fields

Receptive field structure in ganglion cells:
On-center Off-surround



Stimulus condition

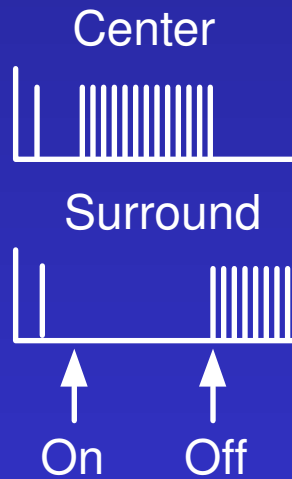


Electrical response

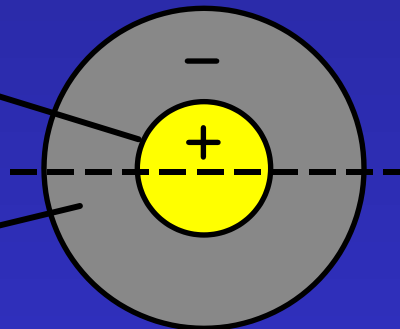
Retinal Receptive Fields

RF of On-center Off-surround cells

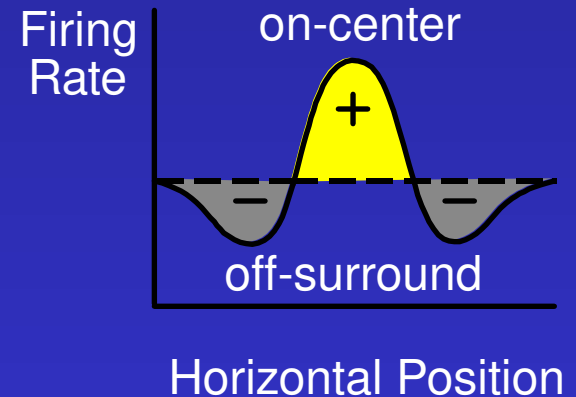
Neural Response



Receptive Field



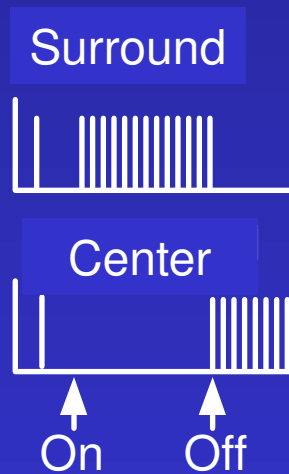
Response Profile



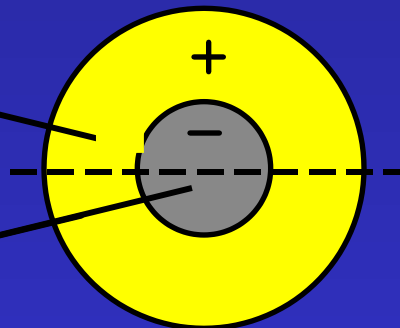
Retinal Receptive Fields

RF of Off-center On-surround cells

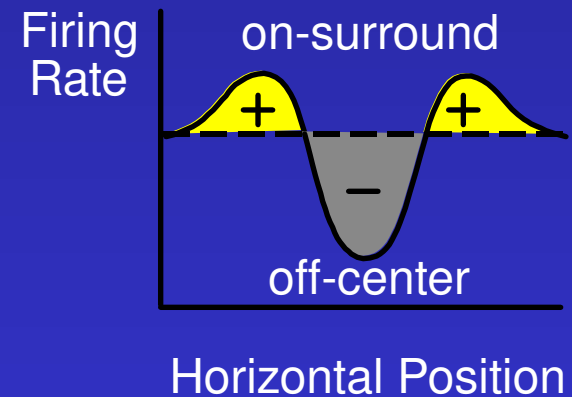
Neural Response



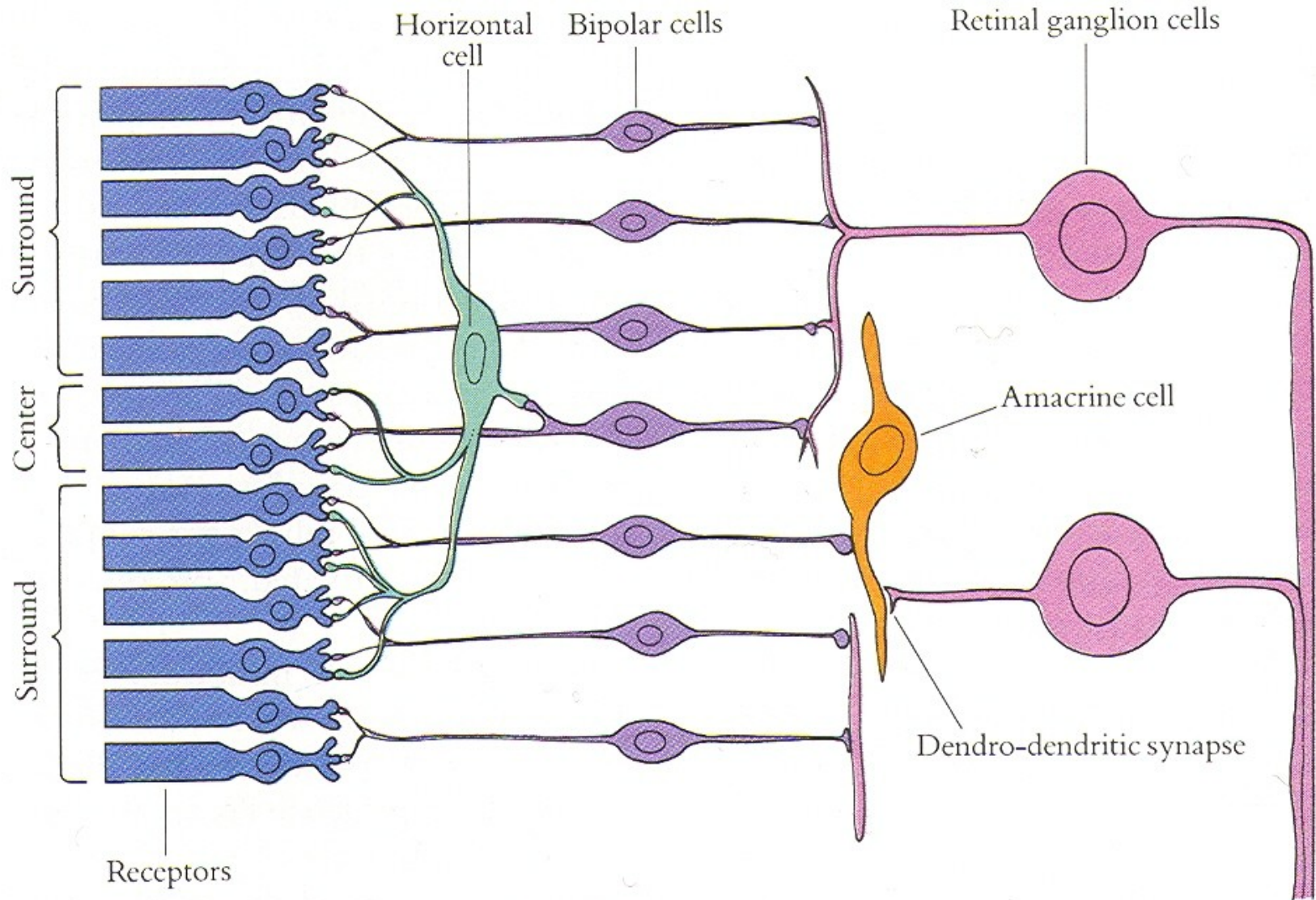
Receptive Field



Response Profile

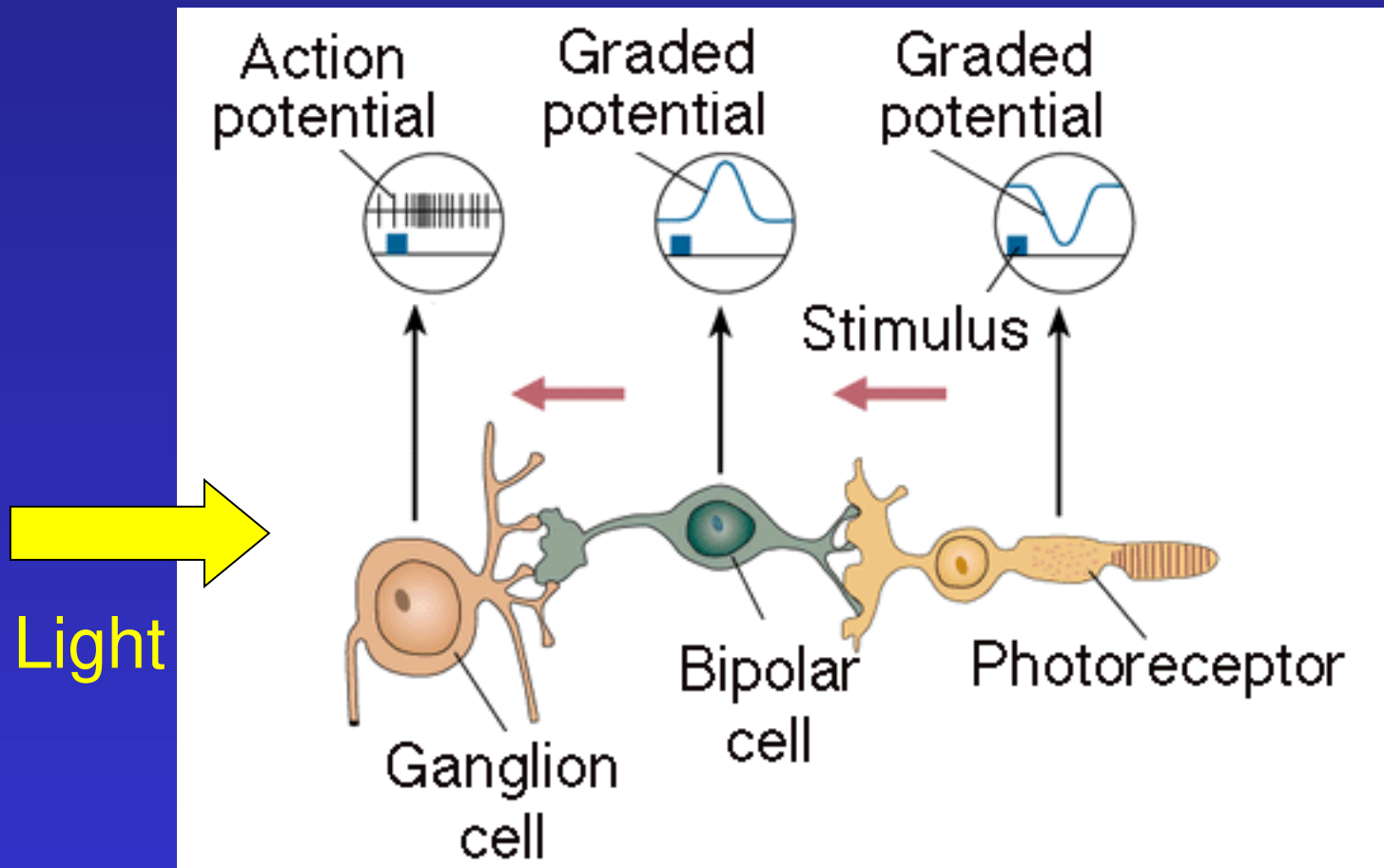


Retinal Receptive Fields



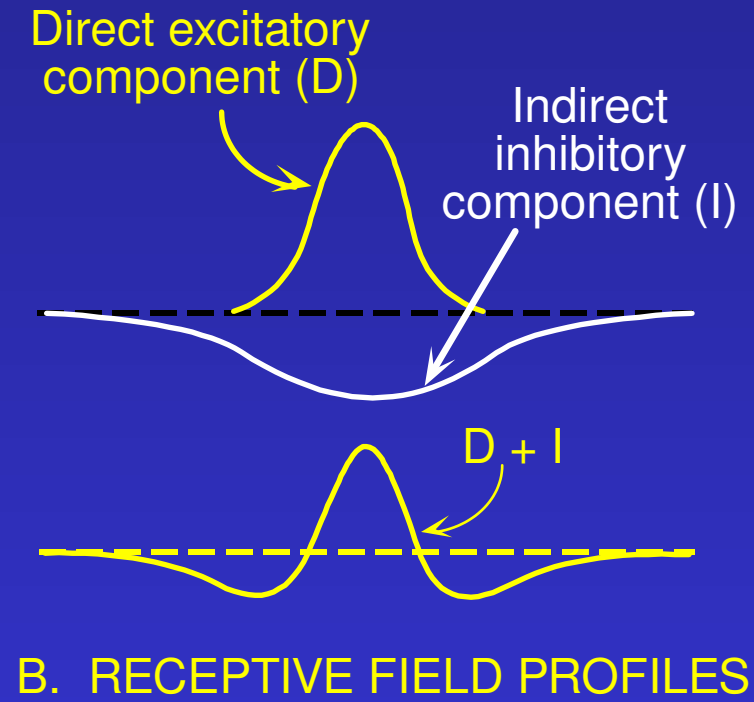
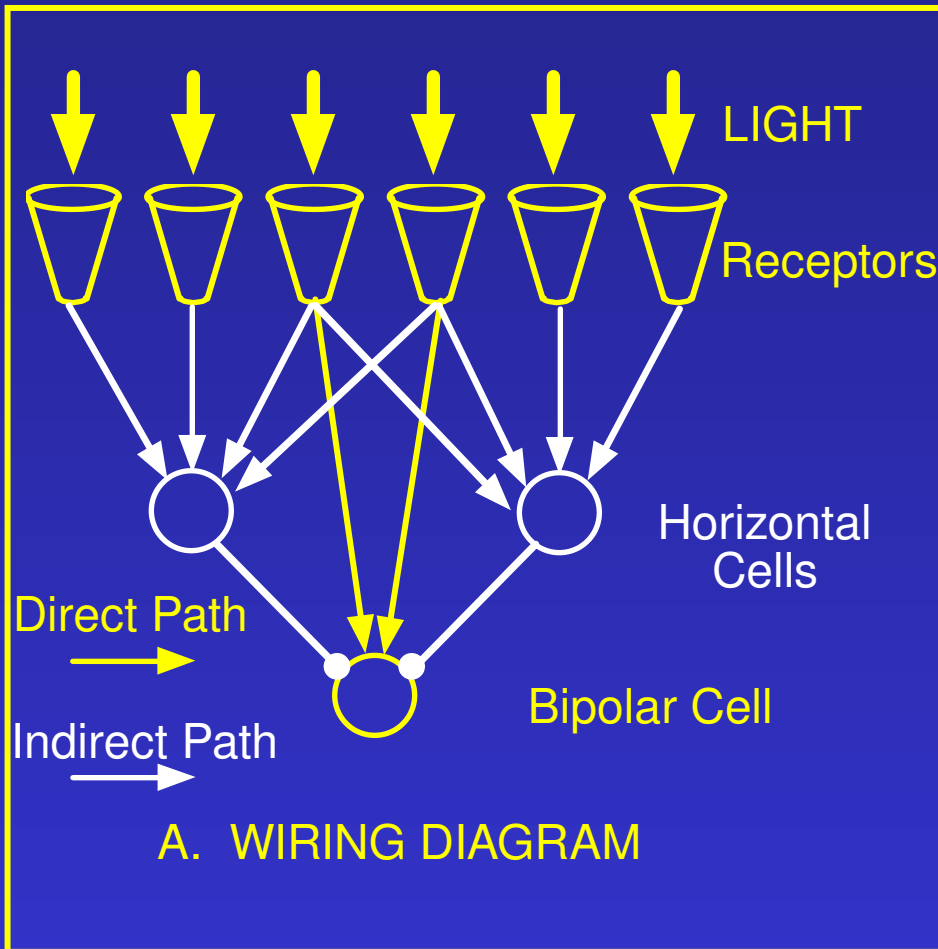
Retinal Receptive Fields

Receptive field structure in bipolar cells



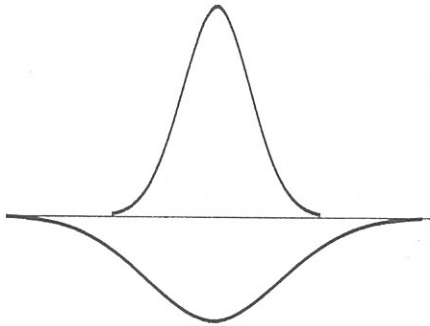
Retinal Receptive Fields

Receptive field structure in bipolar cells

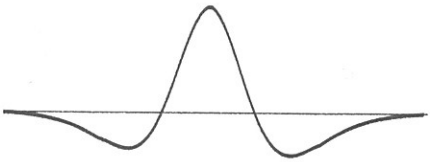


The receptive field of a retinal ganglion cell can be modeled as a “Difference of Gaussians”

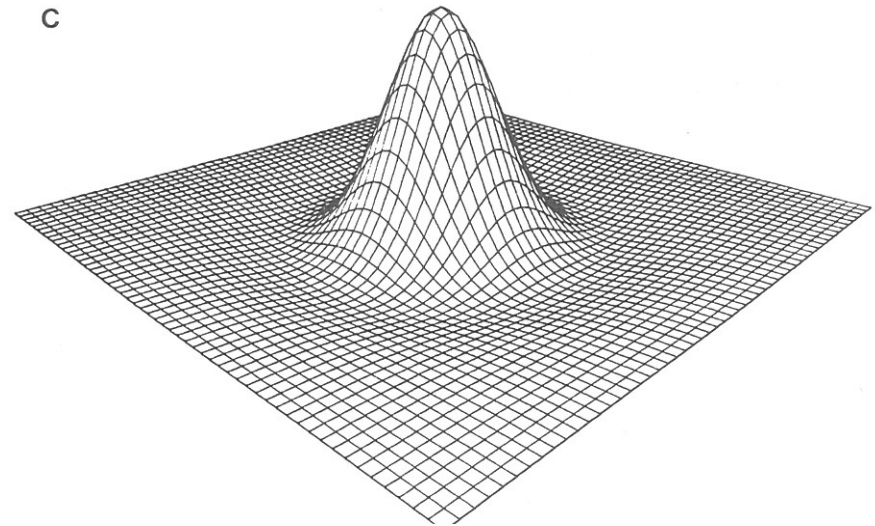
A



B



C



$$G_{\sigma}(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{r^2}{2\sigma^2}}$$

Receptive Fields

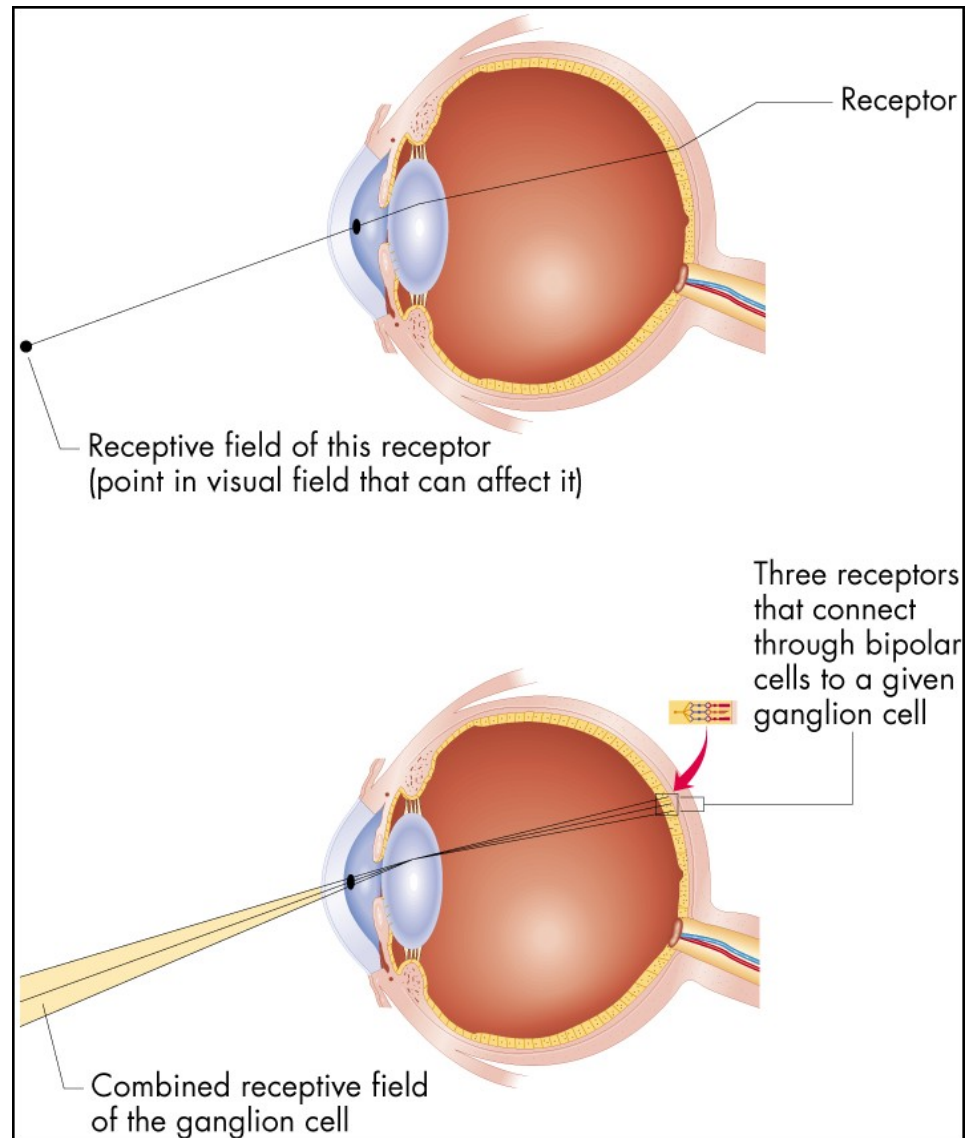
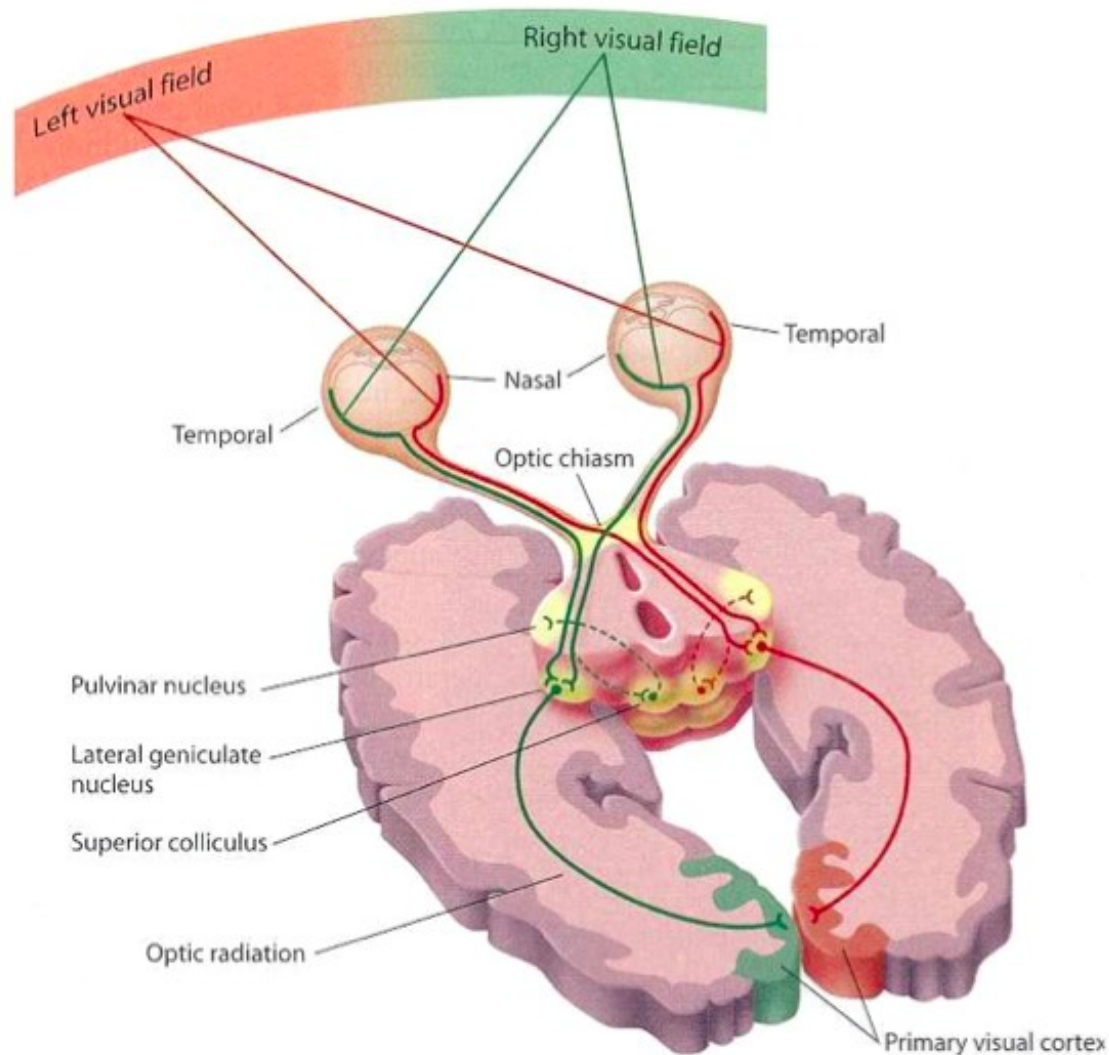


Figure 6.16 Receptive fields

The receptive field of a receptor is simply the area of the visual field from which light strikes that receptor. For any other cell in the visual system, the receptive field is determined by which receptors connect to the cell in question.

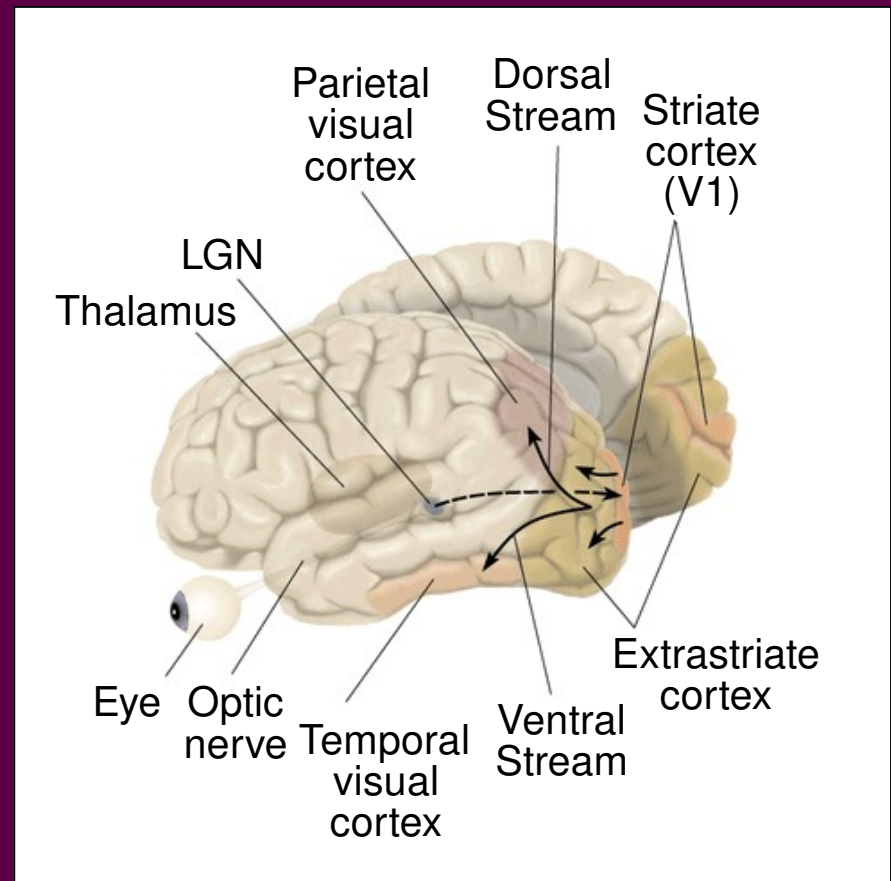
Anatomy of Pathway to Visual Cortex



Visual Cortex

Cortical Area V1

aka:
Primary visual cortex
Striate cortex
Brodman's area 17



Cortical Receptive Fields

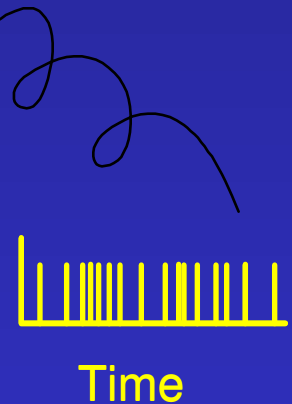
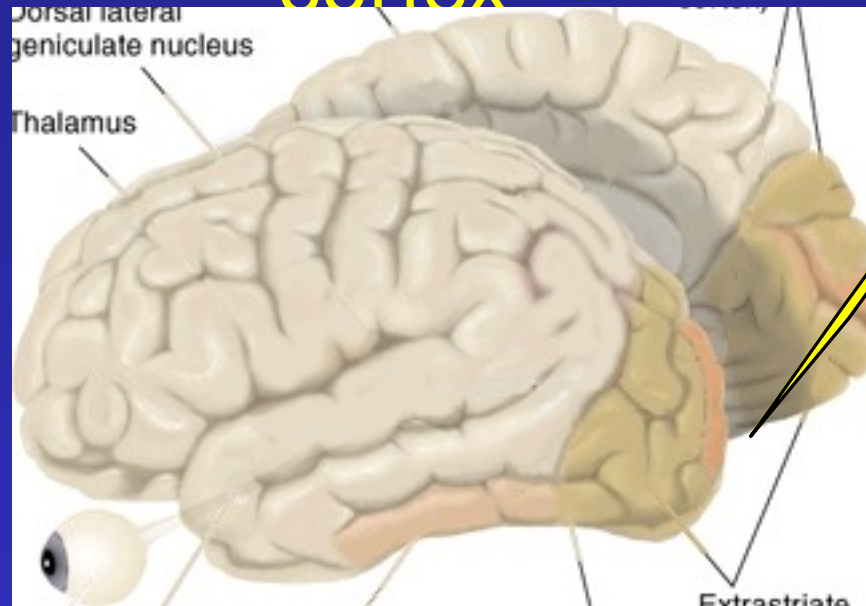
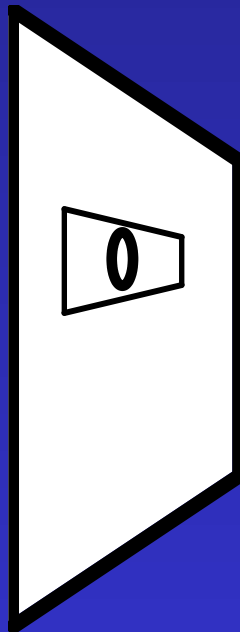
Single-cell recording from visual cortex



David Hubel & Thorston Wiesel

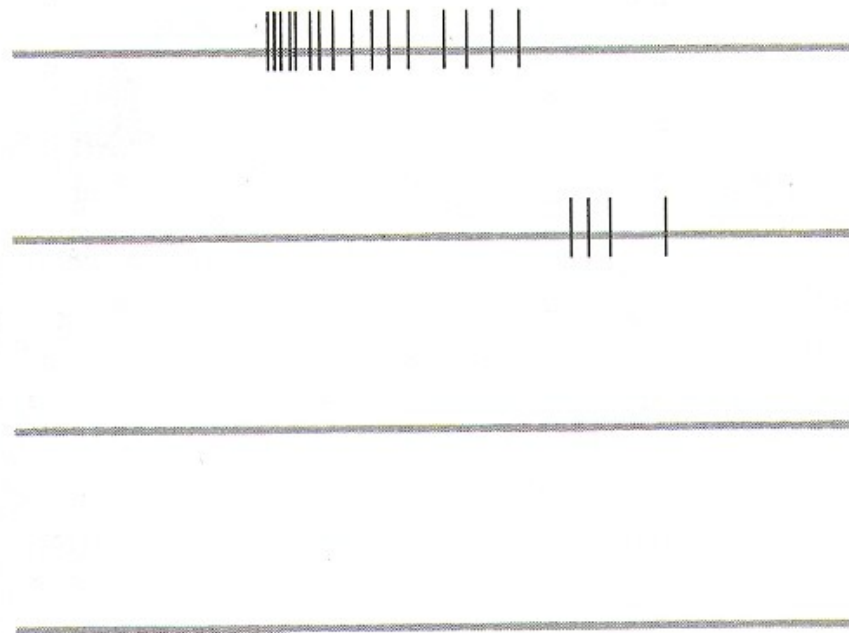
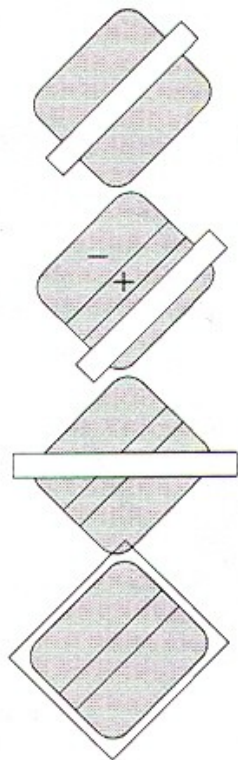
Cortical Receptive Fields

Single-cell recording from visual cortex





<https://www.youtube.com/watch?v=IOHayh06LJ4>



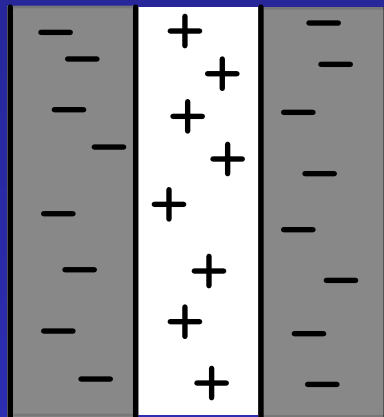
Stimulus:

on off

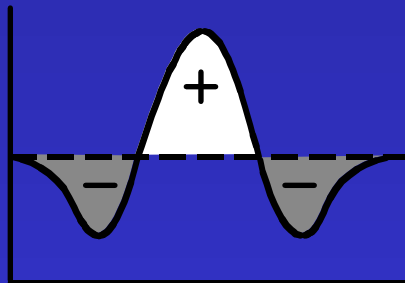
Cortical Receptive Fields

Simple Cells: “Line Detectors”

A. Light Line Detector

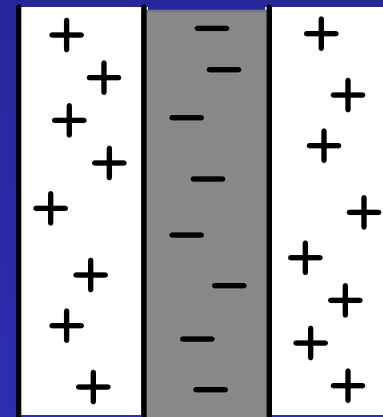


Firing Rate

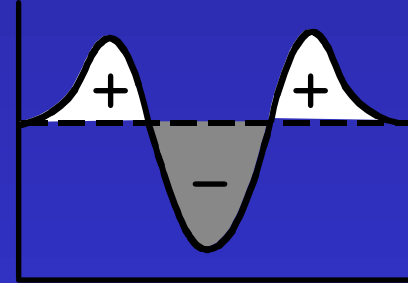


Horizontal Position

B. Dark Line Detector



Firing Rate

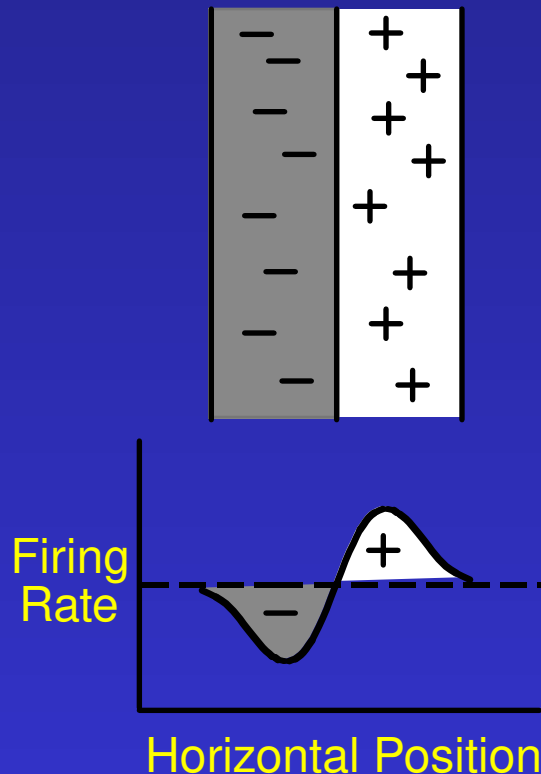


Horizontal Position

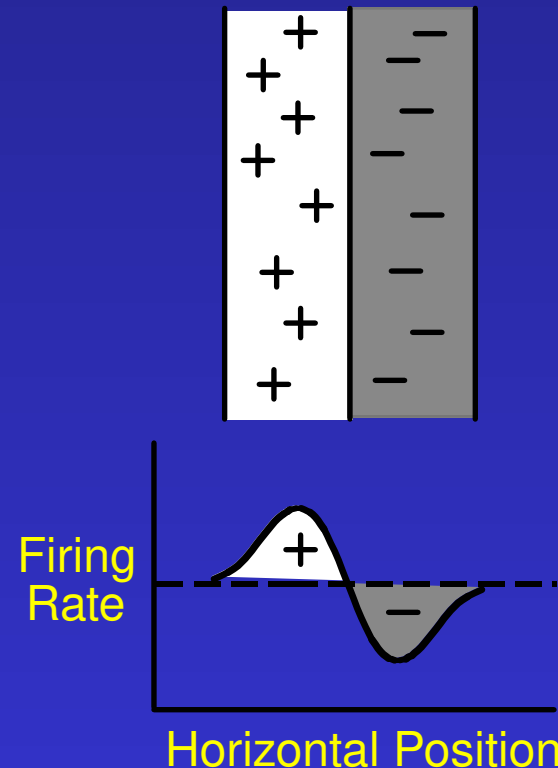
Cortical Receptive Fields

Simple Cells: “Edge Detectors”

C. Dark-to-light Edge Detector

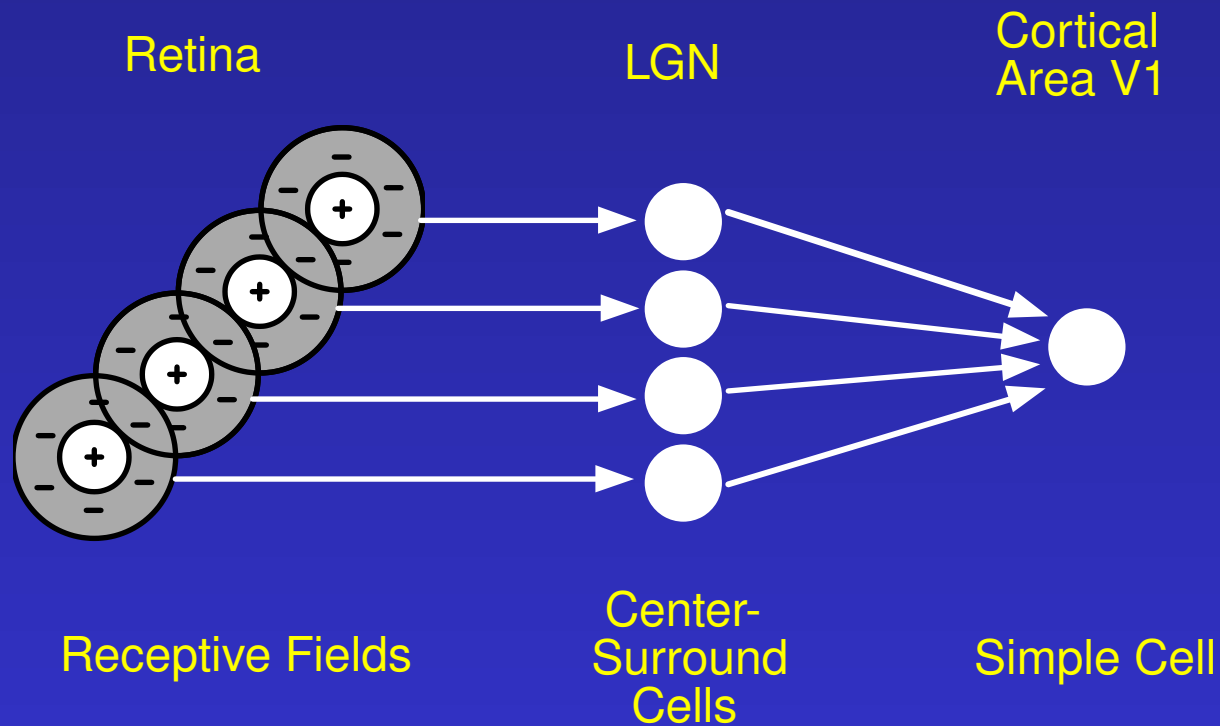


D. Light-to-dark Edge Detector

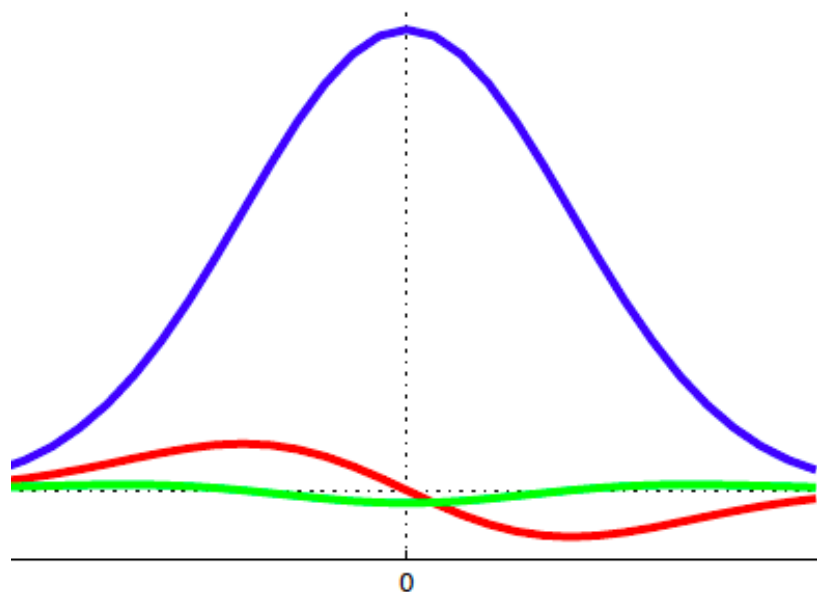


Cortical Receptive Fields

Constructing a line detector



The 1D Gaussian and its derivatives



$$G_\sigma(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$$

$$G'_\sigma(x) = \frac{d}{dx} G_\sigma(x) = -\frac{1}{\sigma} \left(\frac{x}{\sigma} \right) G_\sigma(x)$$

$$G''_\sigma(x) = \frac{d^2}{dx^2} G_\sigma(x) = \frac{1}{\sigma^2} \left(\frac{x^2}{\sigma^2} - 1 \right) G_\sigma(x)$$

$G'_\sigma(x)$'s maxima/minima occur at $G''_\sigma(x)$'s zeros. And, we can see that $G'_\sigma(x)$ is an odd symmetric function and $G''_\sigma(x)$ is an even symmetric function.

Oriented Gaussian Derivatives in 2D

$$f_1(x, y) = G'_{\sigma_1}(x)G_{\sigma_2}(y) \quad (10.4)$$

$$f_2(x, y) = G''_{\sigma_1}(x)G_{\sigma_2}(y) \quad (10.5)$$

We also consider rotated versions of these Gaussian derivative functions.

$$Rot_{\theta}f_1 = G'_{\sigma_1}(u)G_{\sigma_2}(v) \quad (10.6)$$

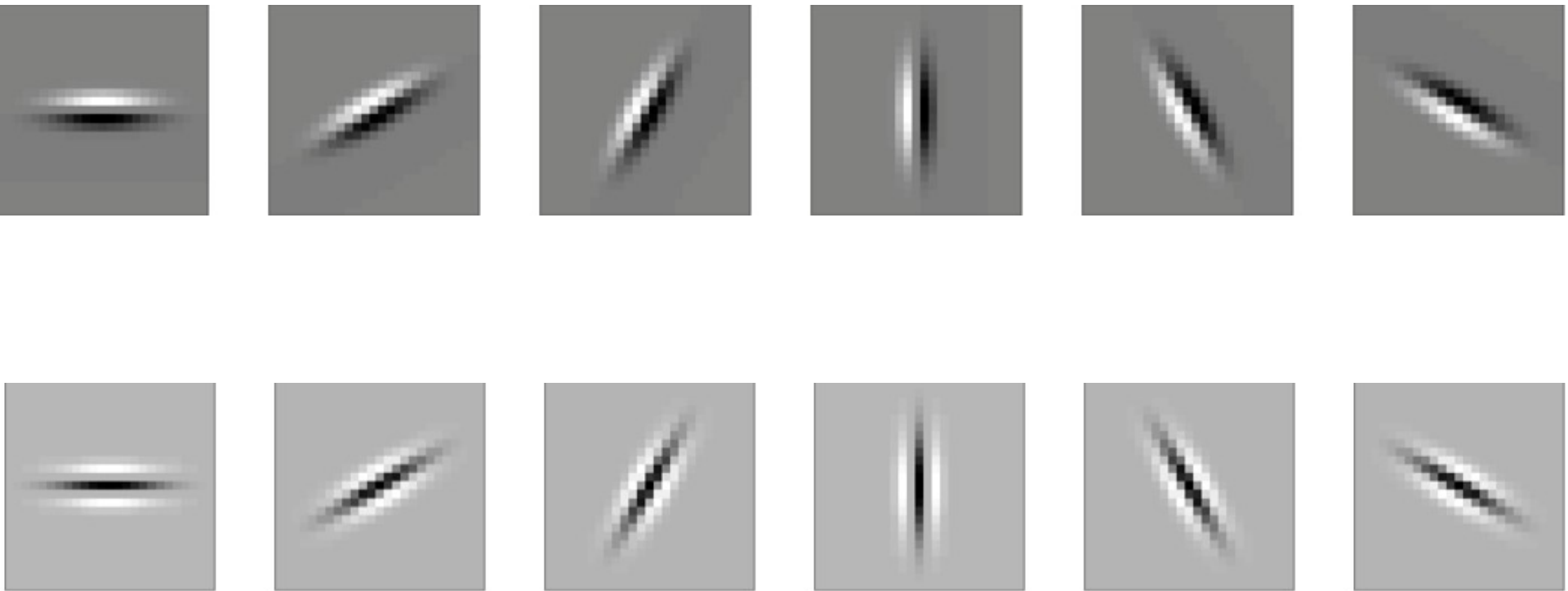
$$Rot_{\theta}f_2 = G''_{\sigma_1}(u)G_{\sigma_2}(v) \quad (10.7)$$

where we set

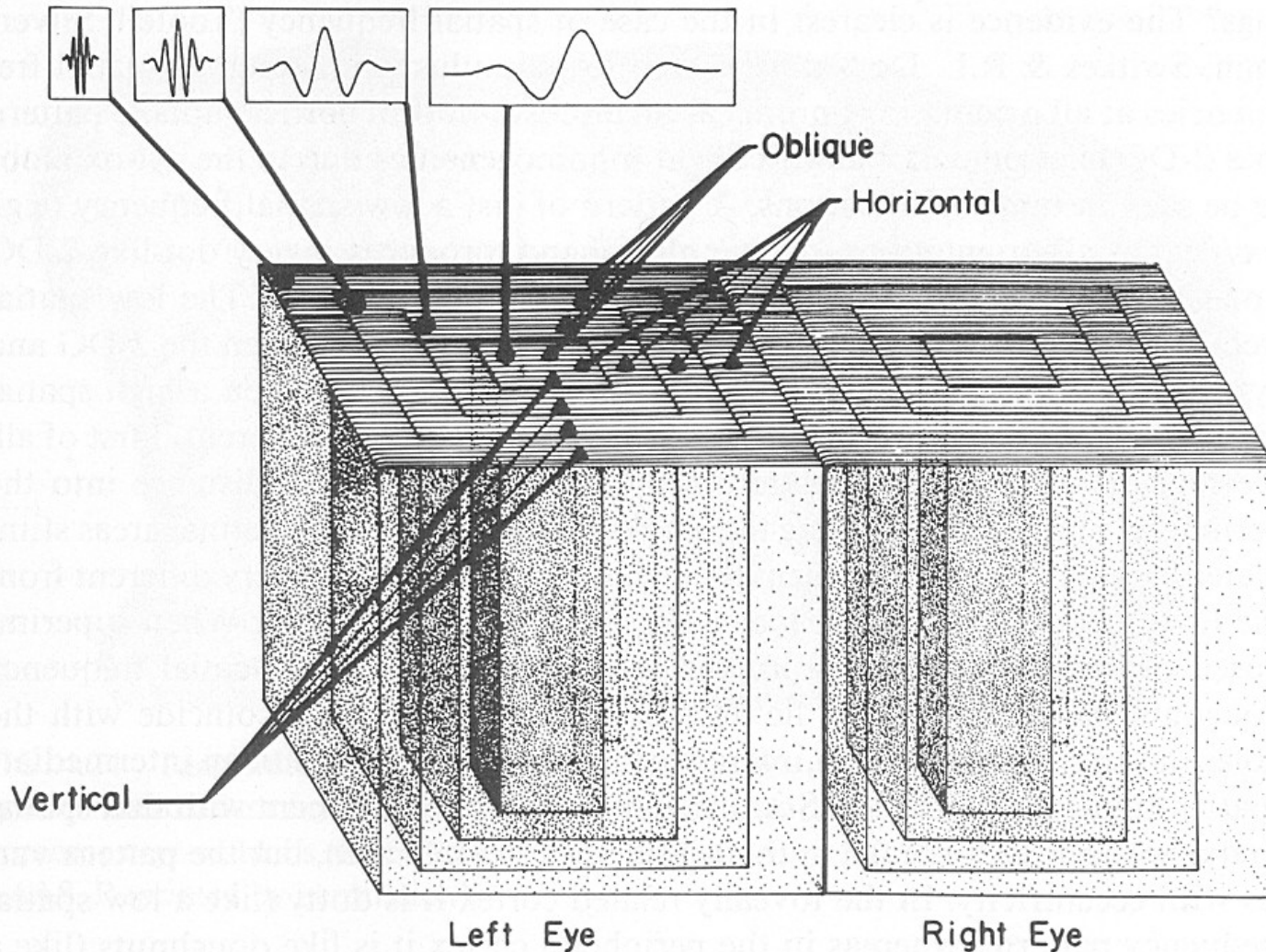
$$\begin{pmatrix} u \\ v \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

These are useful when we convolve with 2D images, e.g. to detect edges at different orientations.

Oriented Gaussian First and Second Derivatives



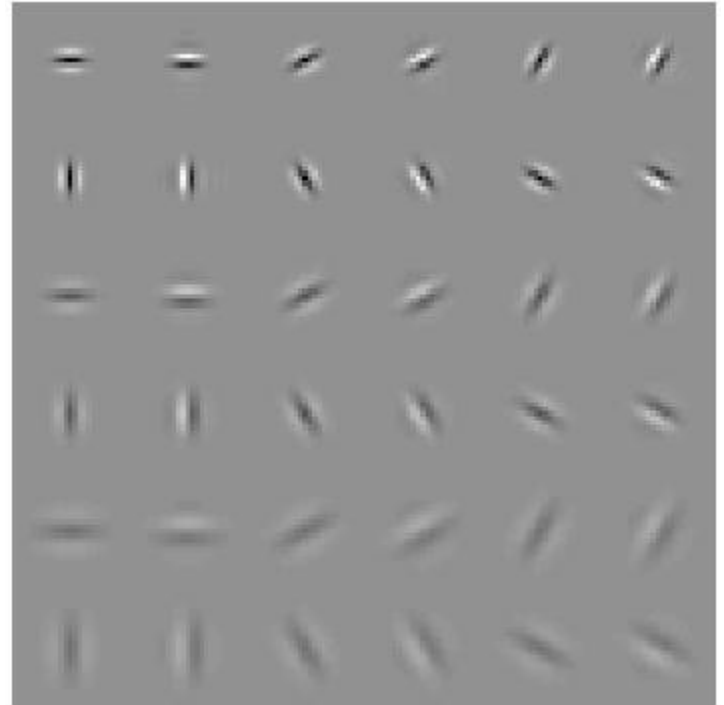
Hypercolumns in visual cortex



Model of Striate Module in Monkeys

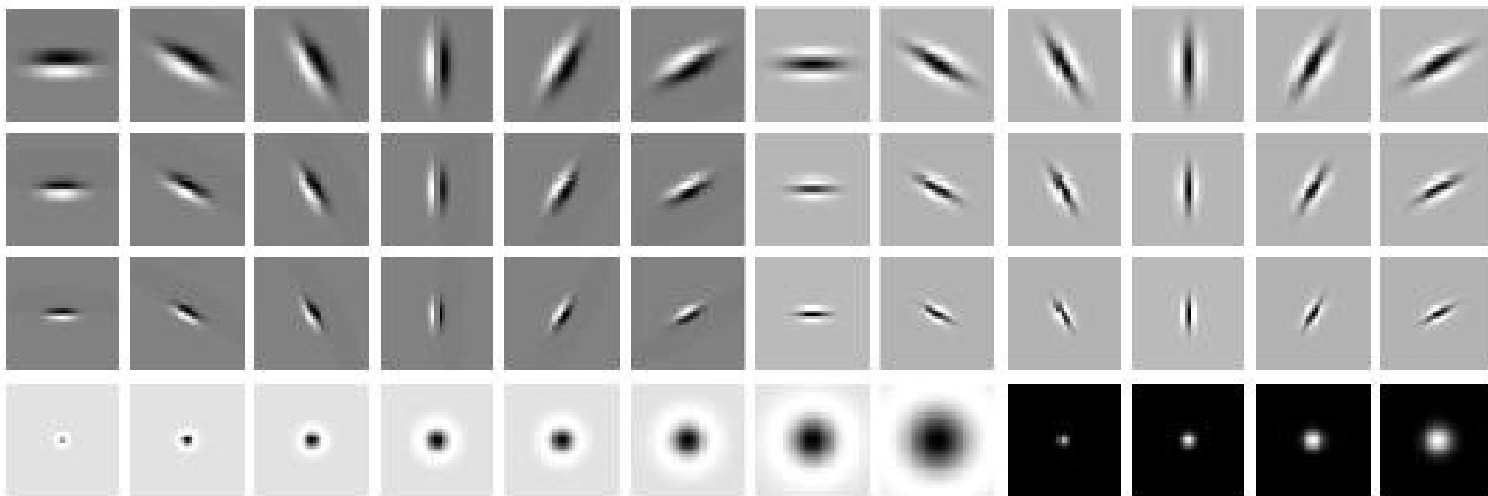
Modeling hypercolumns

- Elongated directional Gaussian derivatives
- Gabor filters could be used instead
- Multiple orientations, scales



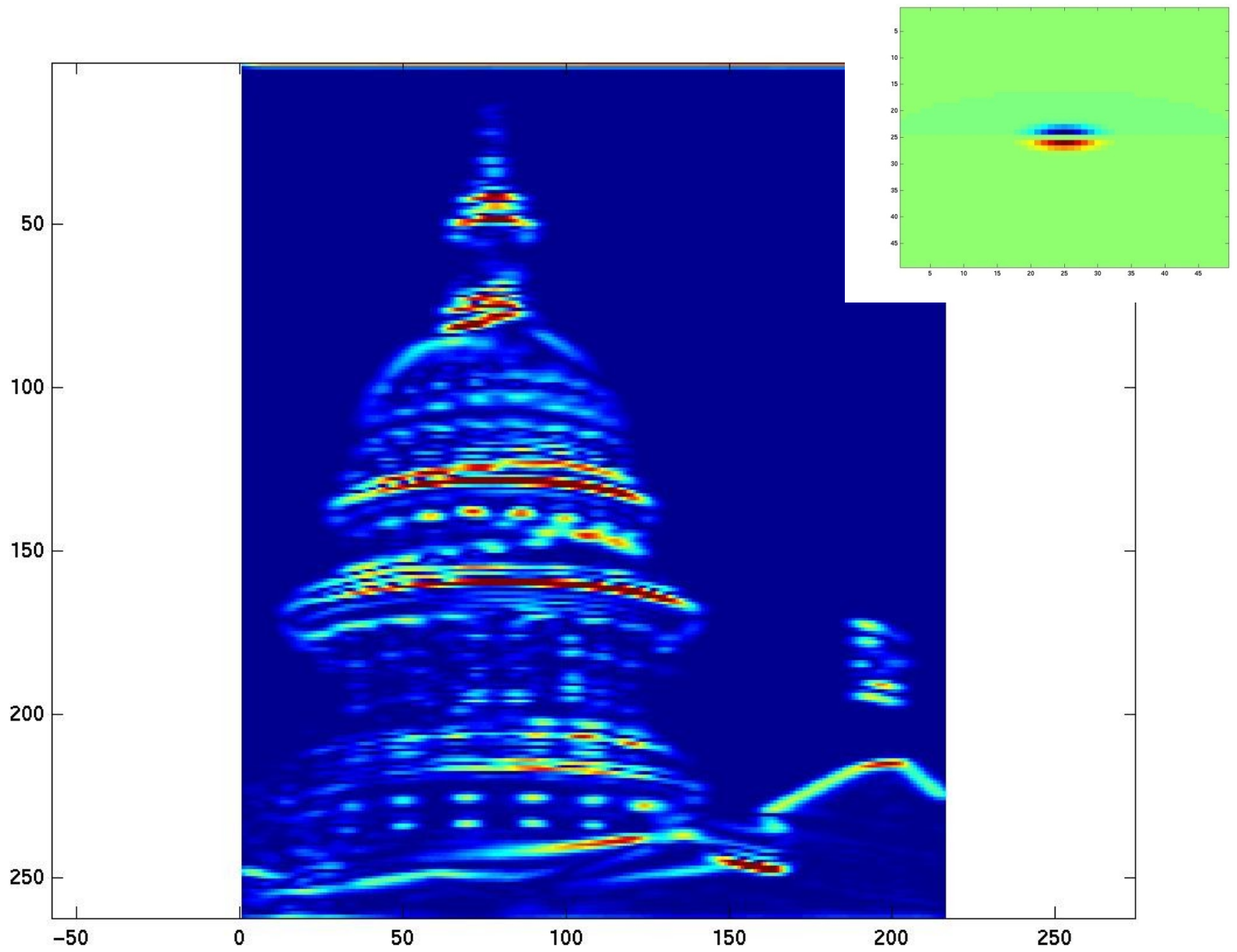
Overcomplete representation: filter banks

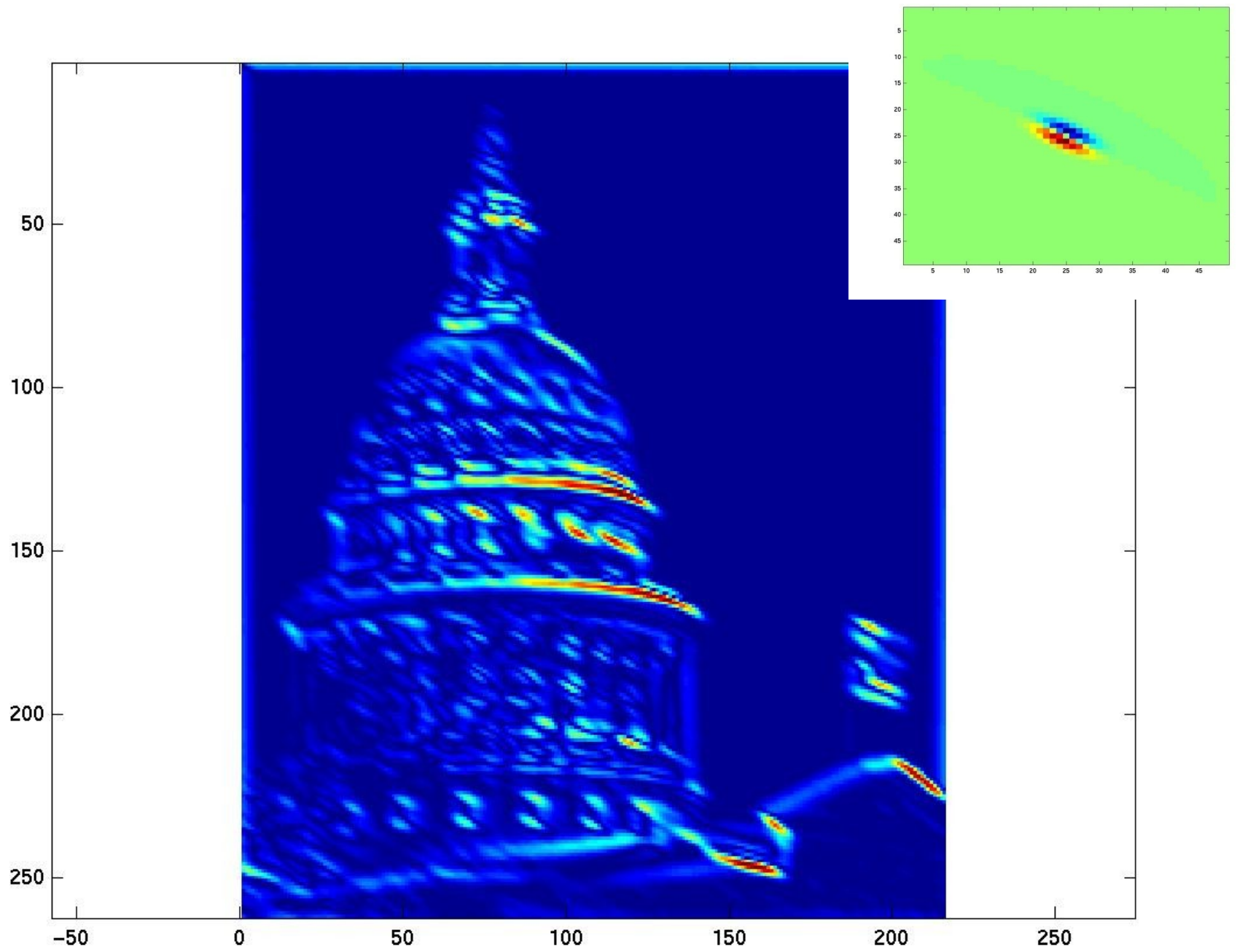
LM Filter Bank

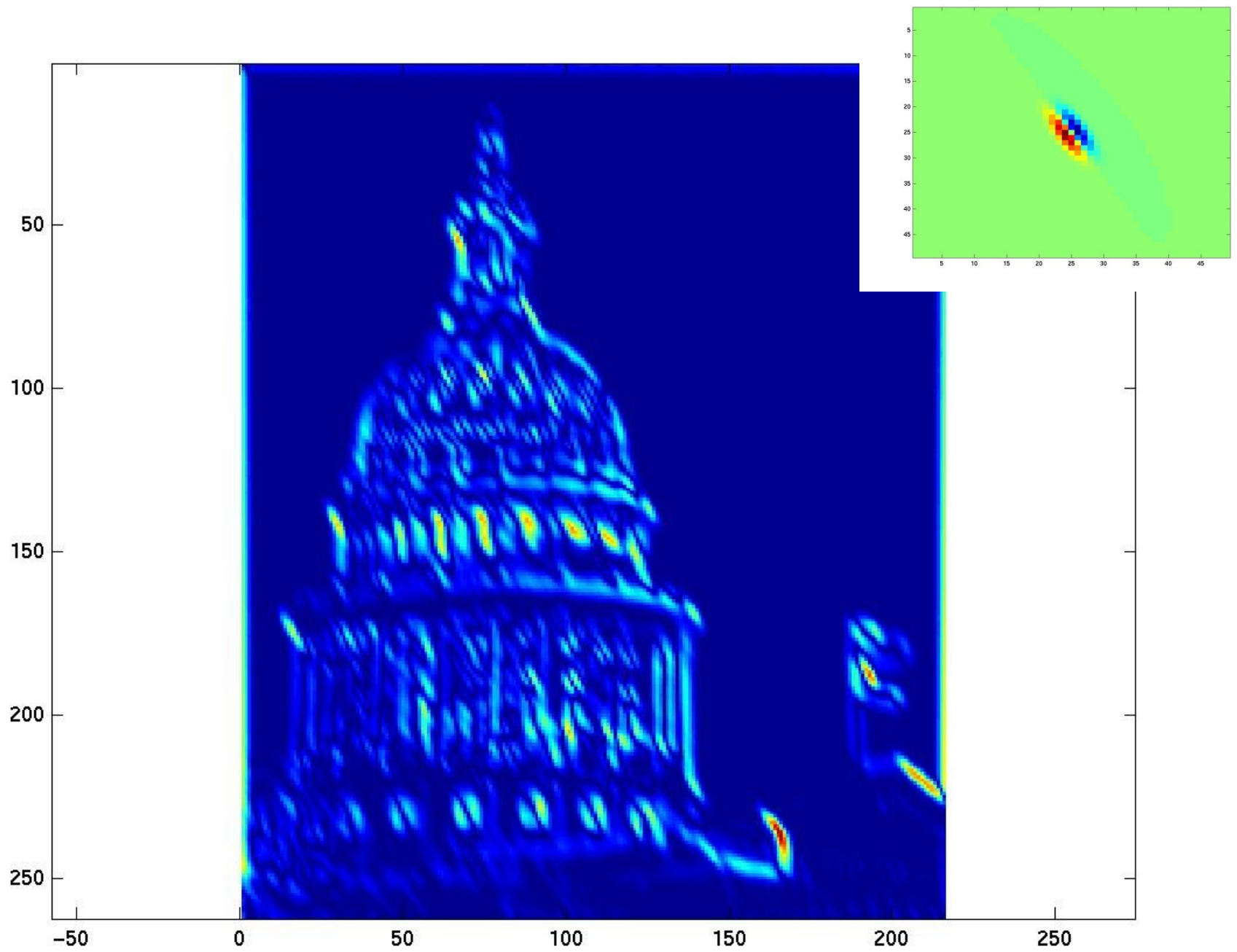


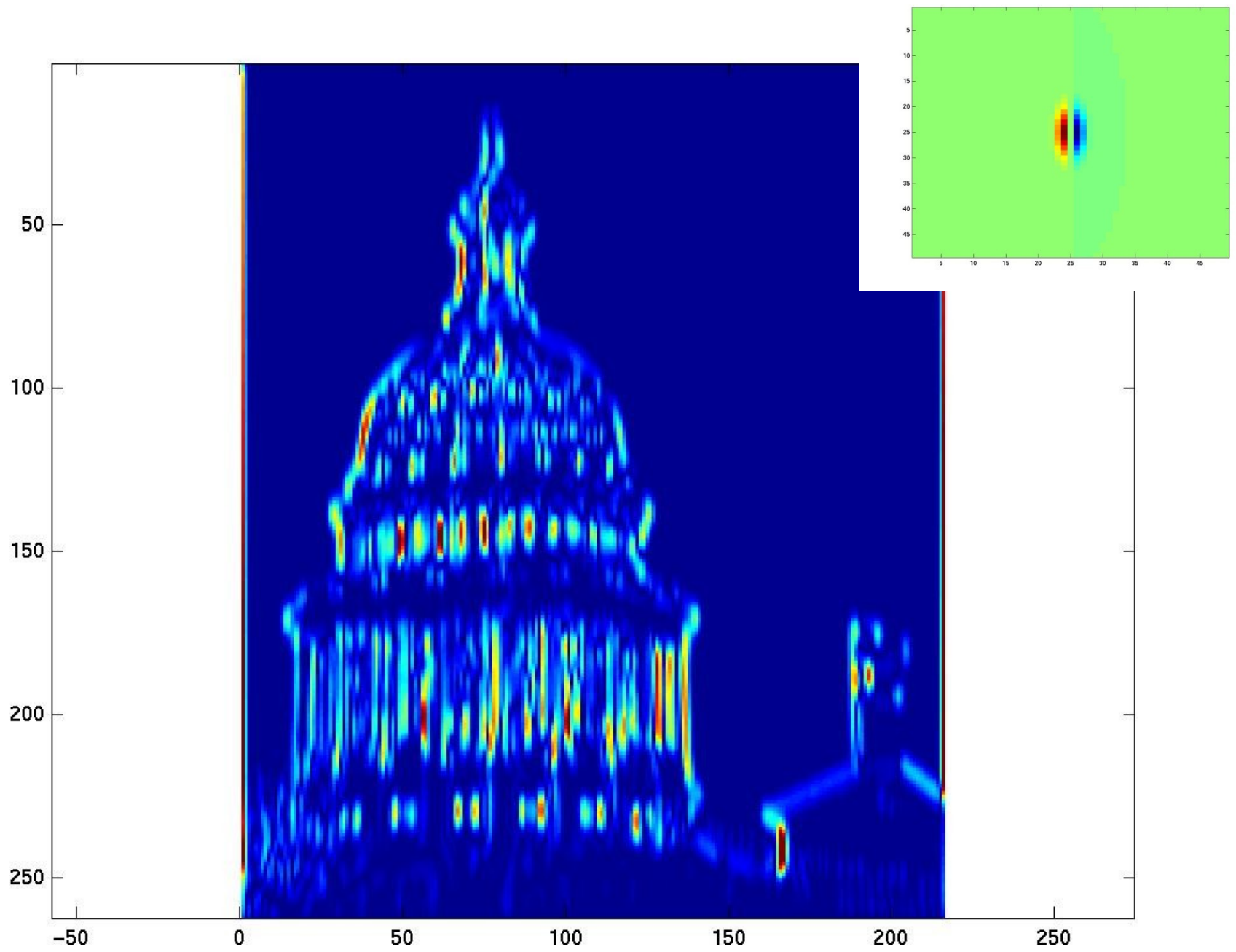
Code for filter banks: www.robots.ox.ac.uk/~vgg/research/texclass/filters.html

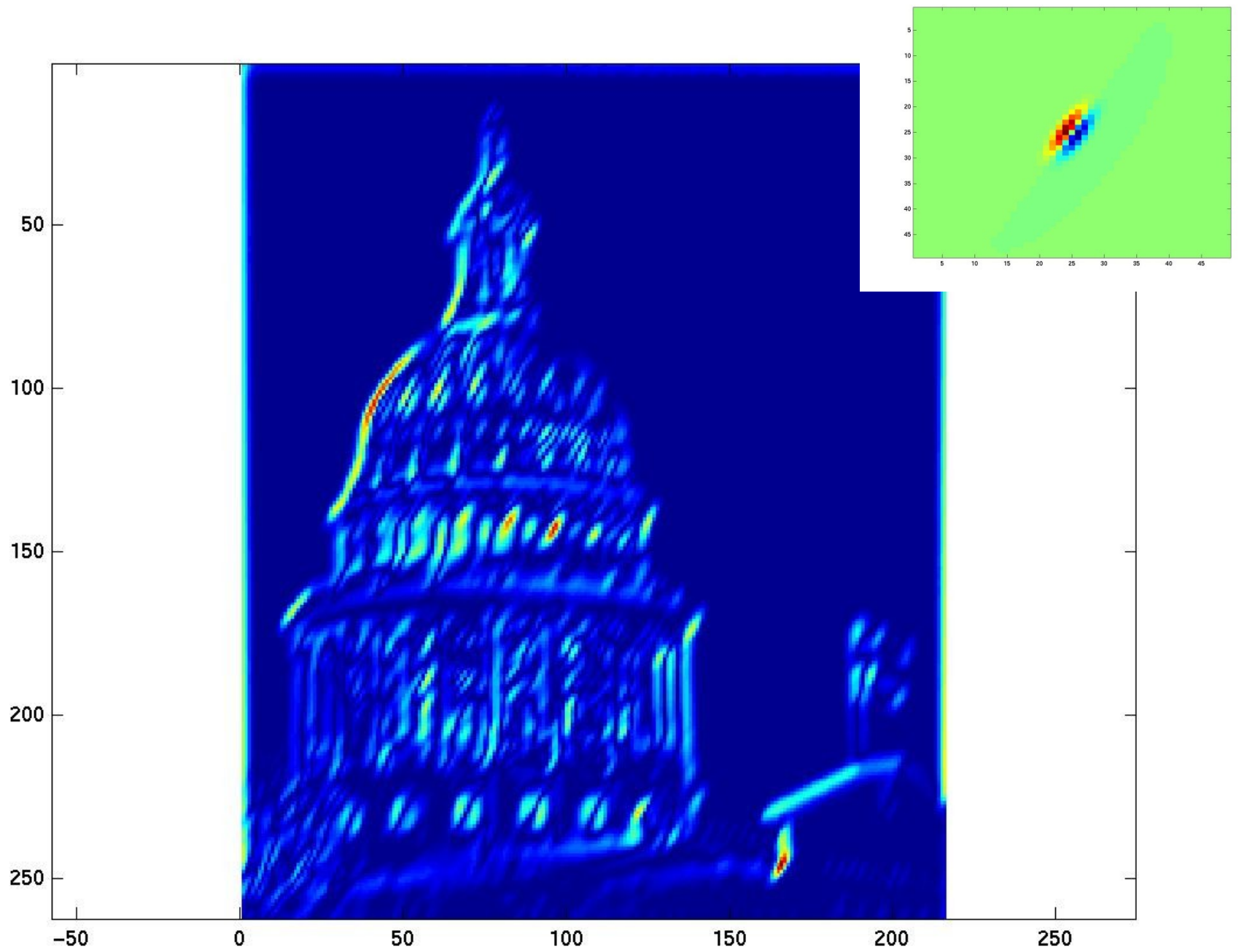


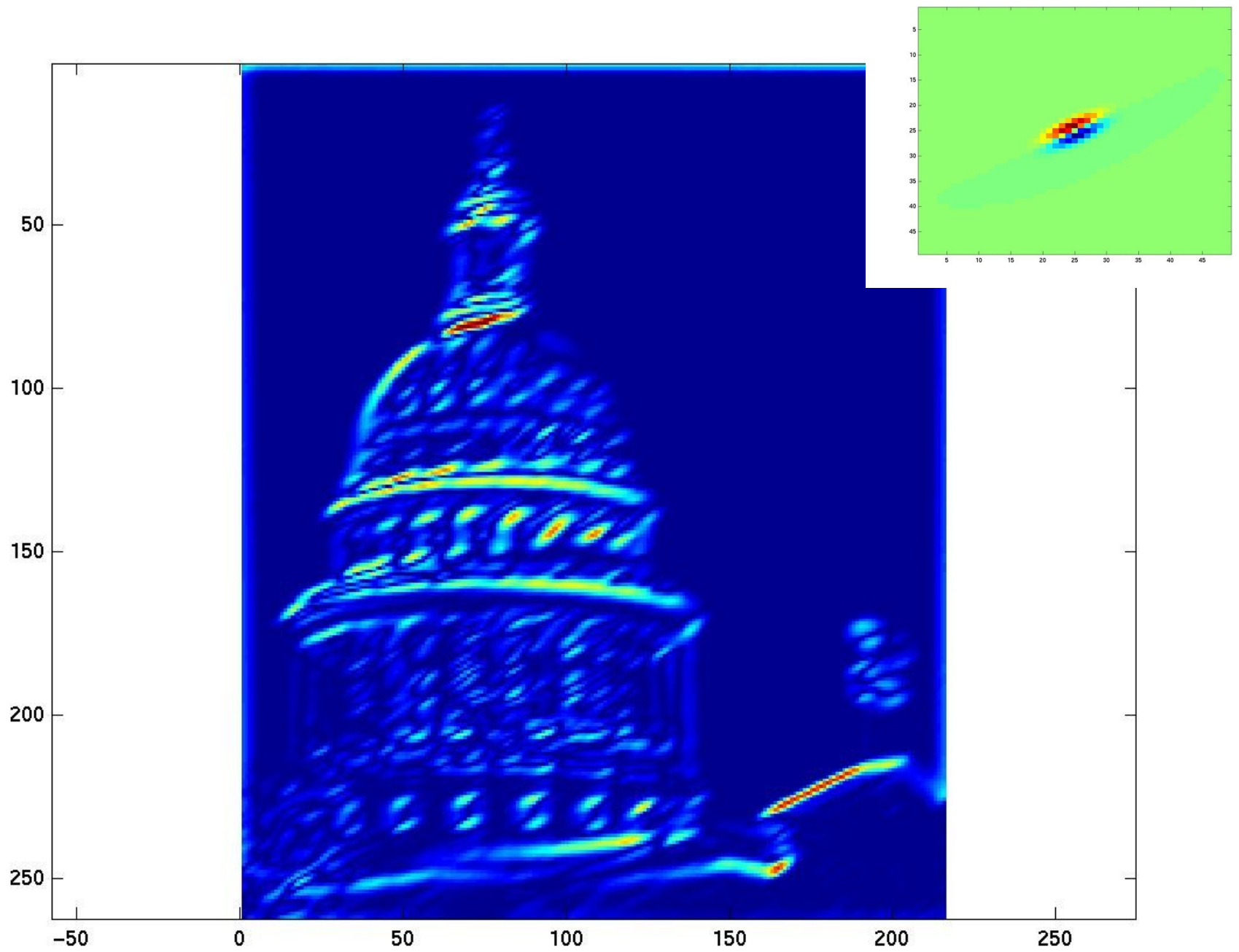


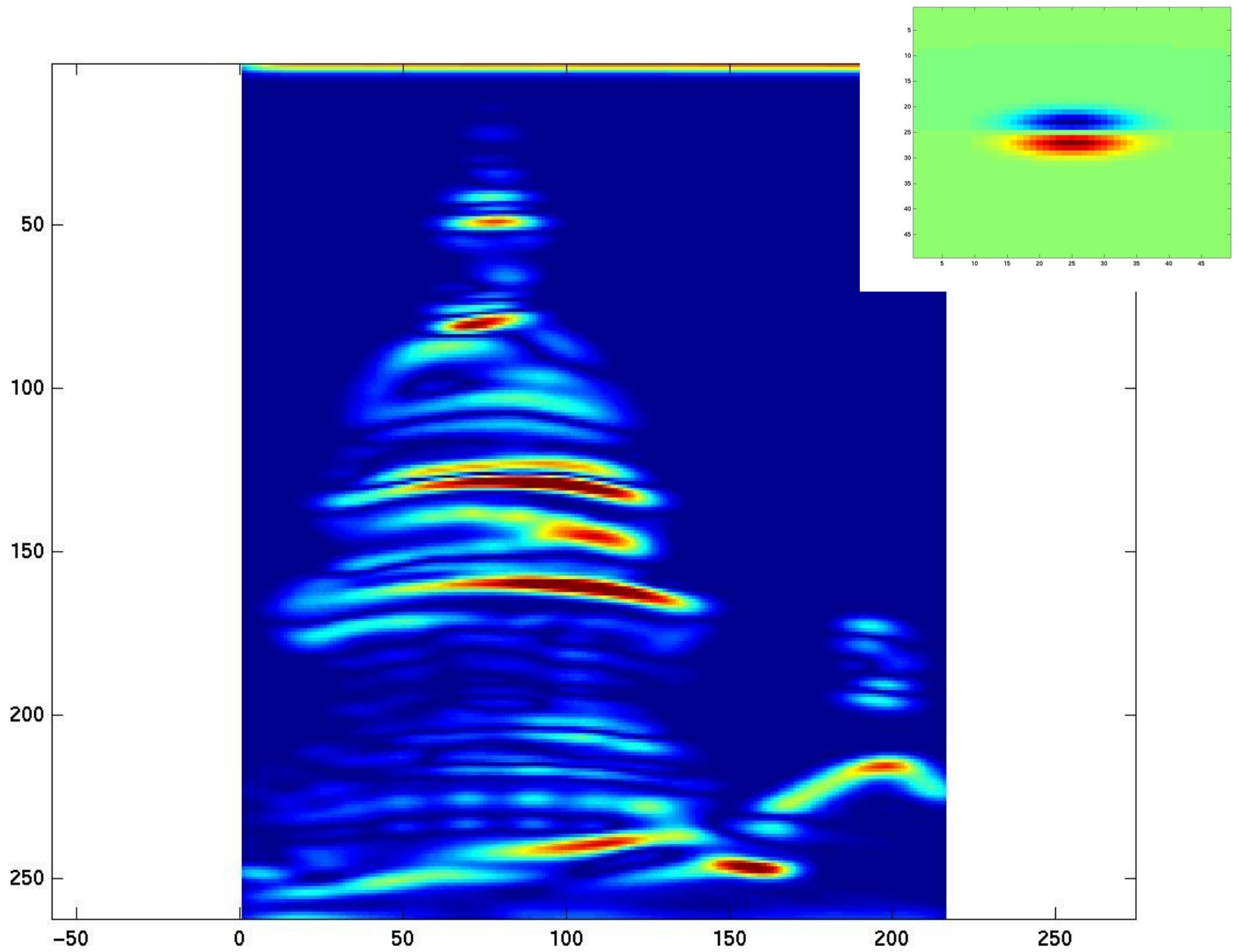


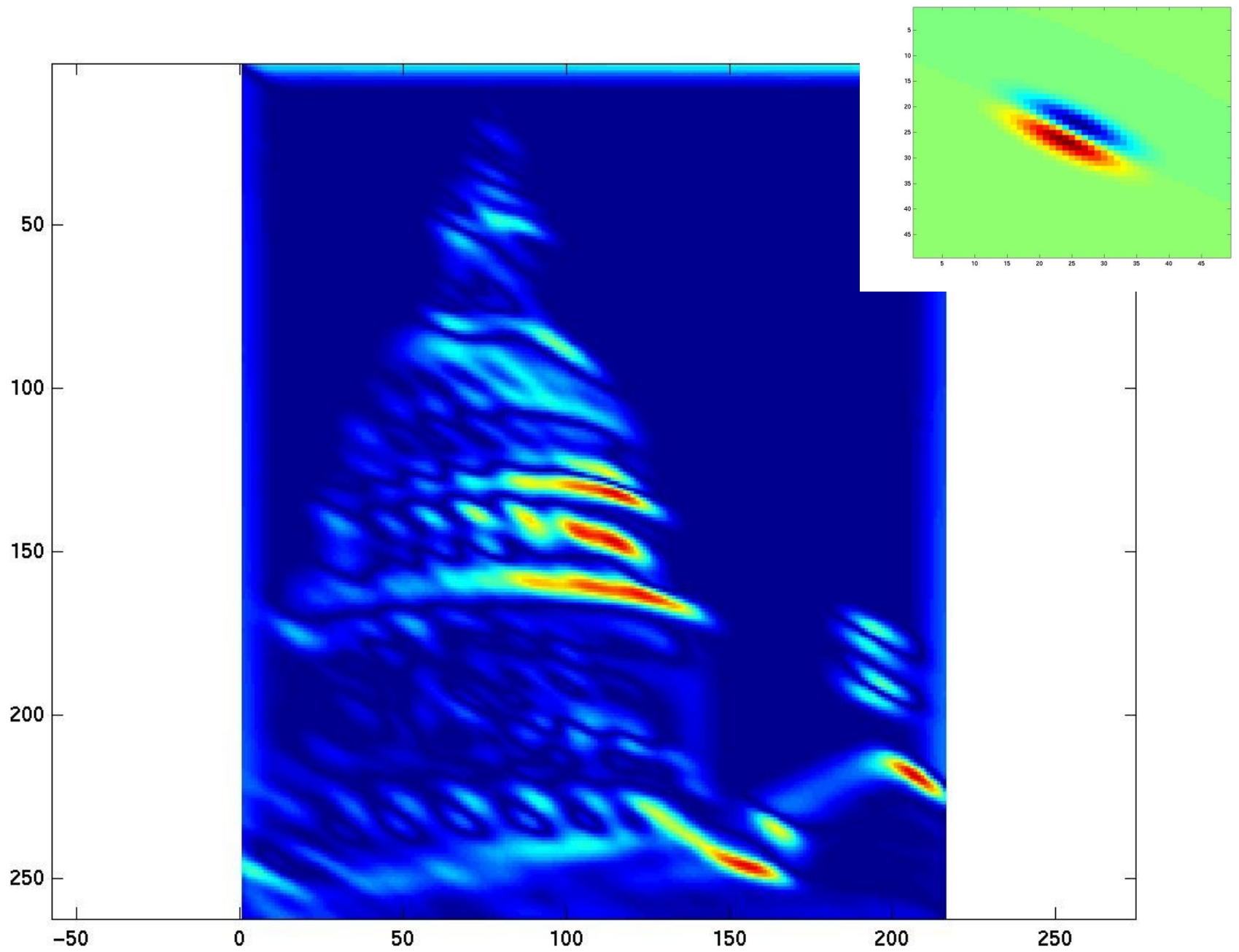


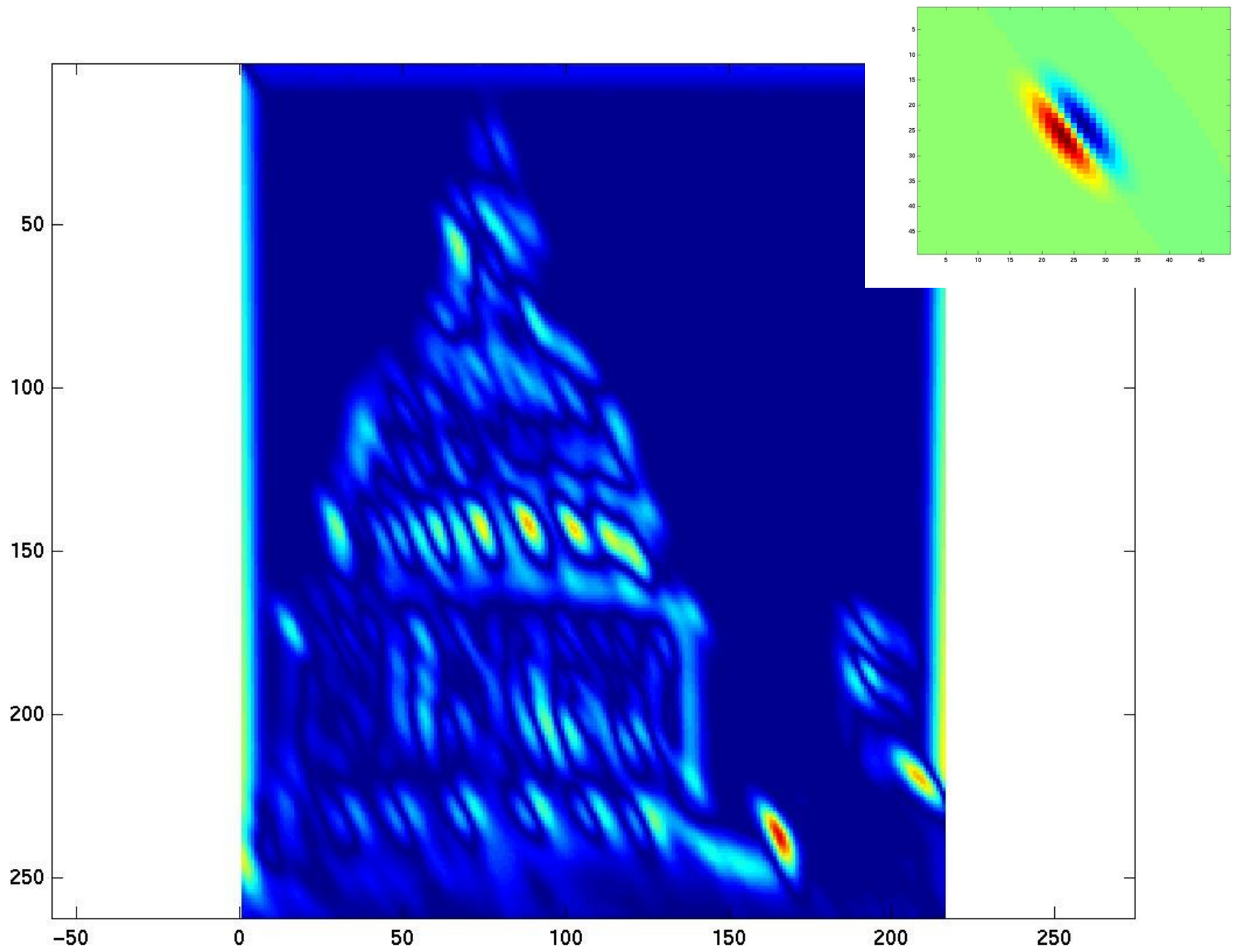


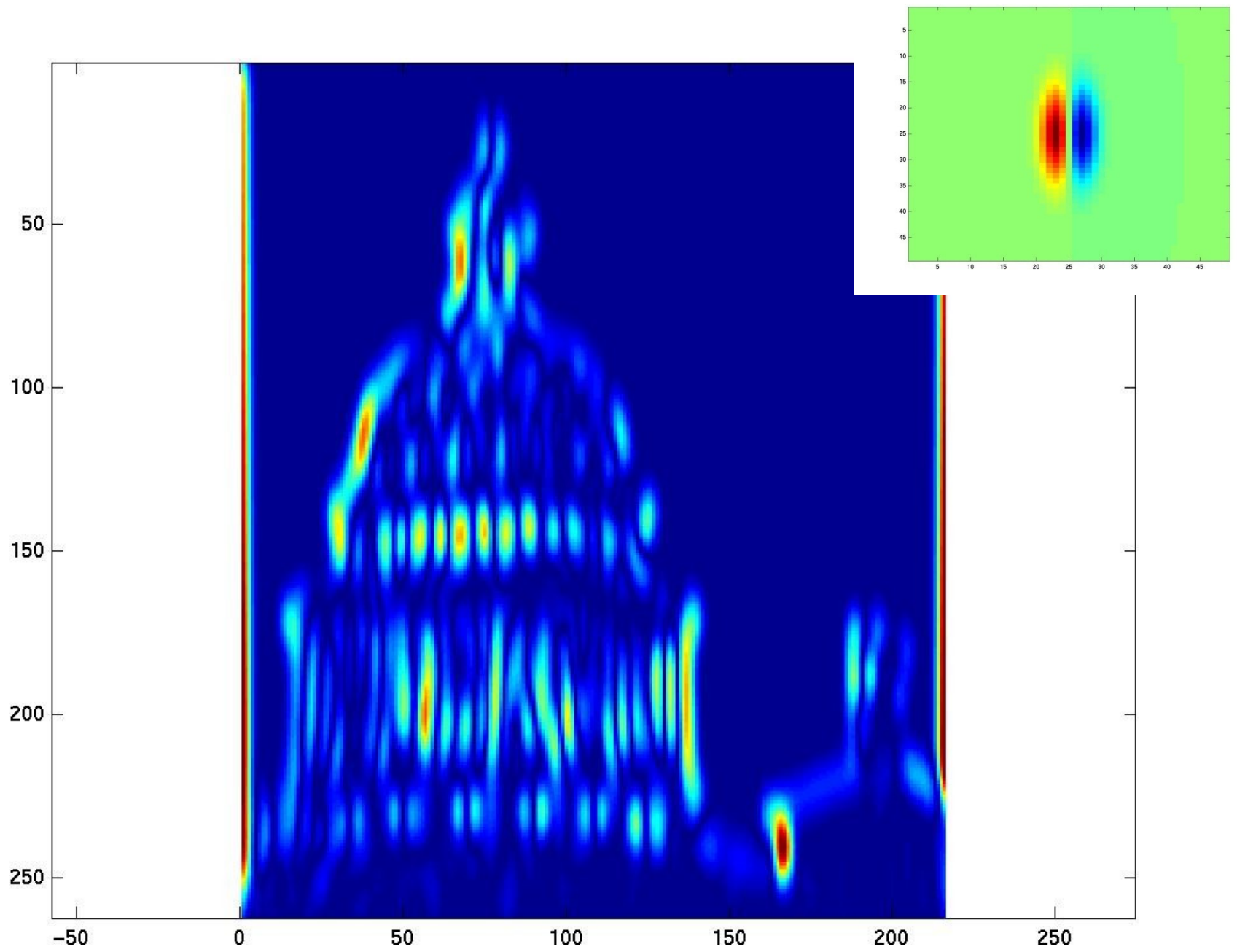


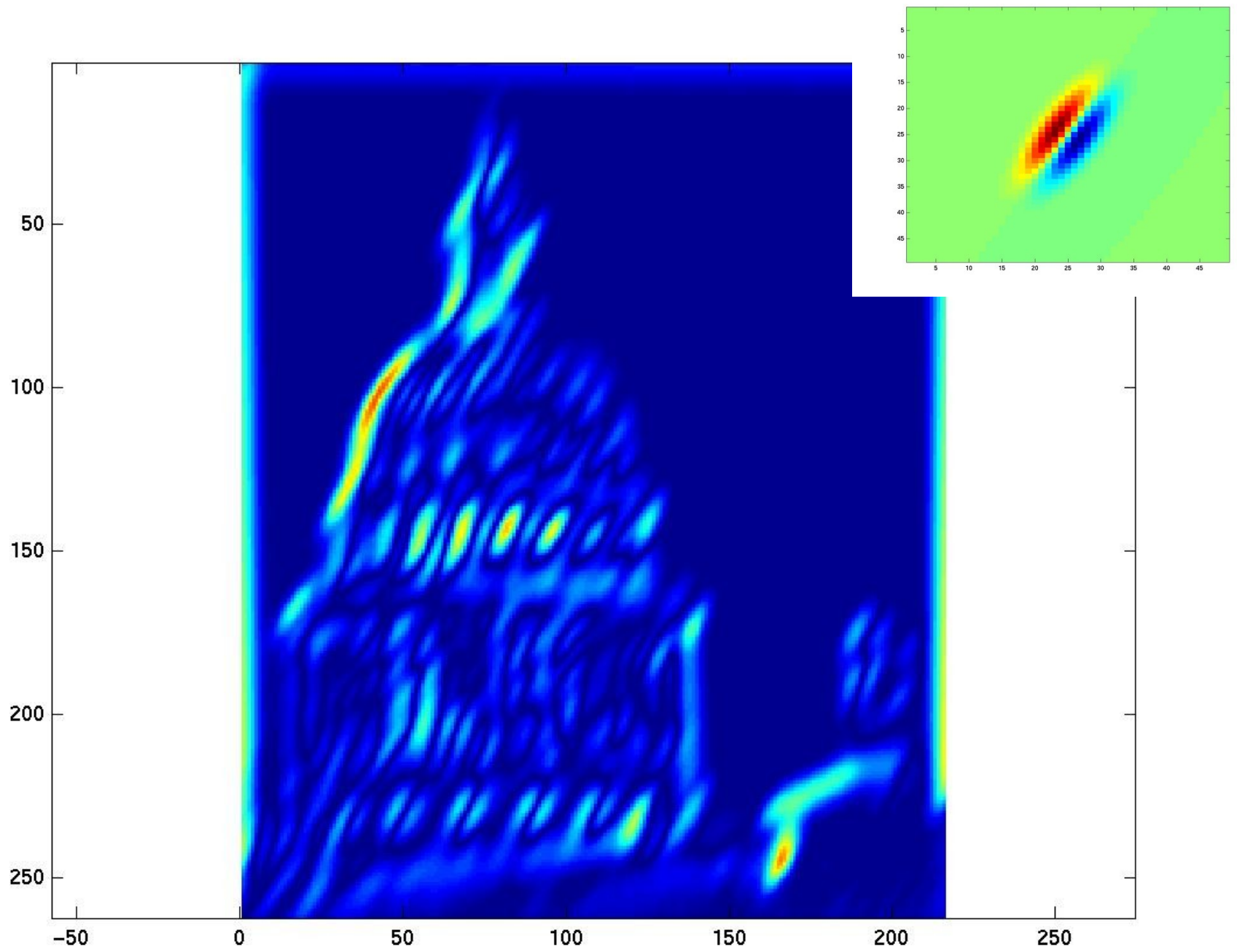


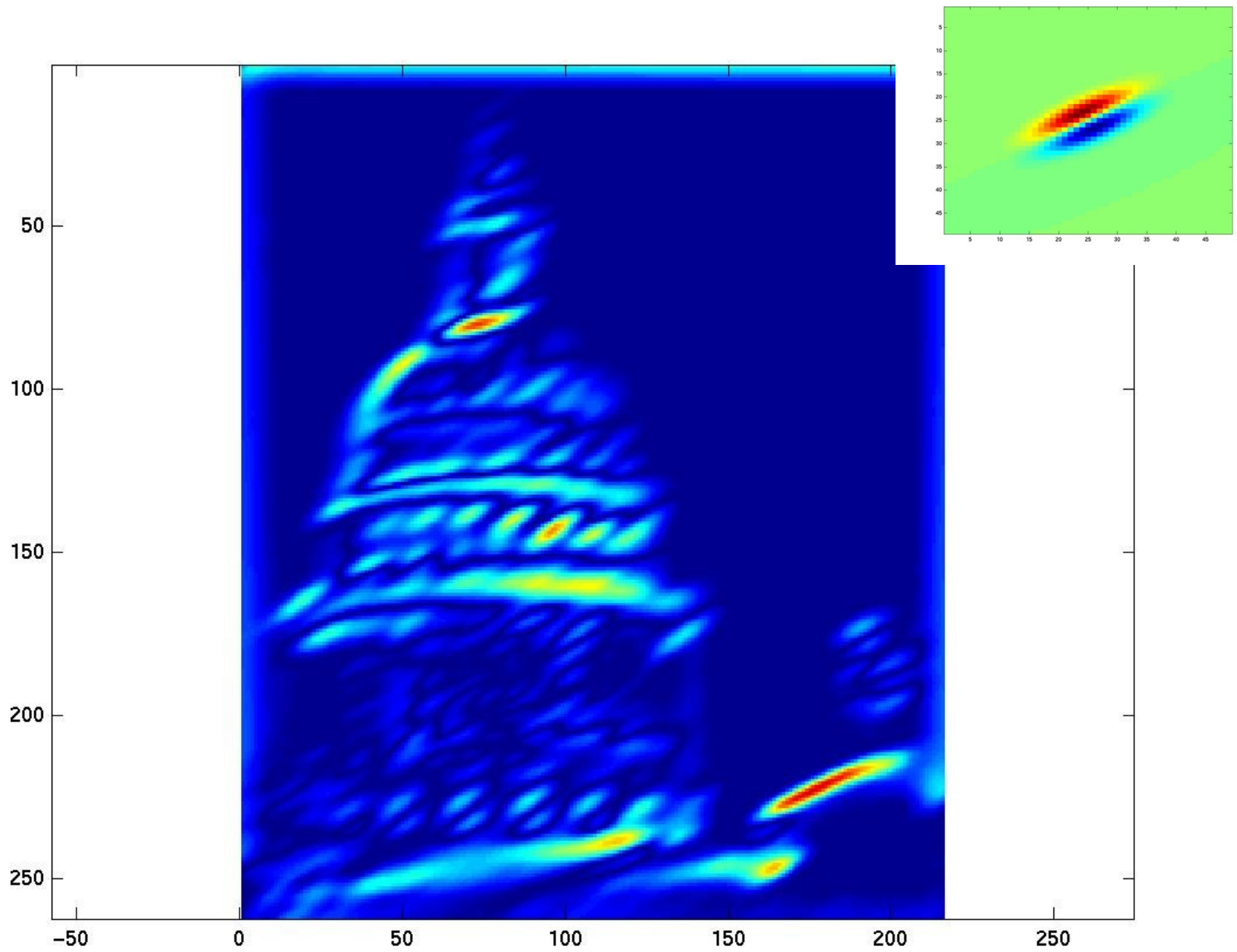


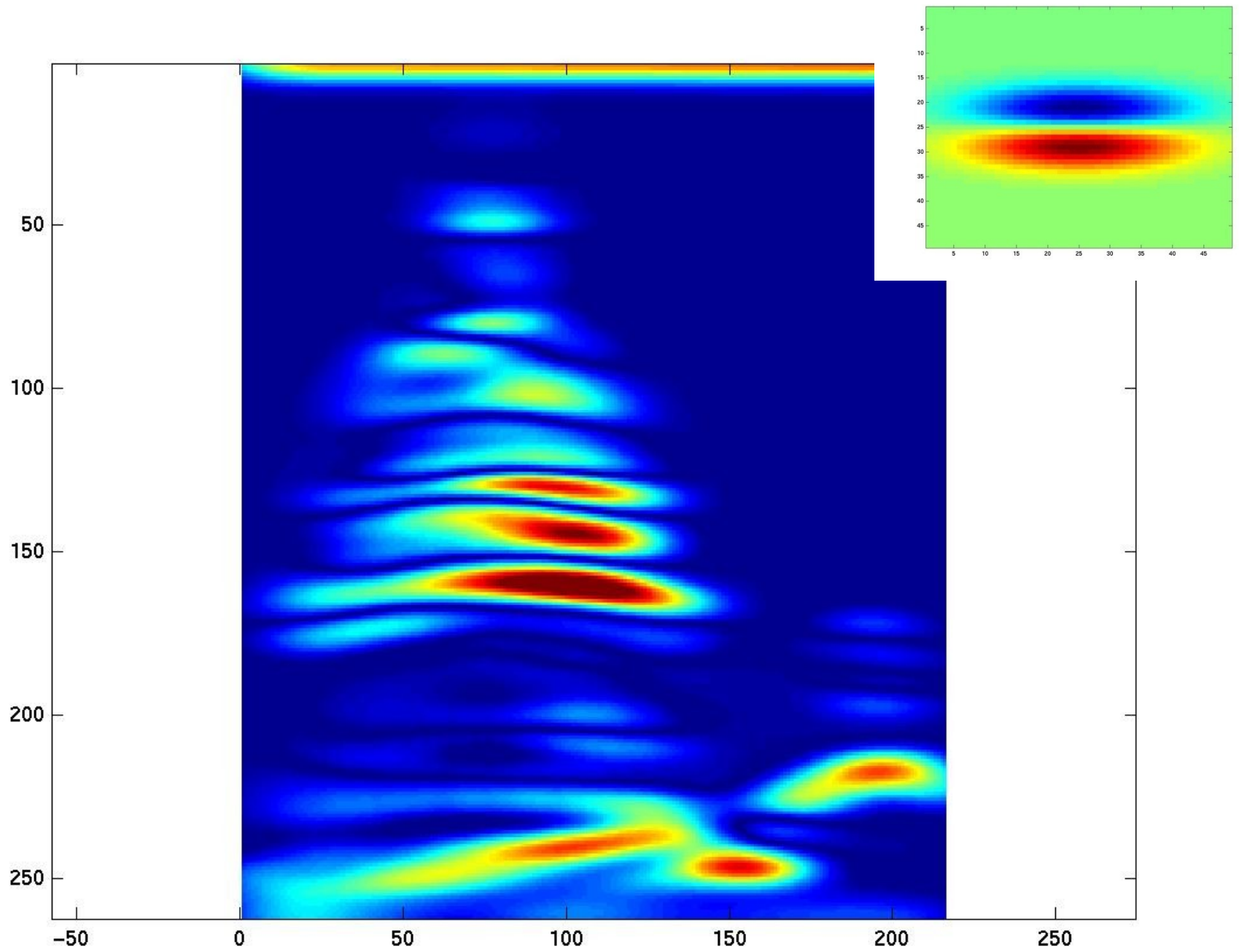


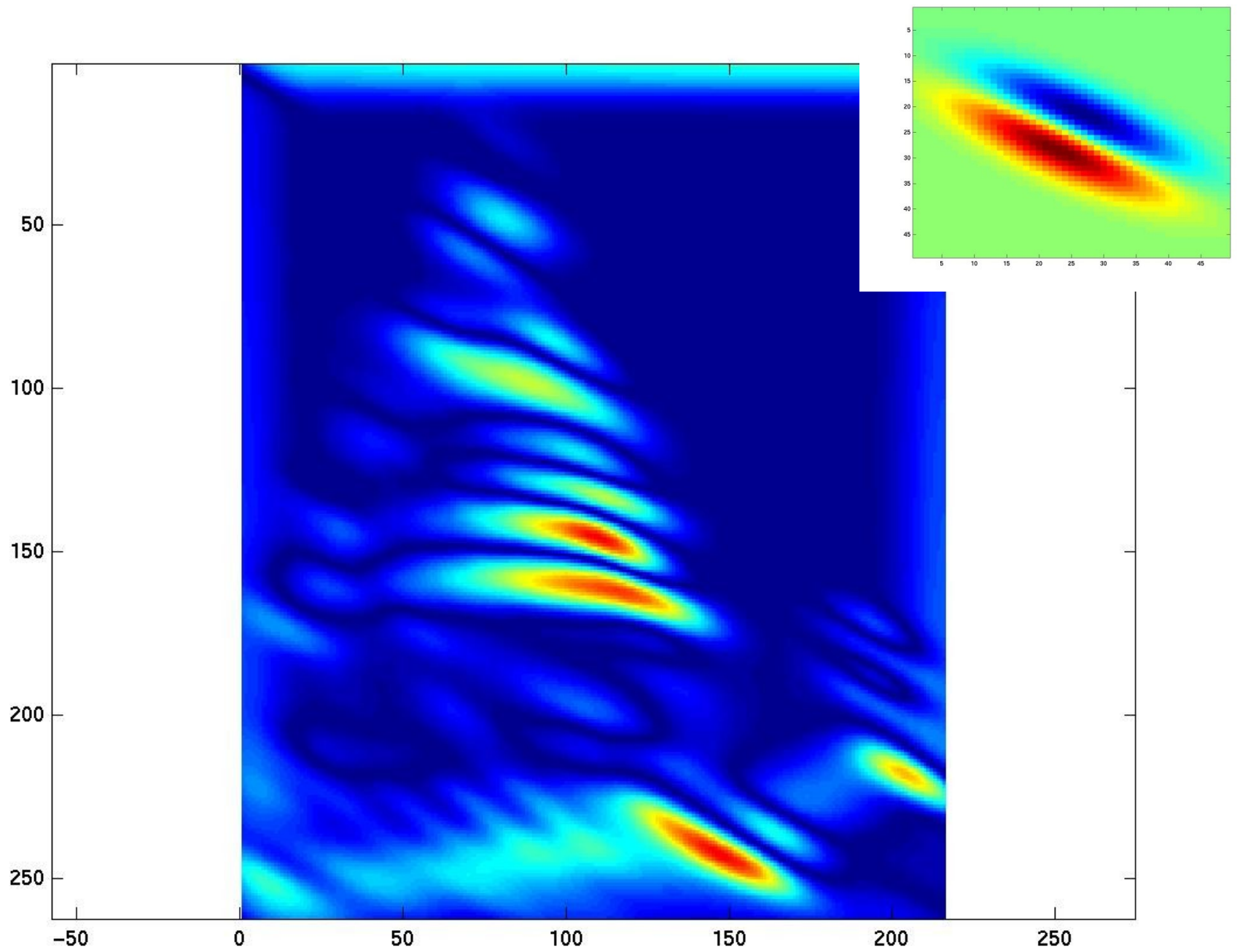


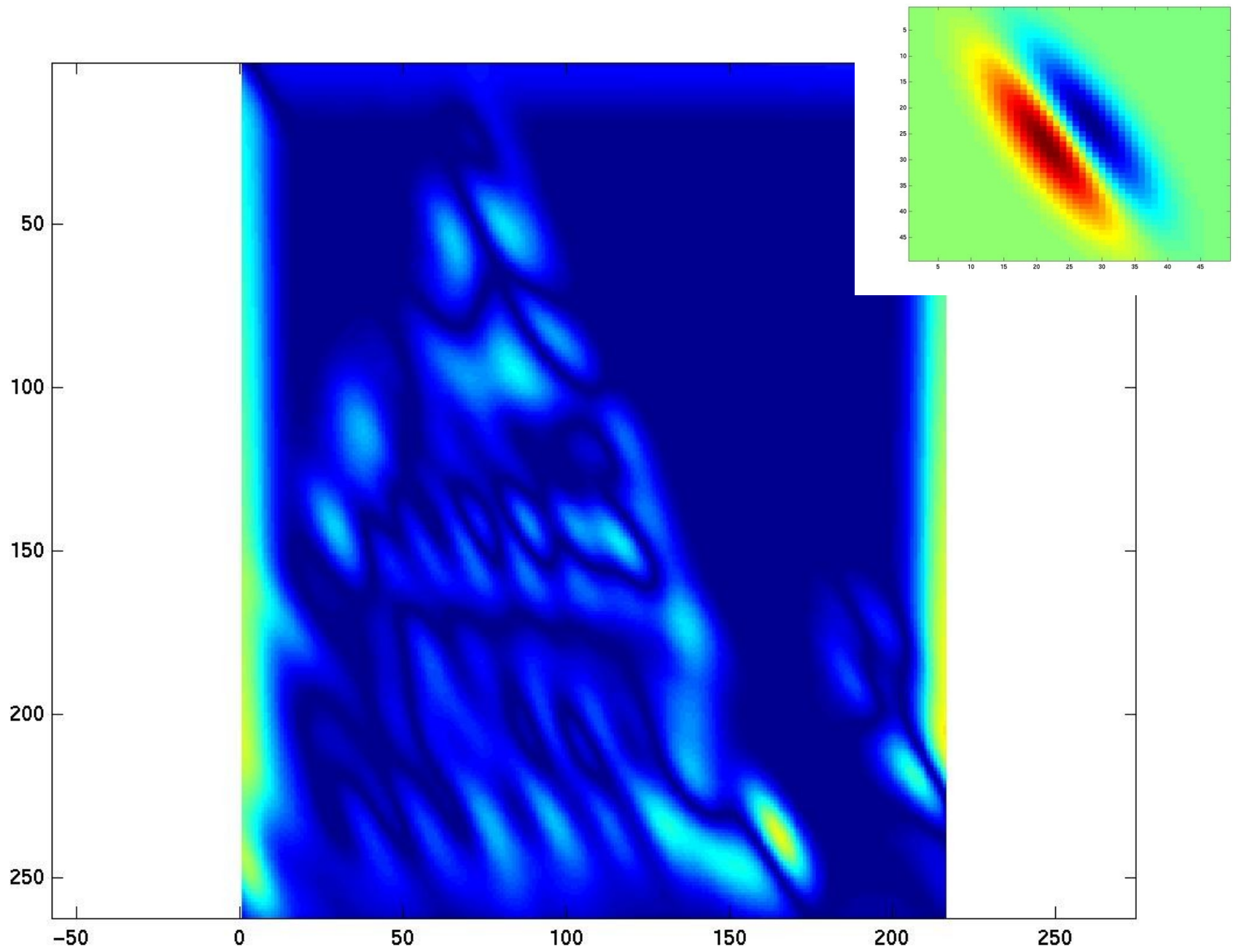


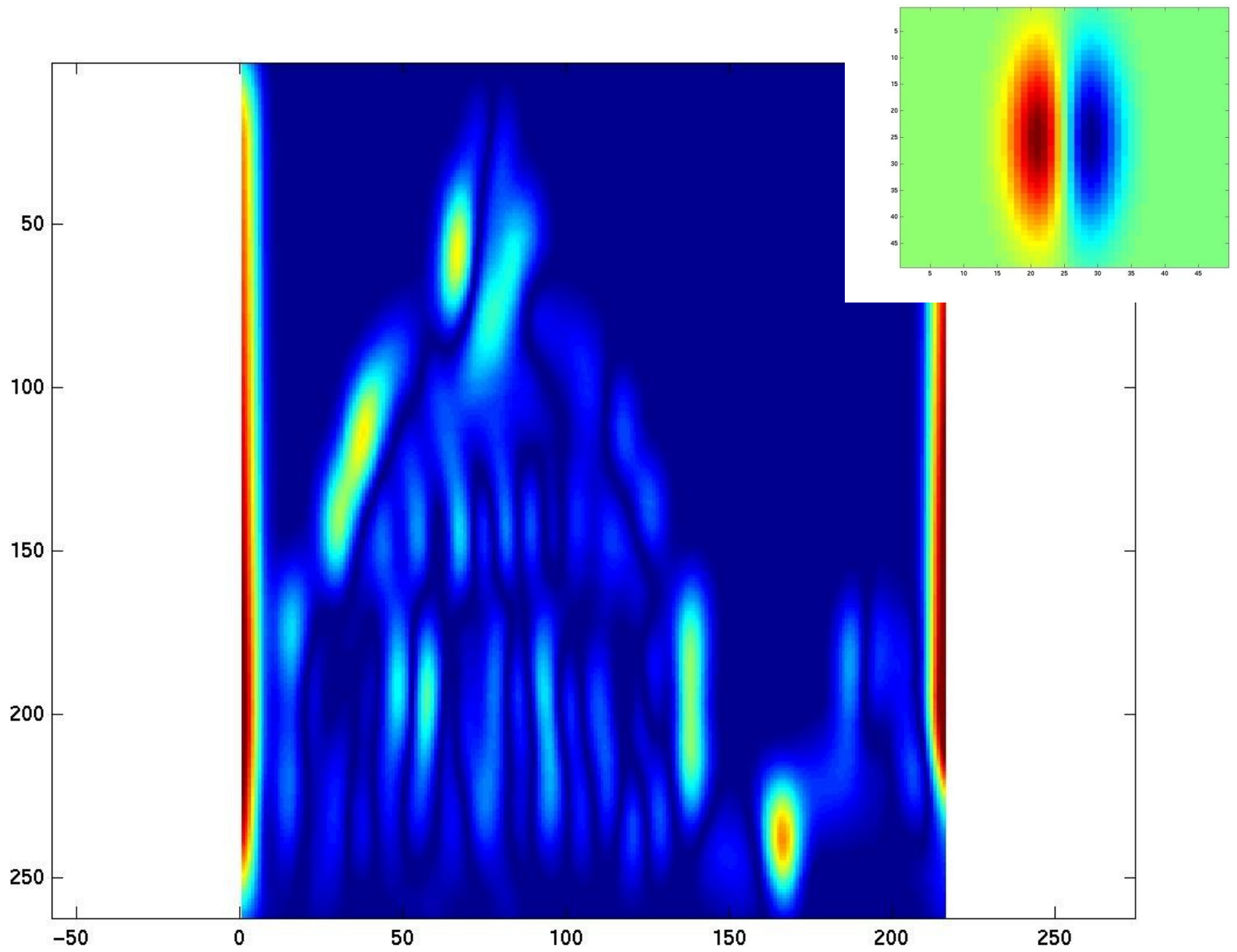


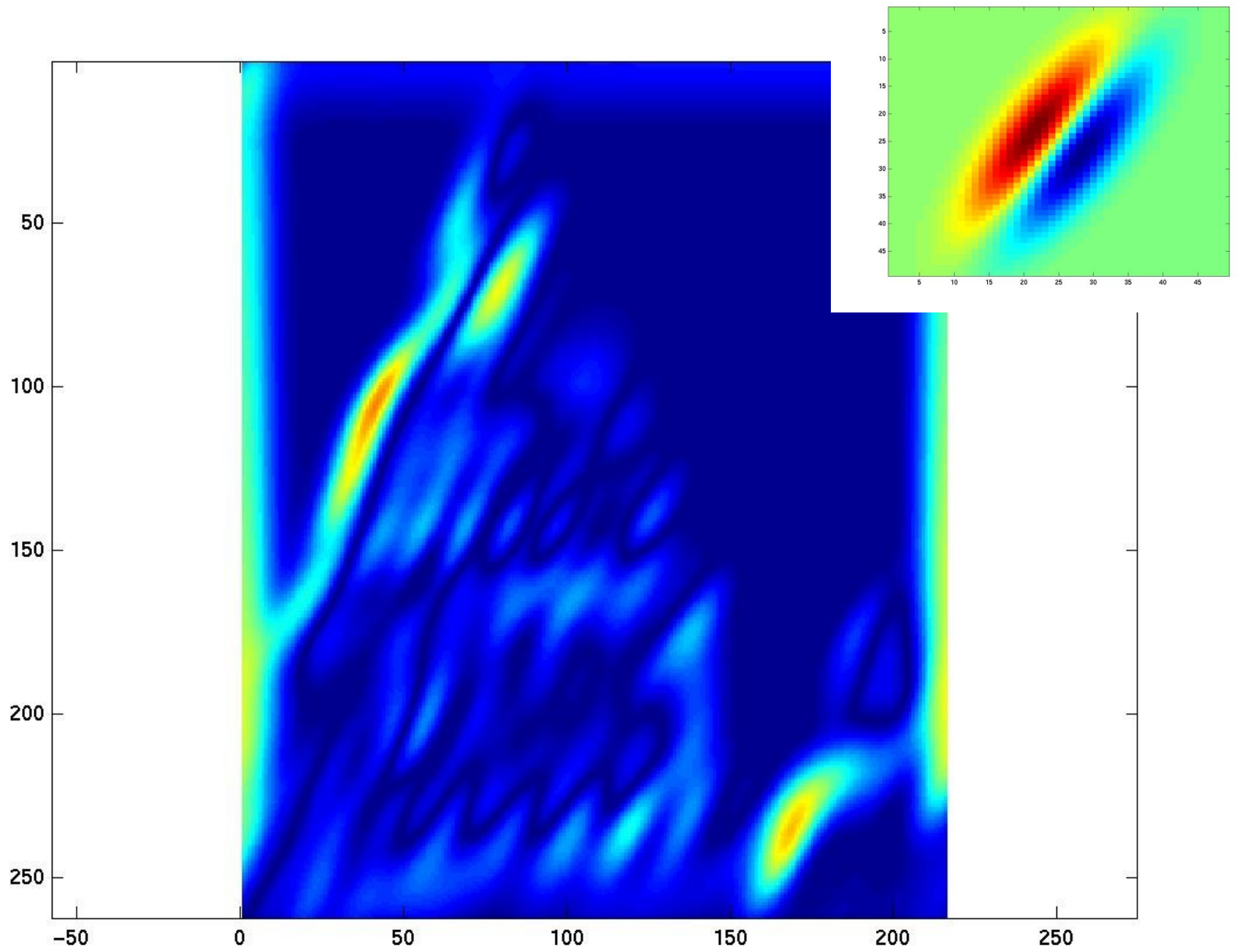


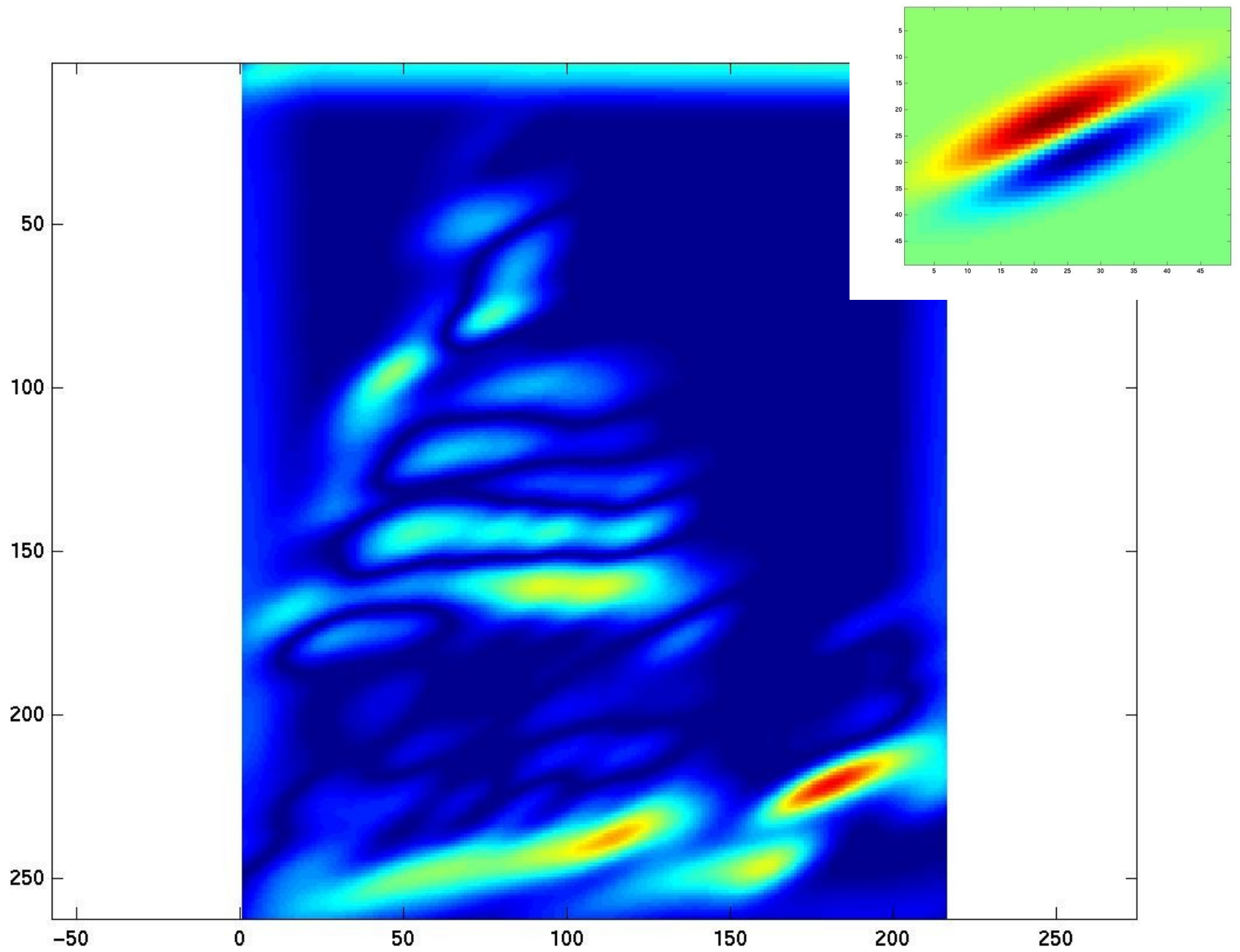


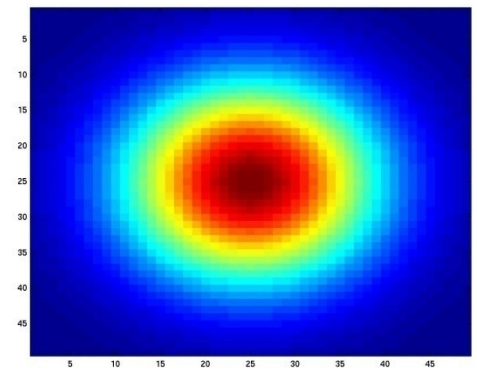
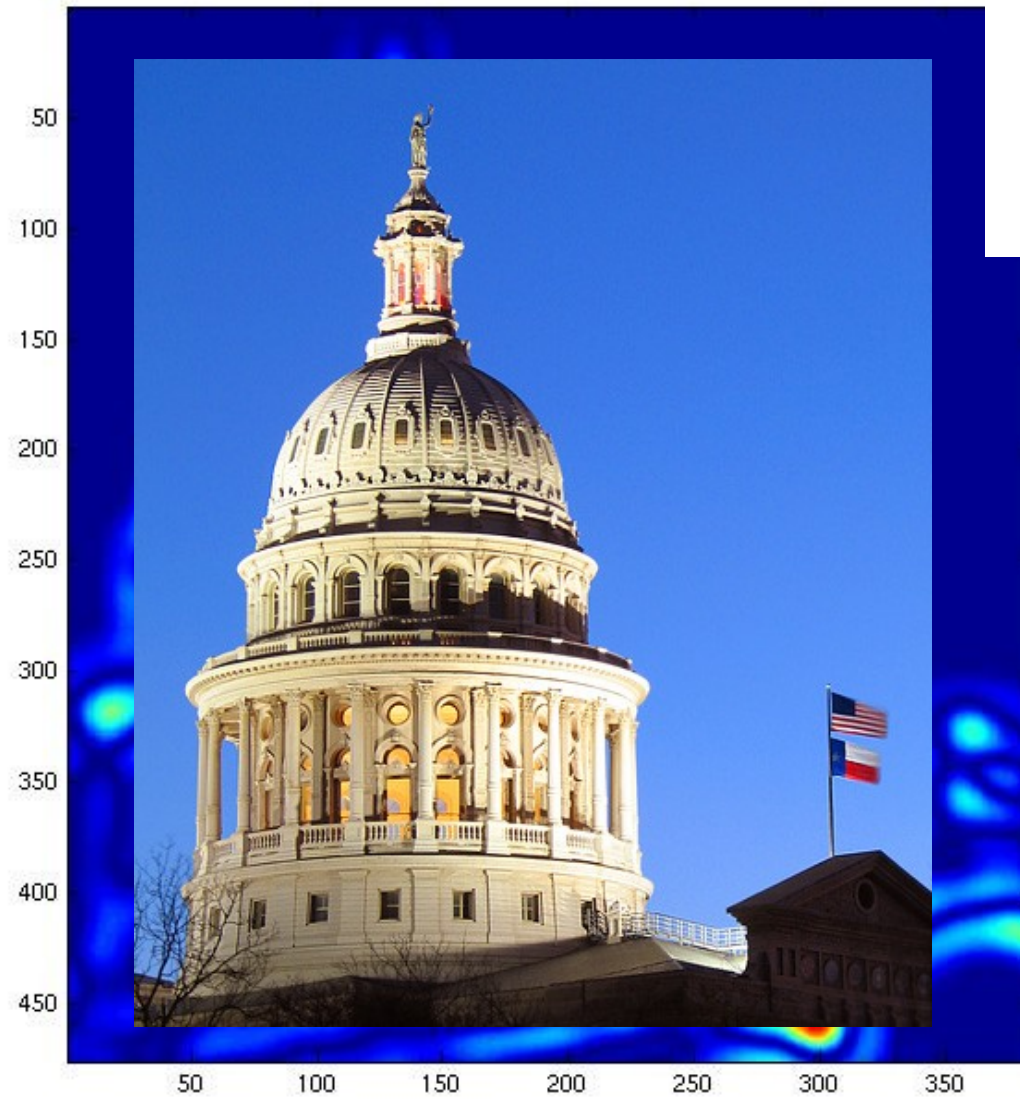








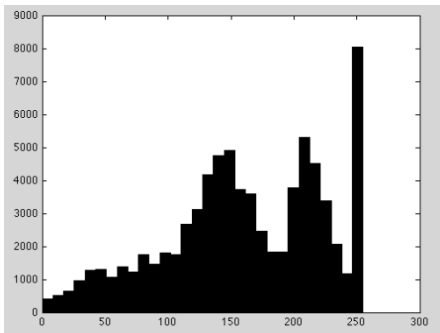




Two questions of texture modeling

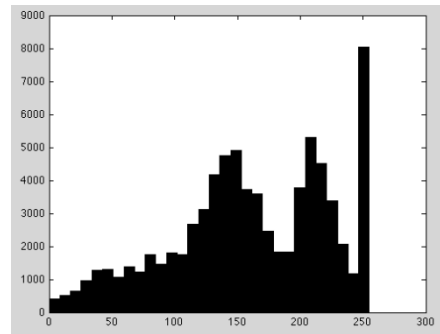
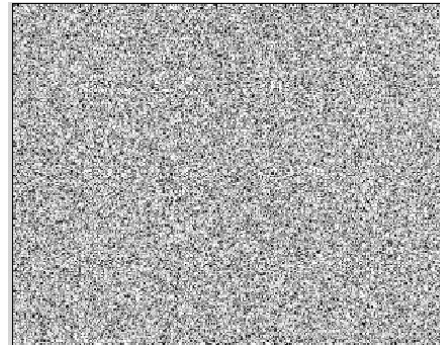
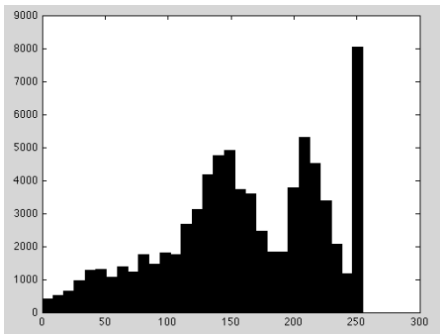
- What are the texture features (textons)?
 - Pixels
 - Pixel patches
 - Outputs of V1-like filters
 - Clusters of patches / filter outputs
 - CNN features
 - Etc.
- How do we aggregate statistics
 - Various types of histograms
 - Implicit or explicit

Pixel Histograms



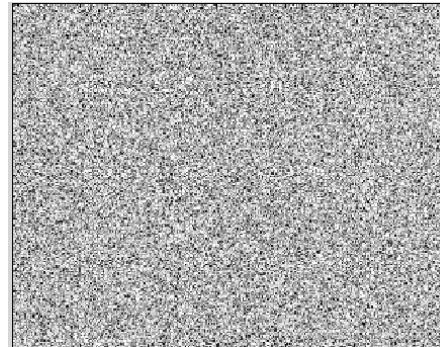
Slide by Erik Learned-Miller

Pixel Histograms



Slide by Erik Learned-Miller

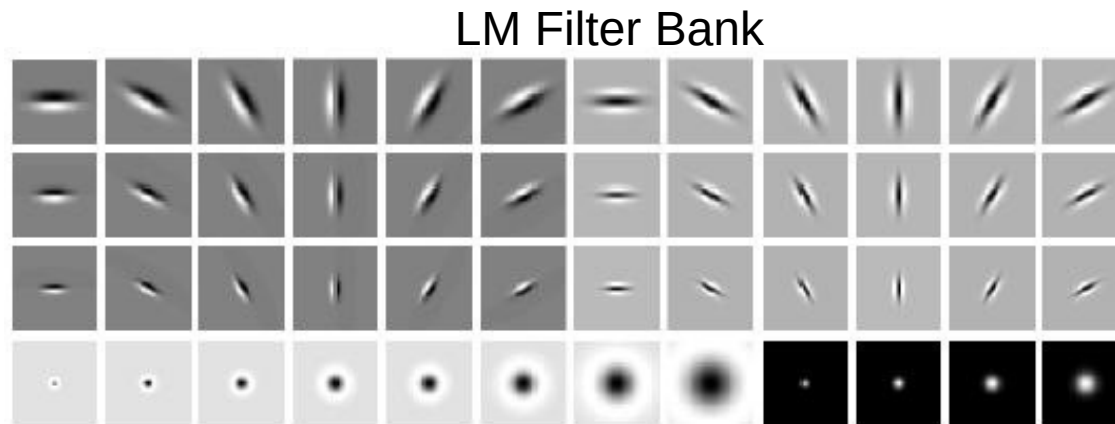
Gray value histogram comparisons



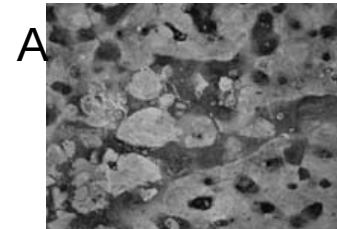
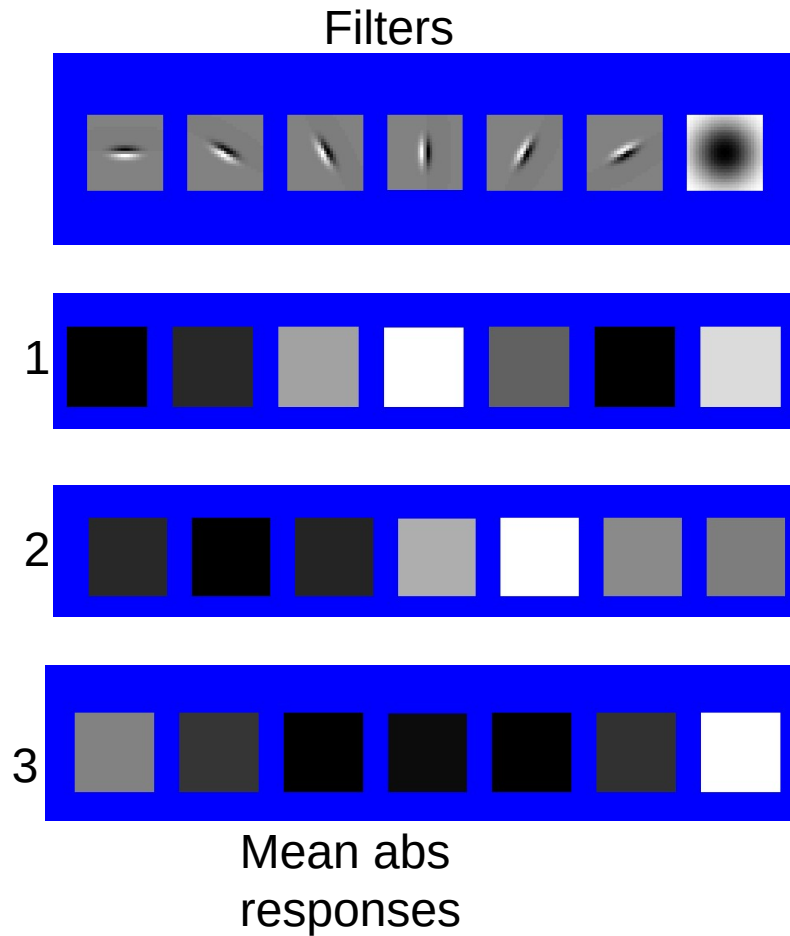
They're equal

Slide by Erik Learned-Miller

Going up from pixels: V1 filter-banks

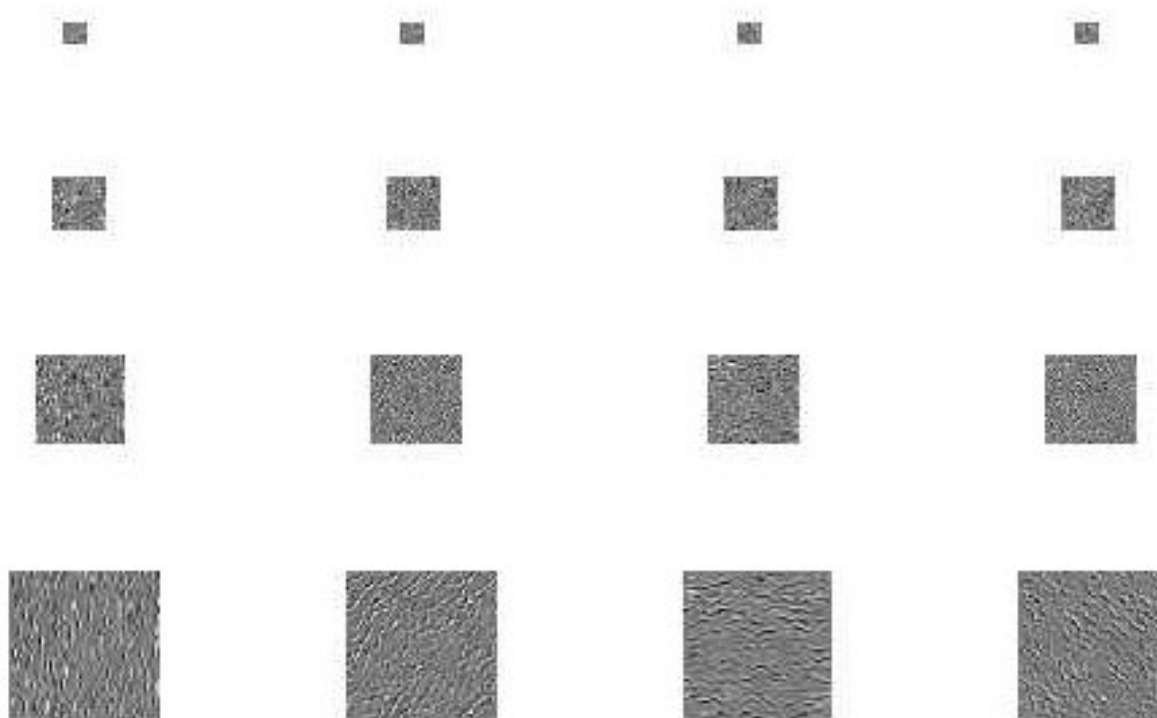
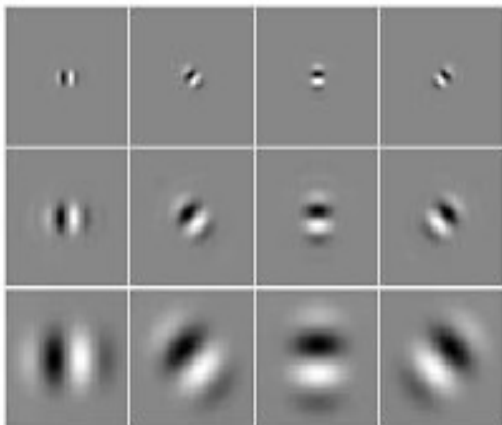


Can you match the texture to its histogram?

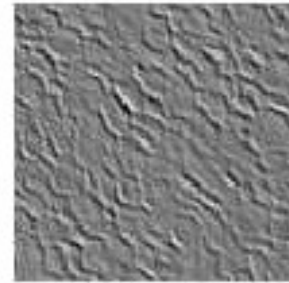
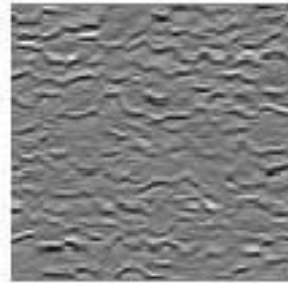
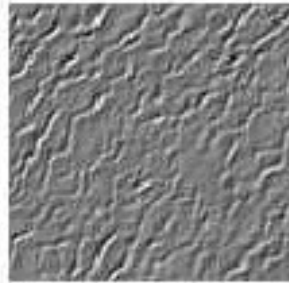
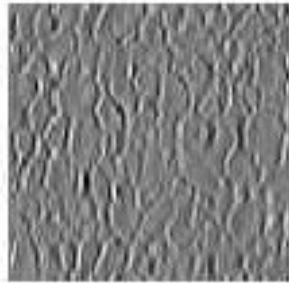


Slightly fancier: histogram for each filter

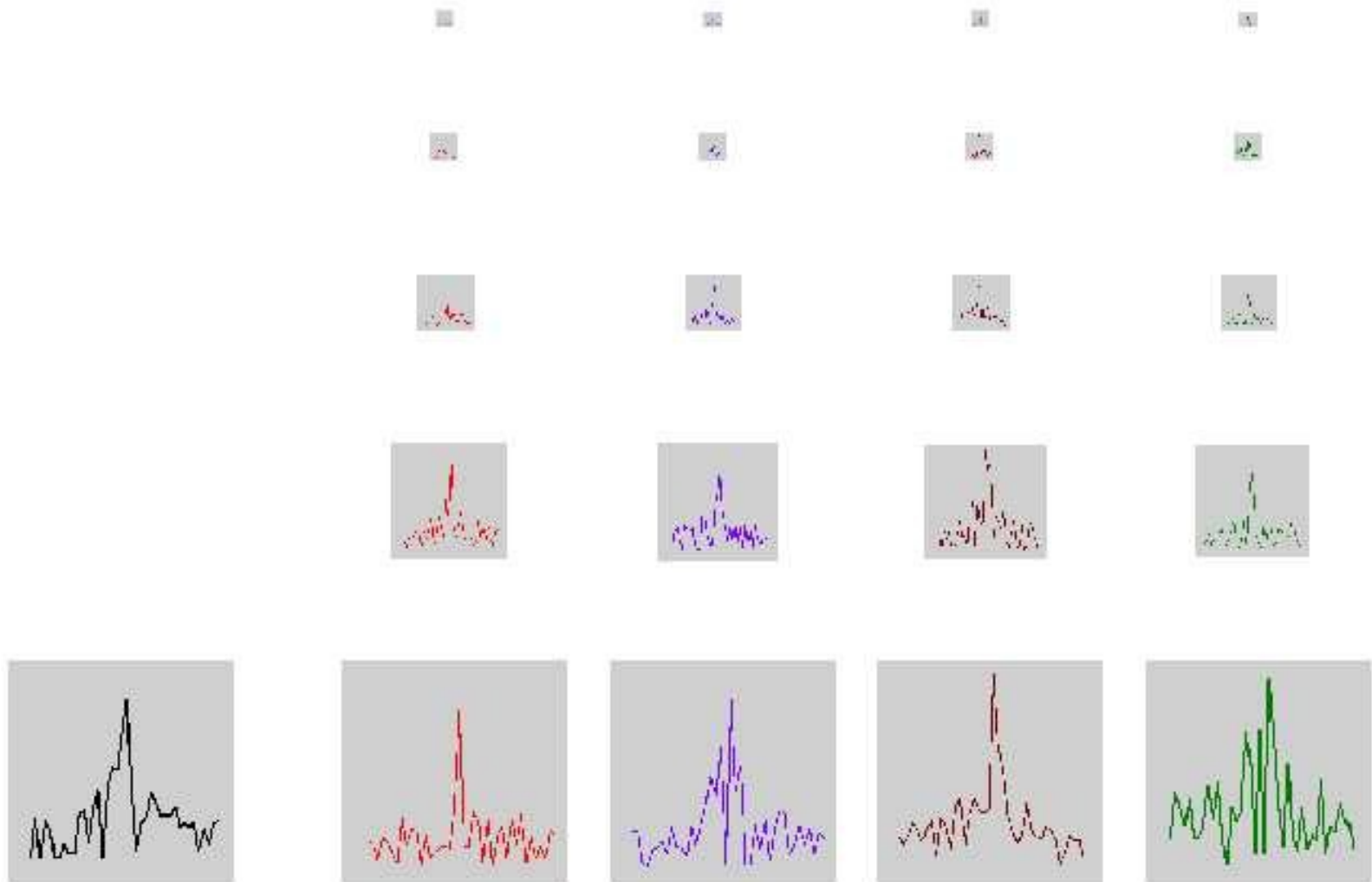
Filter bank



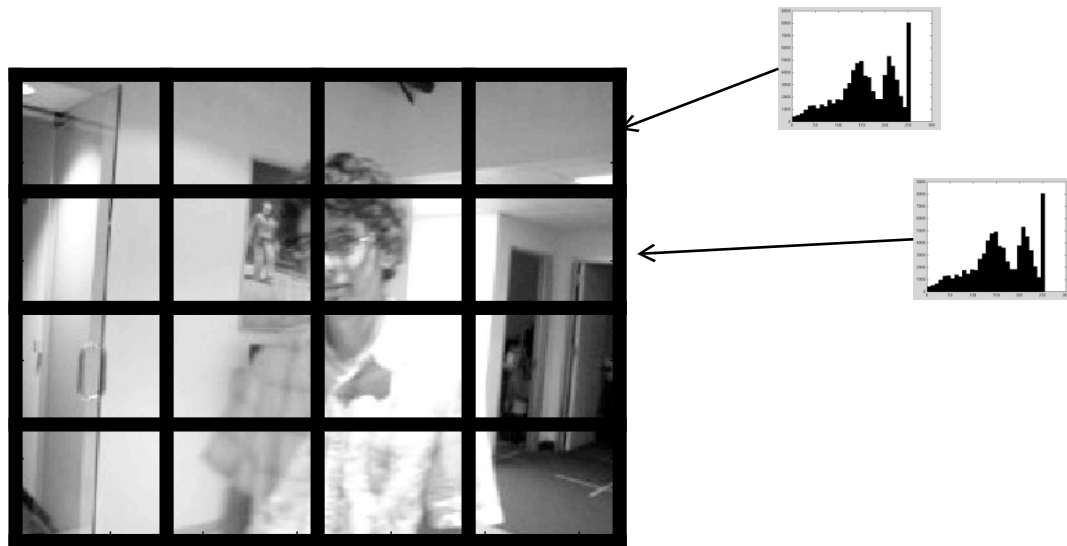
Input image



Filter response histograms

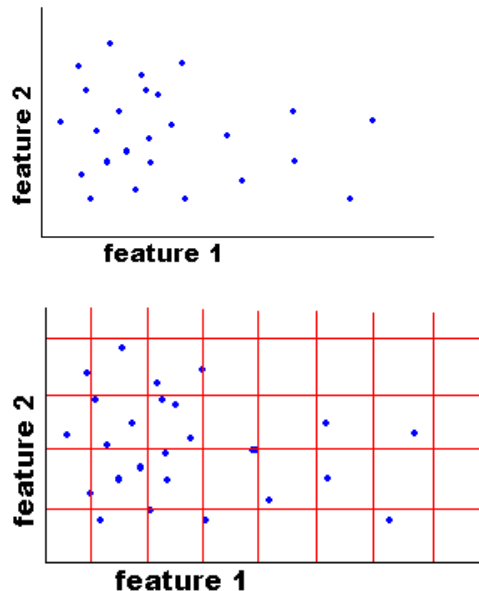


Adding spatial structure



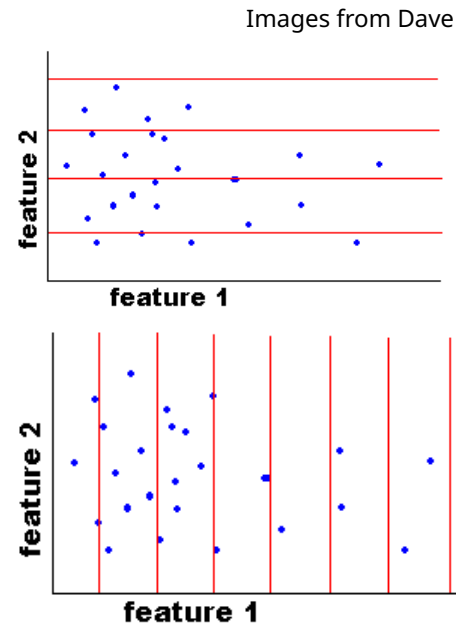
A separate histogram for each region.

Image Representations: Histograms



Joint histogram

- Requires lots of data
- Loss of resolution to avoid empty bins

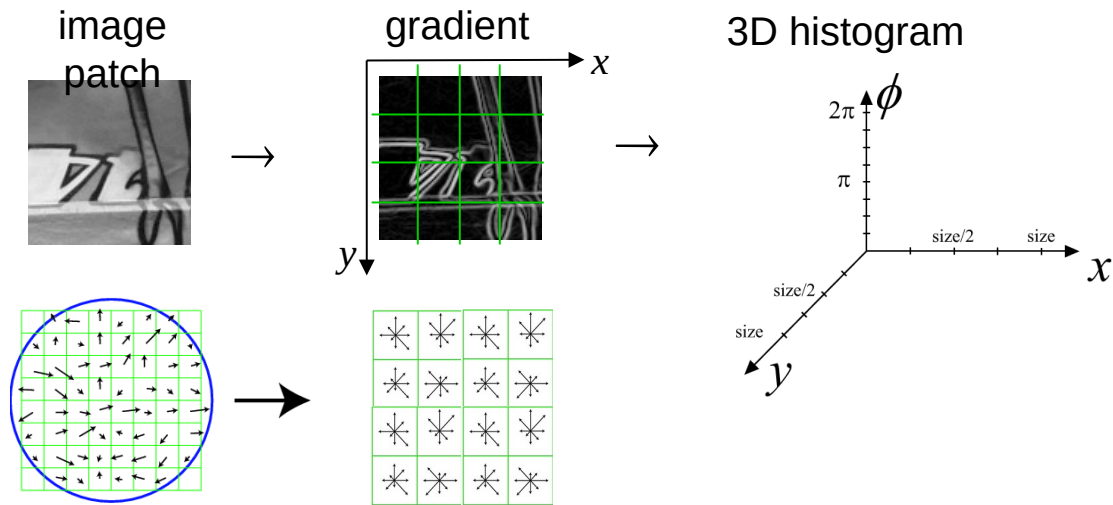


Marginal histogram

- Requires independent features
- More data/bin than joint histogram

Ex: SIFT descriptor [Lowe'99]

distribution of the gradient over an image patch



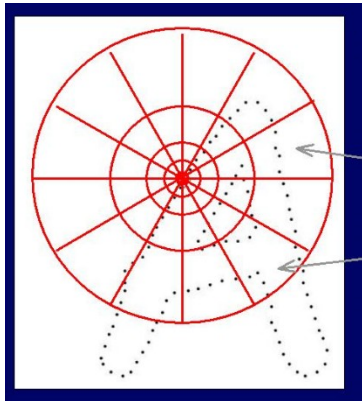
4x4 location grid and 8 orientations (128 dimensions)

very good performance in image matching [Mikolaczyk and Schmid'03]

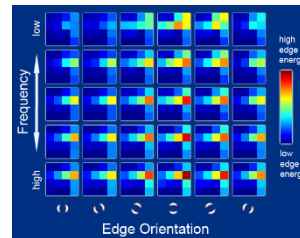
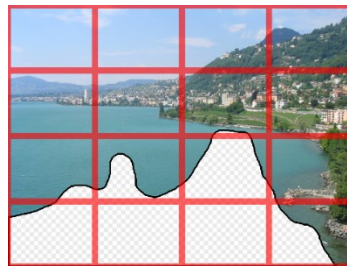
Gradient Histograms pop-up everywhere



HOG descriptor



Generalized Shape Context



Gist Descriptor

Freeman and Roth IAFGR 1995

Lowe ICCV1999

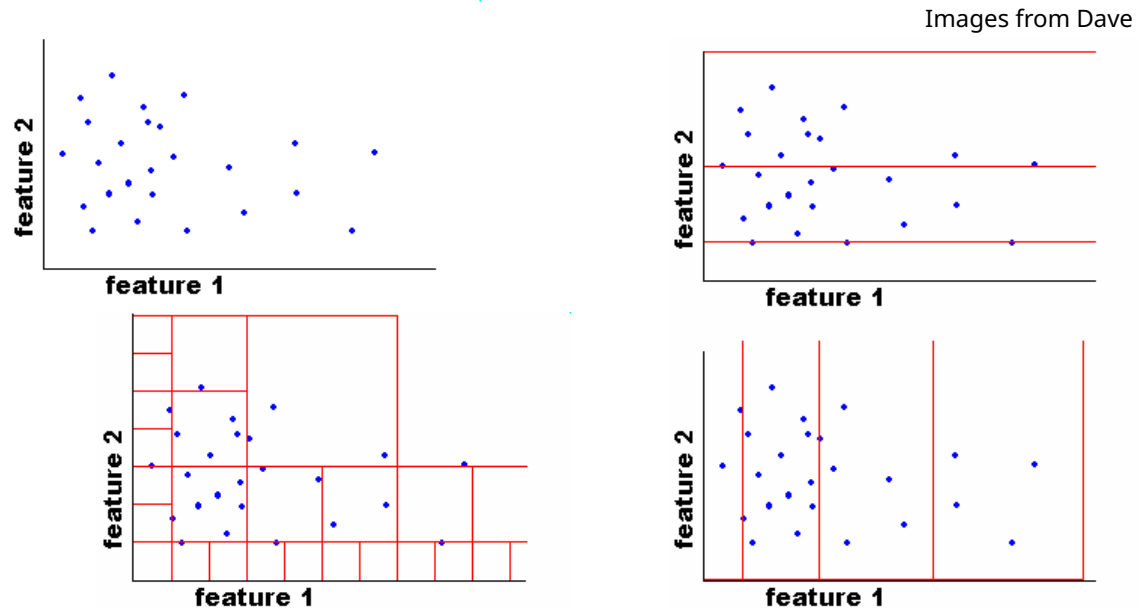
Oliva & Torralba, 2001

Belongie et al, 2001

Dalal & Triggs CVPR05

Binning achieves invariance to small patch offsets

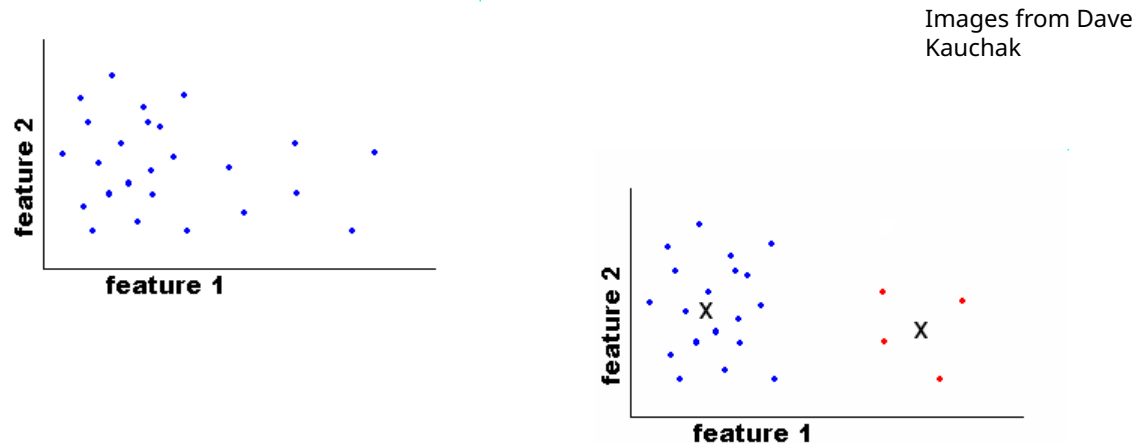
Adaptive Representations



Adaptive binning

- Better data/bin distribution, fewer empty bins
- Can adapt available resolution to relative feature importance

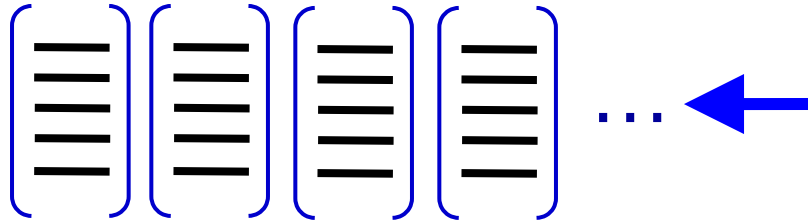
Clustering: very adaptive representations



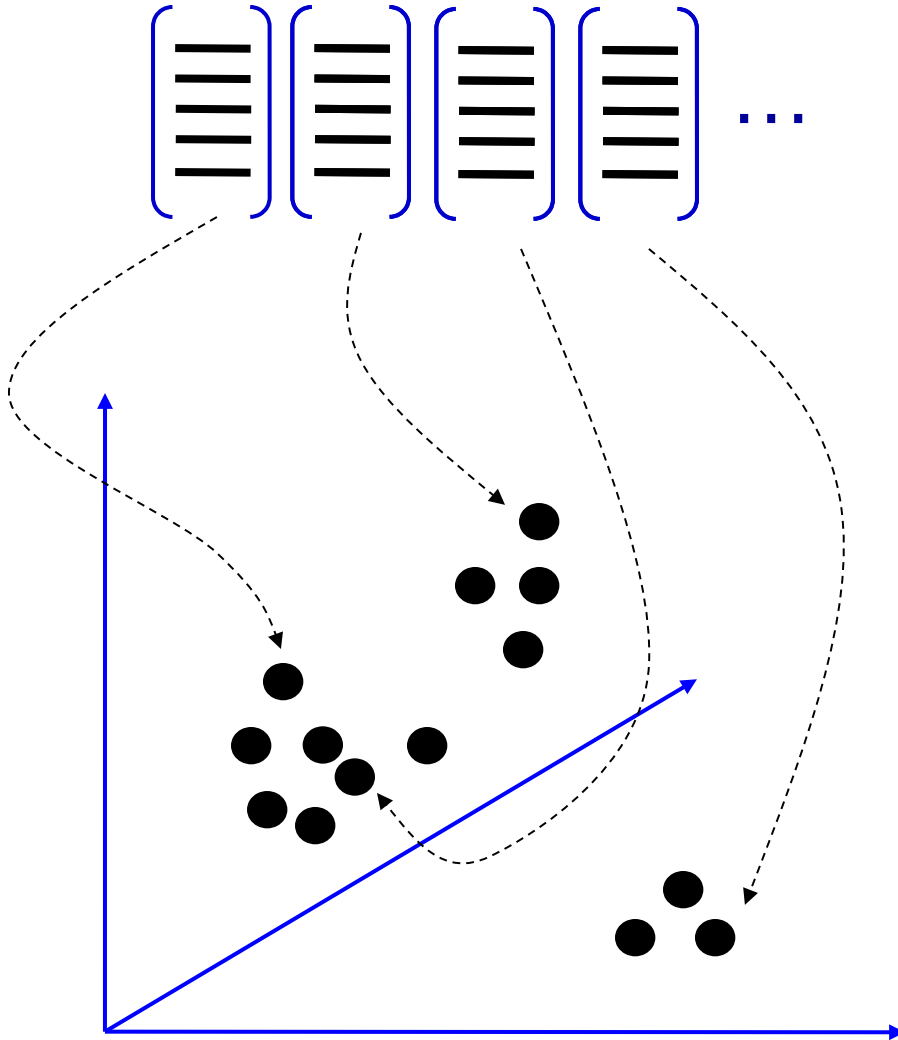
Clusters / Signatures

- “super-adaptive” binning
- Does not require discretization along any fixed axis

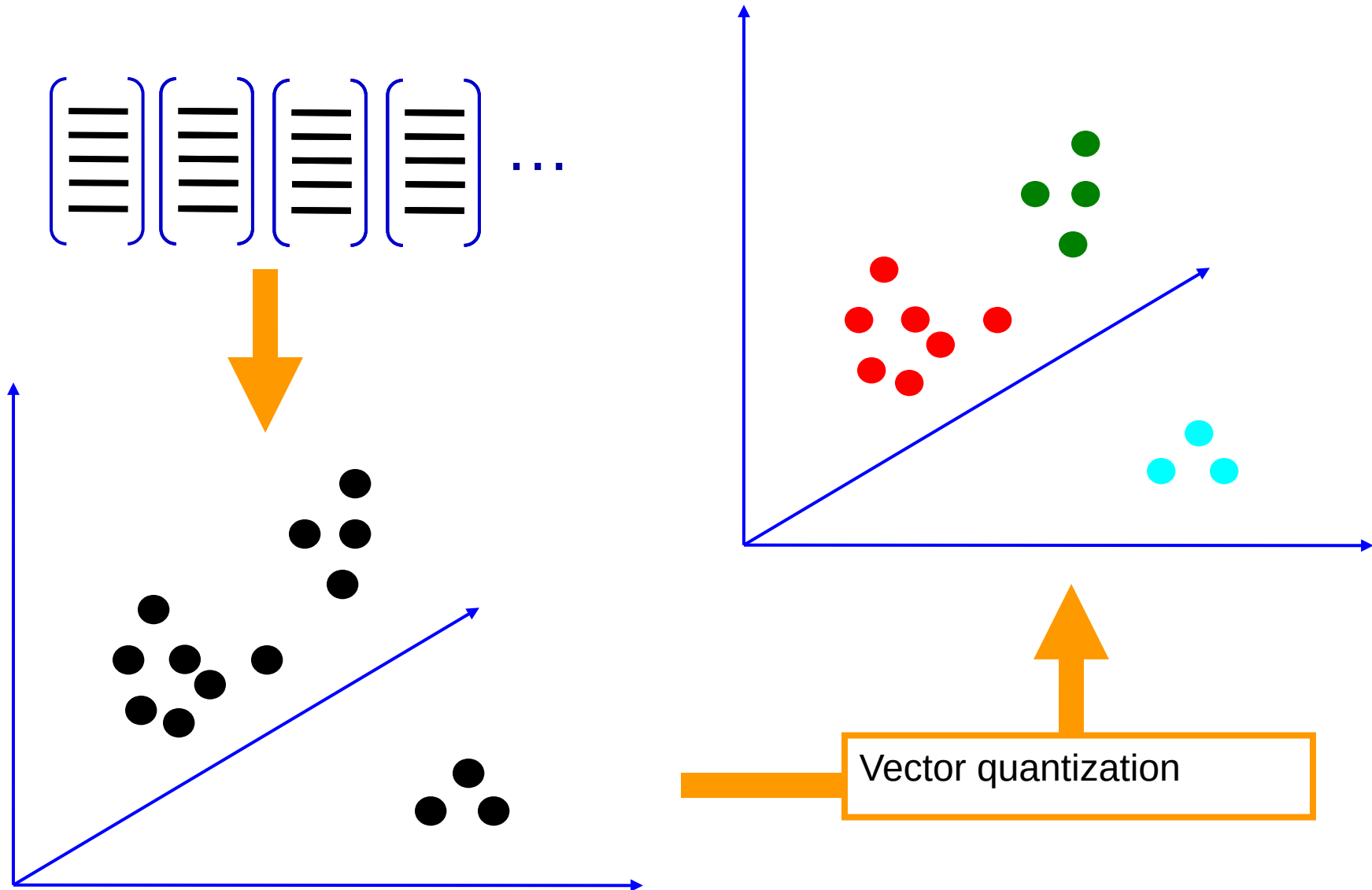
Patch Features



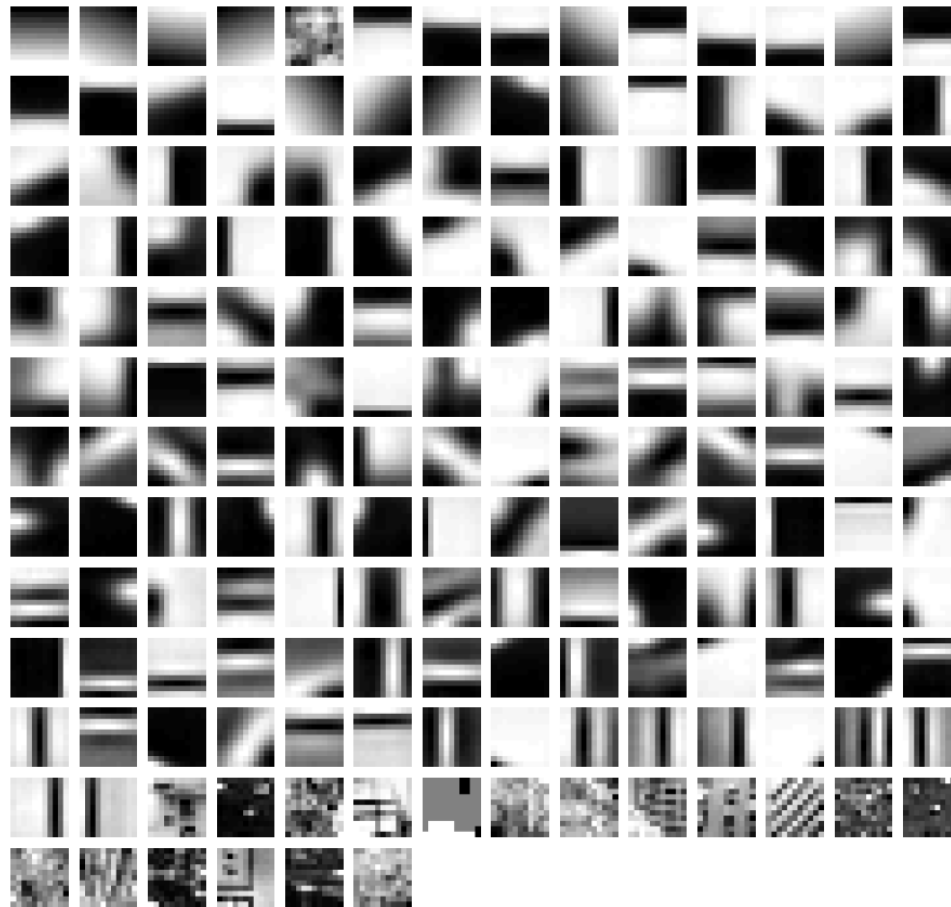
dictionary formation



Clustering (usually k-means)



Clustered Image Patches (“Bag of Visual Words”)



Fei-Fei et al. 2005

Analogy to documents

Of all the sensory impressions proceeding to the brain, the visual experiences are the dominant ones. Our perception of the world around us is based essentially on the messages that reach our eyes.

For a long time, the retinal image was considered as a movie screen. It is now known that the image is processed in a more complex way.

Following the discovery of the visual pathway to the various centers of the brain, Hubel and Wiesel have demonstrated that the message about the image falling on the retina undergoes a step-by-step analysis in a system of nerve cells stored in columns. In this system each cell has its specific function and is responsible for a specific detail in the pattern of the retinal image.

**sensory, brain,
visual, perception,
retinal, cerebral cortex,
eye, cell, optical
nerve, image
Hubel, Wiesel**

China is forecasting a trade surplus of \$90bn (£51bn) to \$100bn this year, a threefold increase on 2004's \$32bn. The Commerce Ministry said the surplus would be created by a predicted 30% increase in exports to \$750bn, compared with \$560bn in 2004.

The increase in exports will be partly due to a rise in the value of exports of manufactured goods, which will be partly offset by a rise in the value of imports of raw materials and semi-finished goods.

China's trade surplus with the US will be \$660bn. The US trade deficit with China will be \$660bn. The US trade deficit with China will be \$660bn. The US trade deficit with China will be \$660bn.

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**China, trade,
surplus, commerce,
exports, imports, US,
yuan, bank, domestic,
foreign, increase,
trade, value**

Object



Bag of 'words'



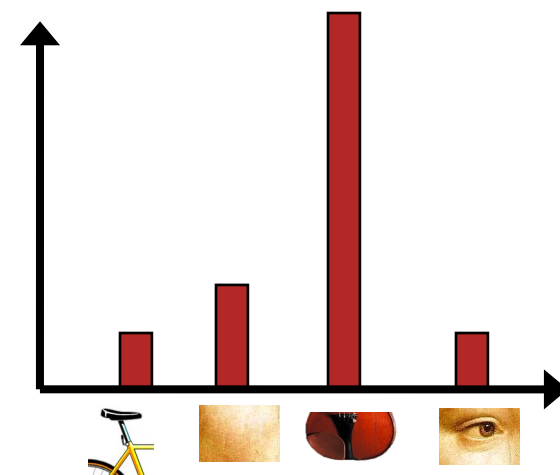
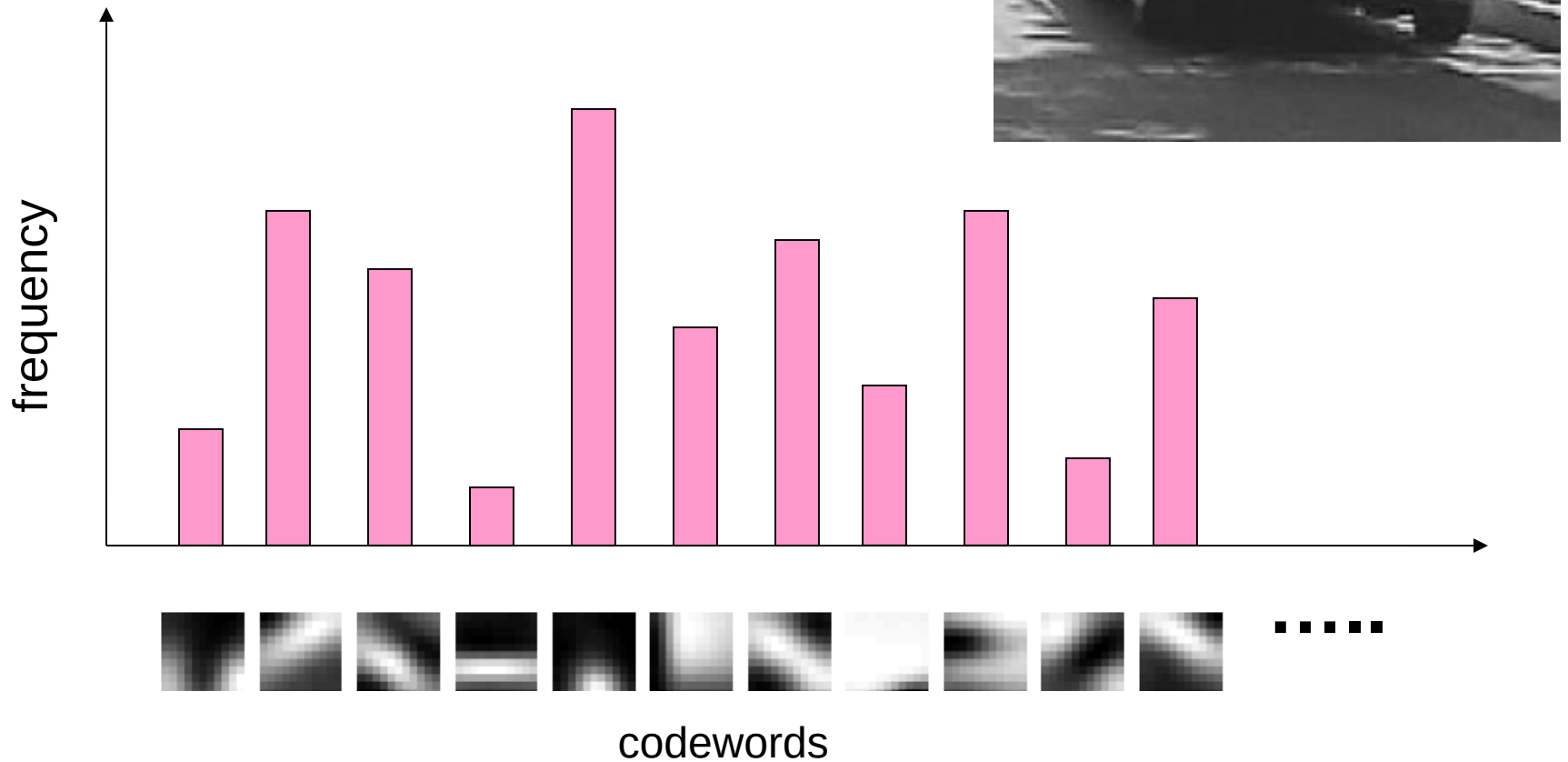


Image representation



Scene Classification (Renninger & Malik)

beach



mountain



forest



city



street



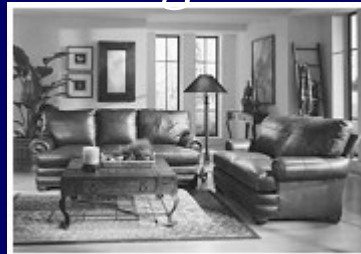
farm



kitchen



livingroom



bedroom

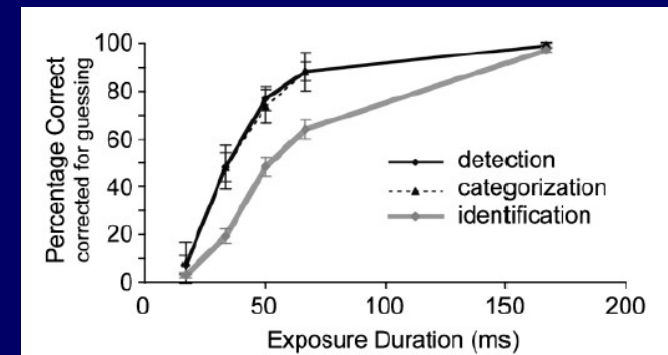


bathroom

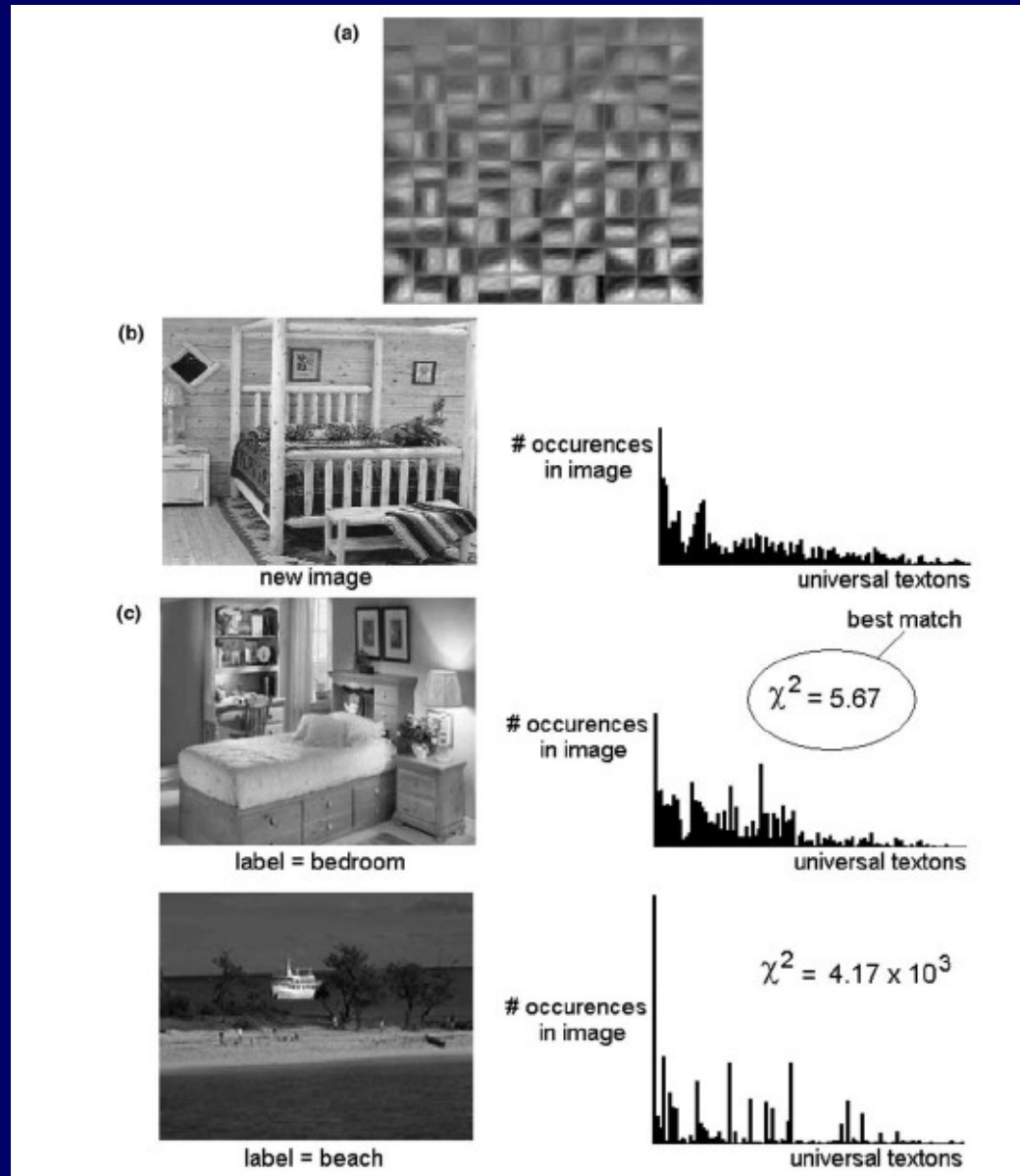


Image classification can be pre-attentive!

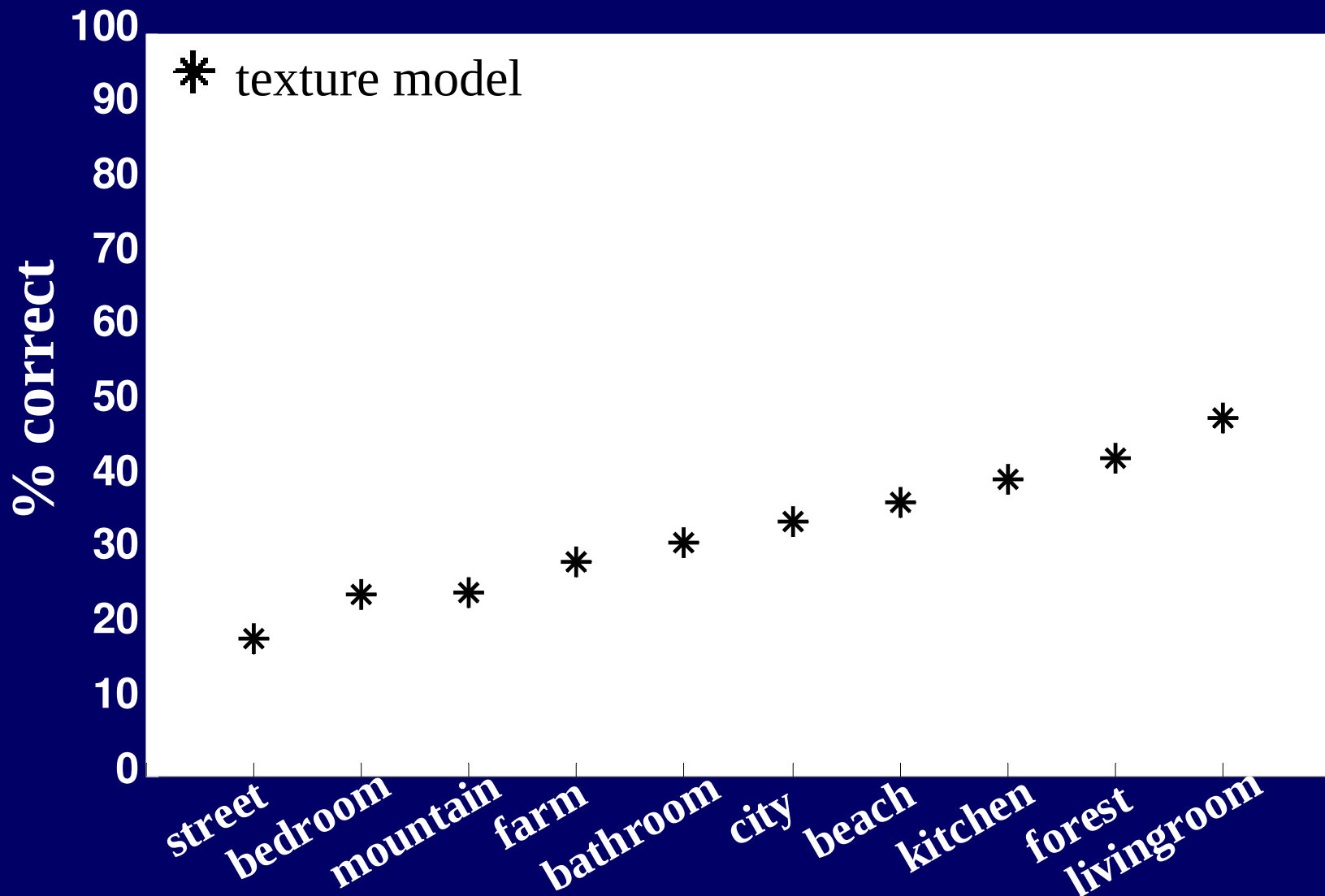
- On a task of judging animal vs no animal, humans can make mostly correct saccades in 150 ms (Kirchner & Thorpe, 2006)
 - Comparable to synaptic delay in the retina, LGN, V1, V2, V4, IT pathway.
 - Doesn't rule out feed back but shows **feed forward only is very powerful**
- Detection and categorization are practically simultaneous (Grill-Spector & Kanwisher, 2005)



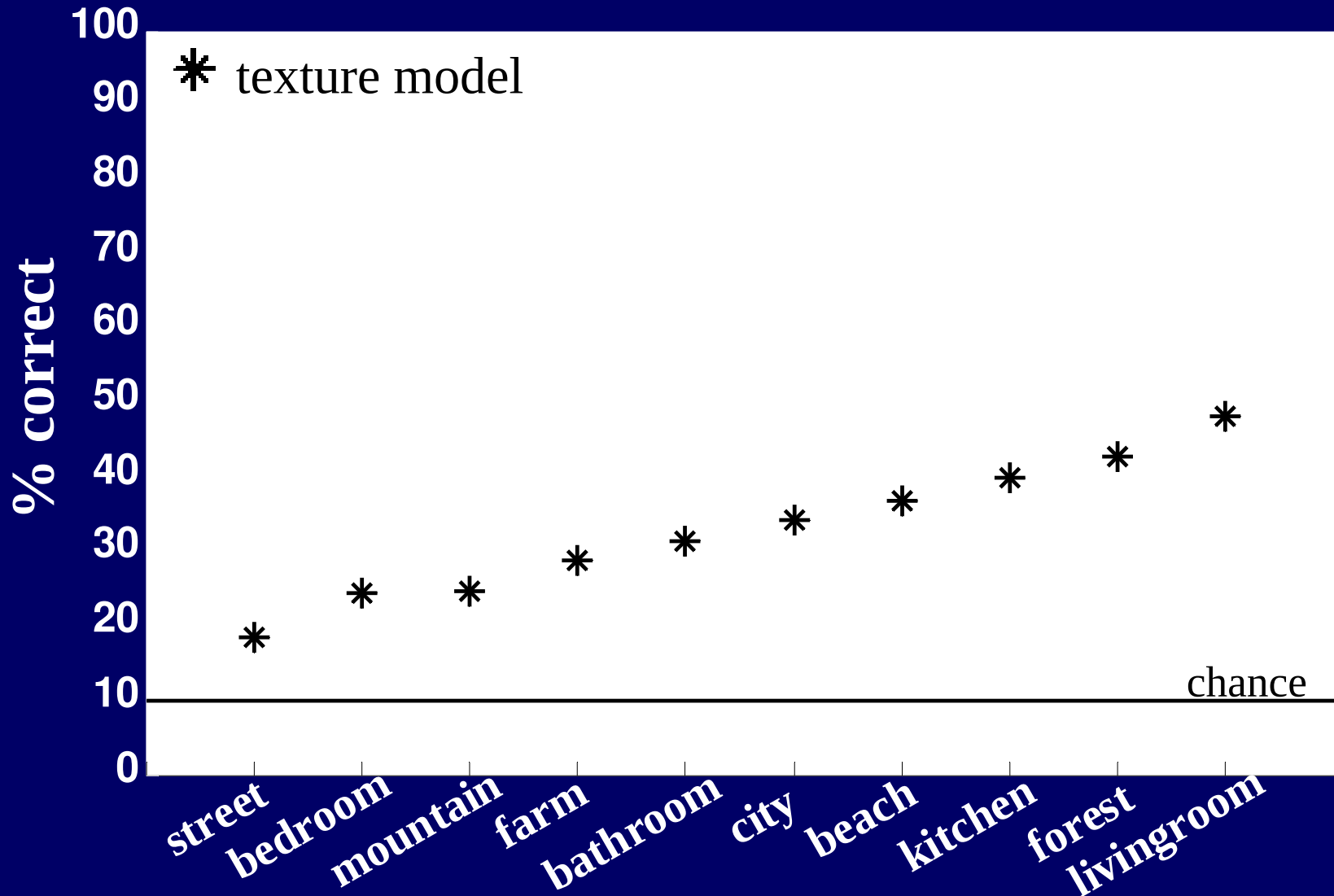
Texton Histogram Matching



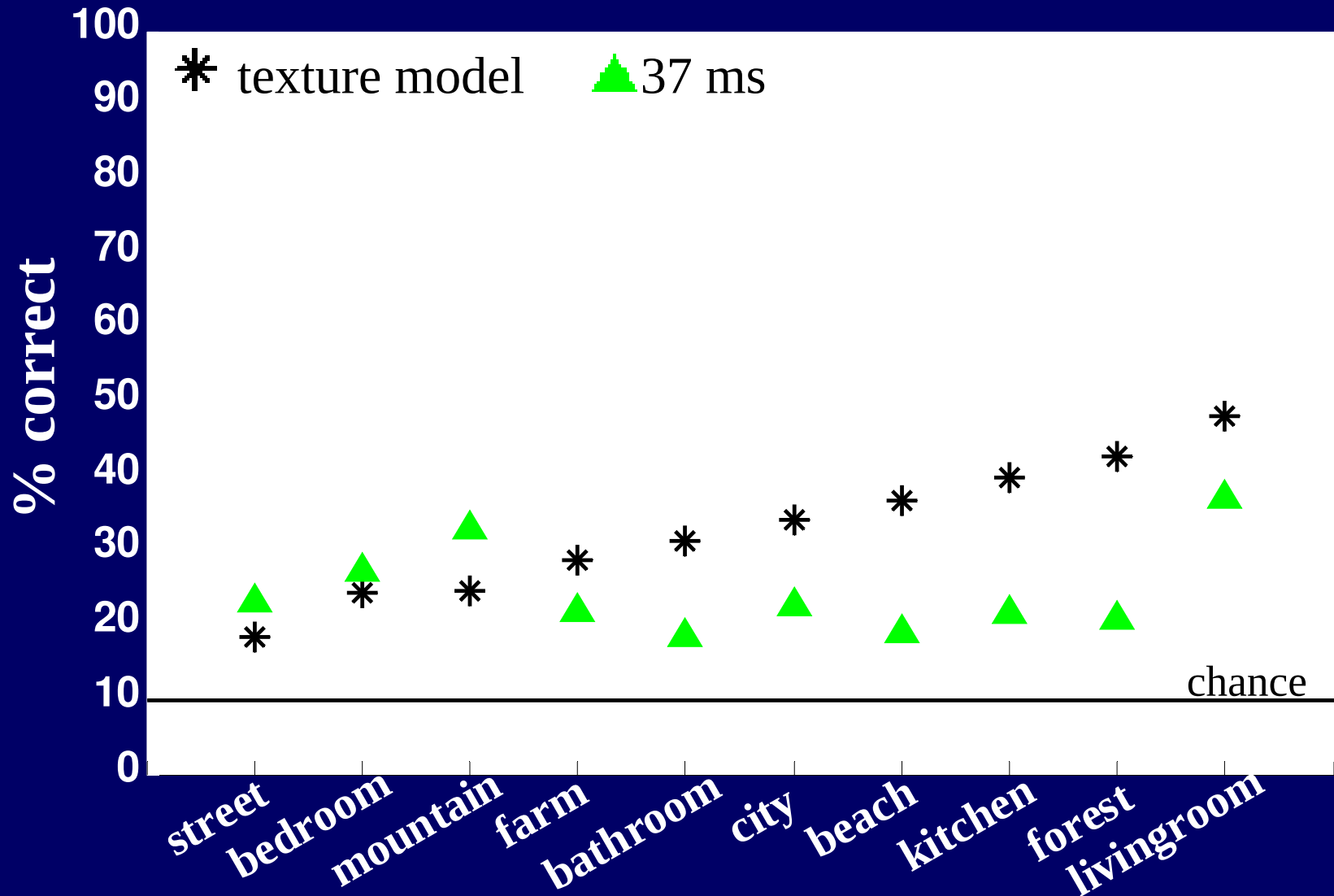
Discrimination of Basic Categories



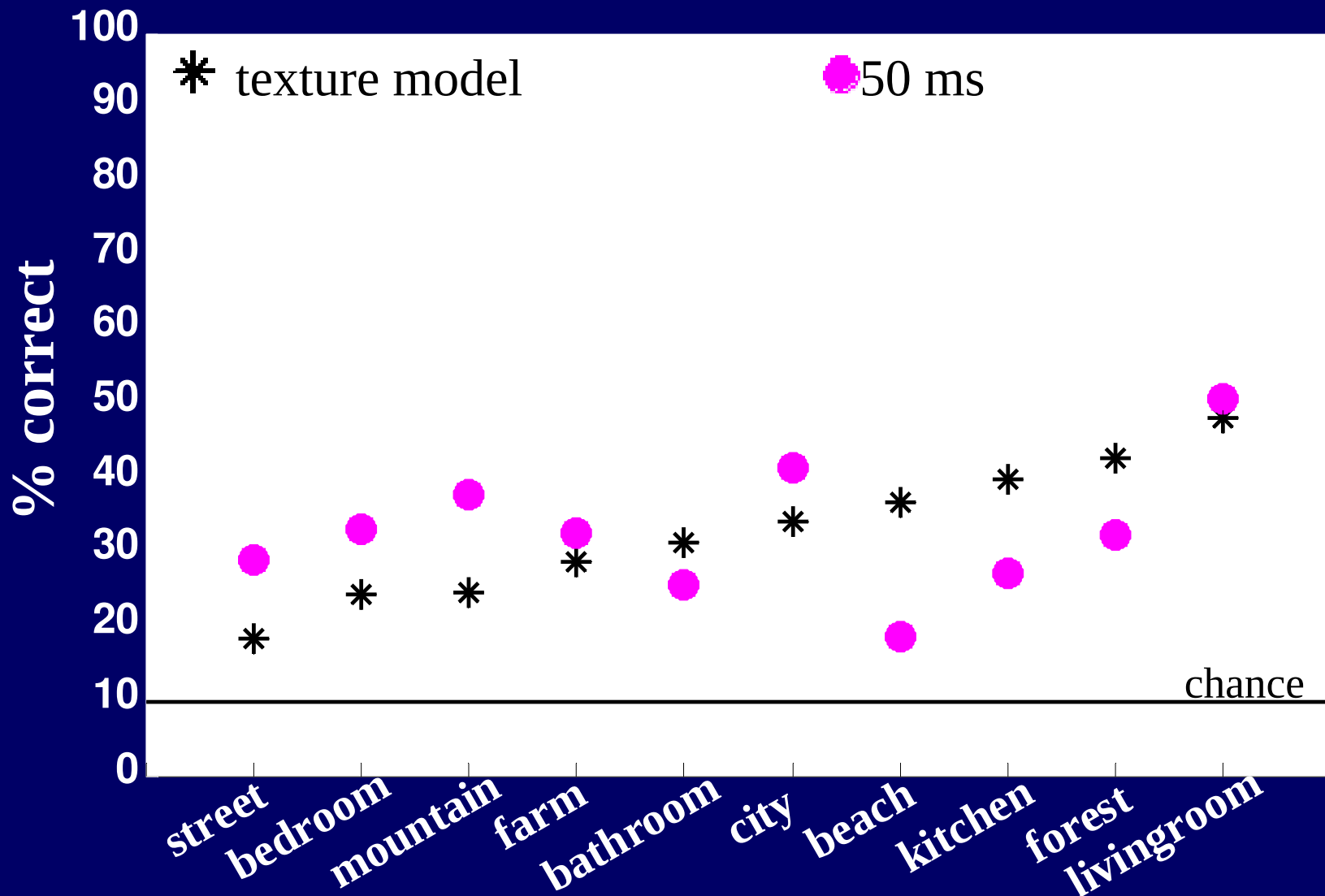
Discrimination of Basic Categories



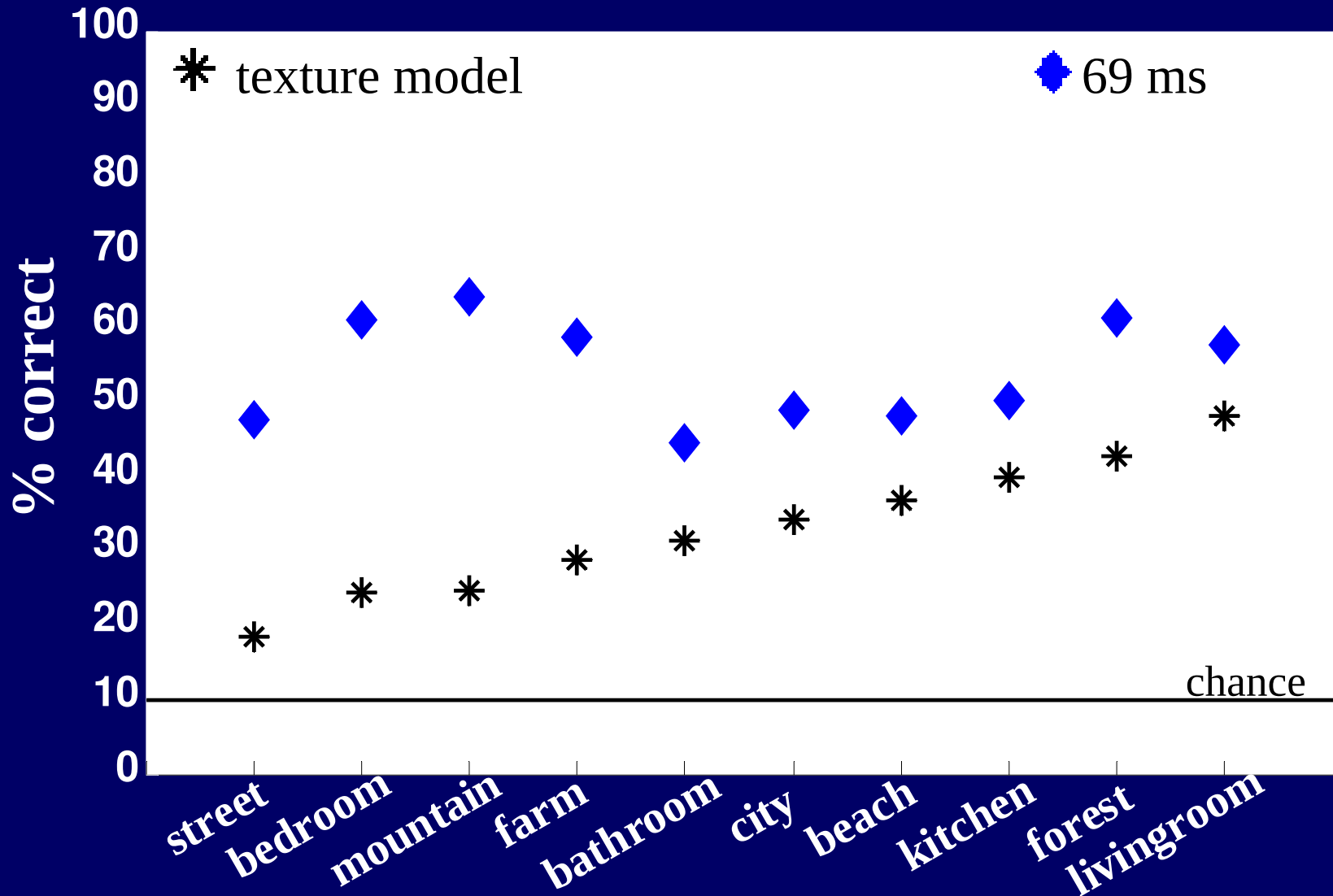
Discrimination of Basic Categories



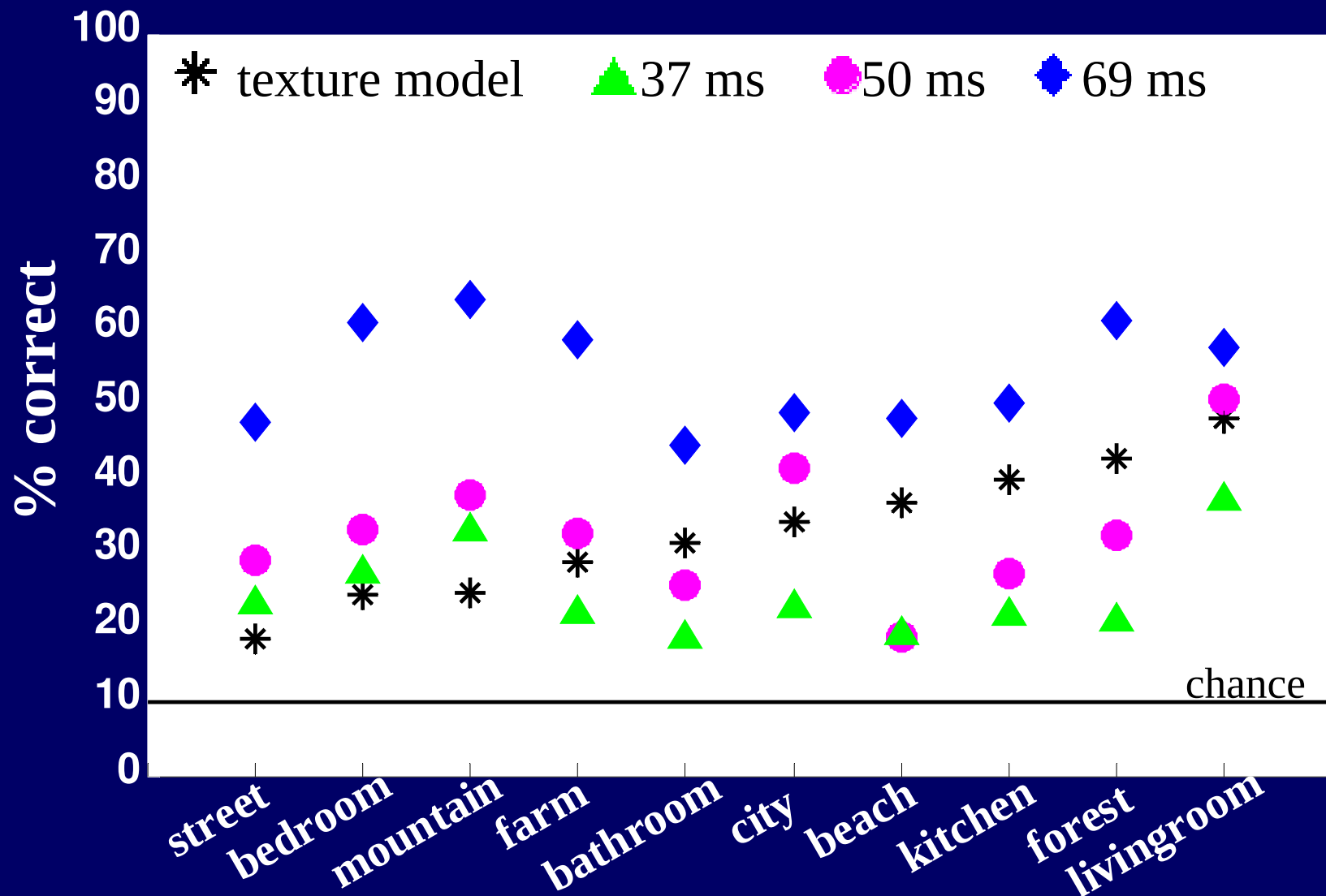
Discrimination of Basic Categories



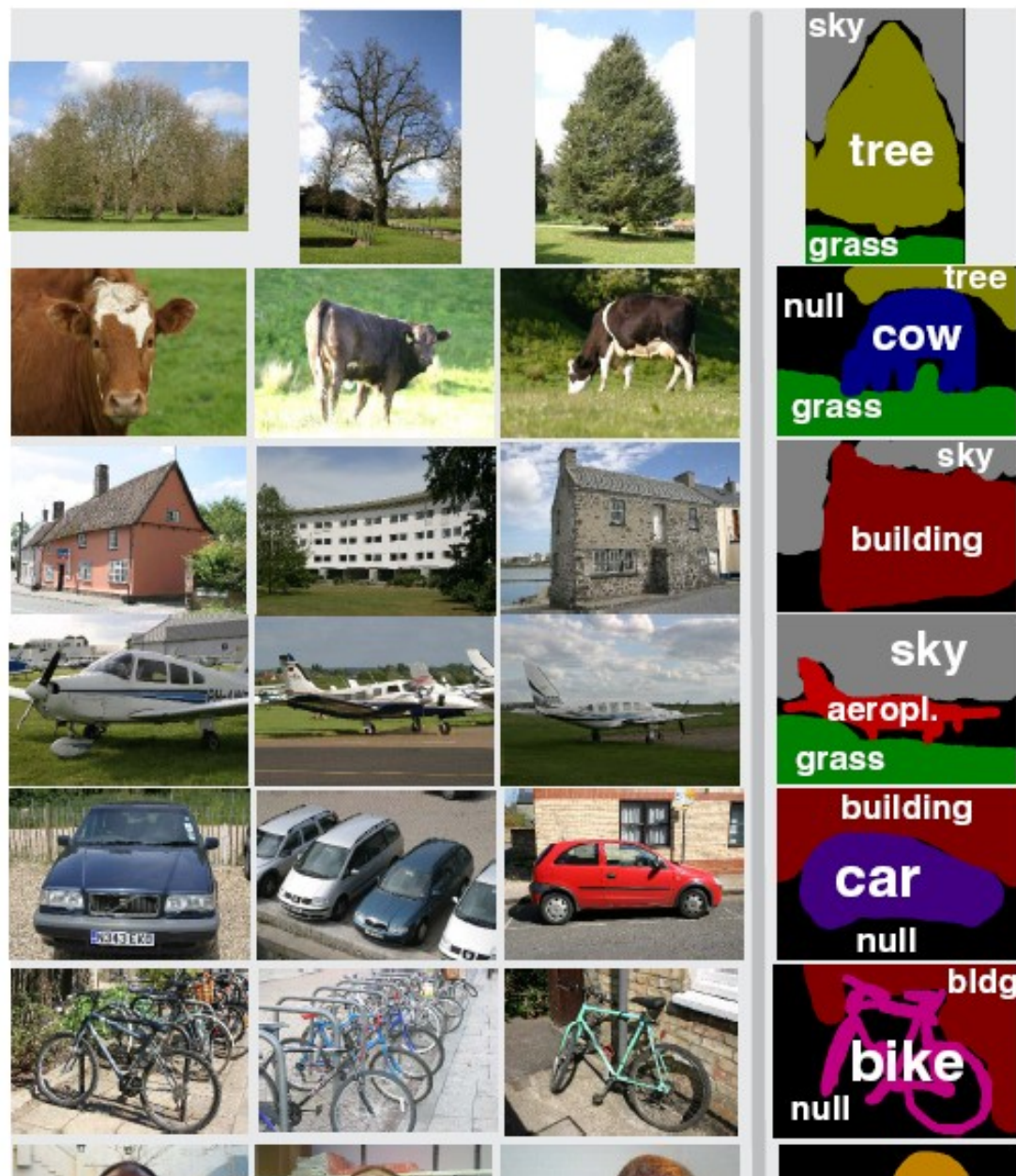
Discrimination of Basic Categories



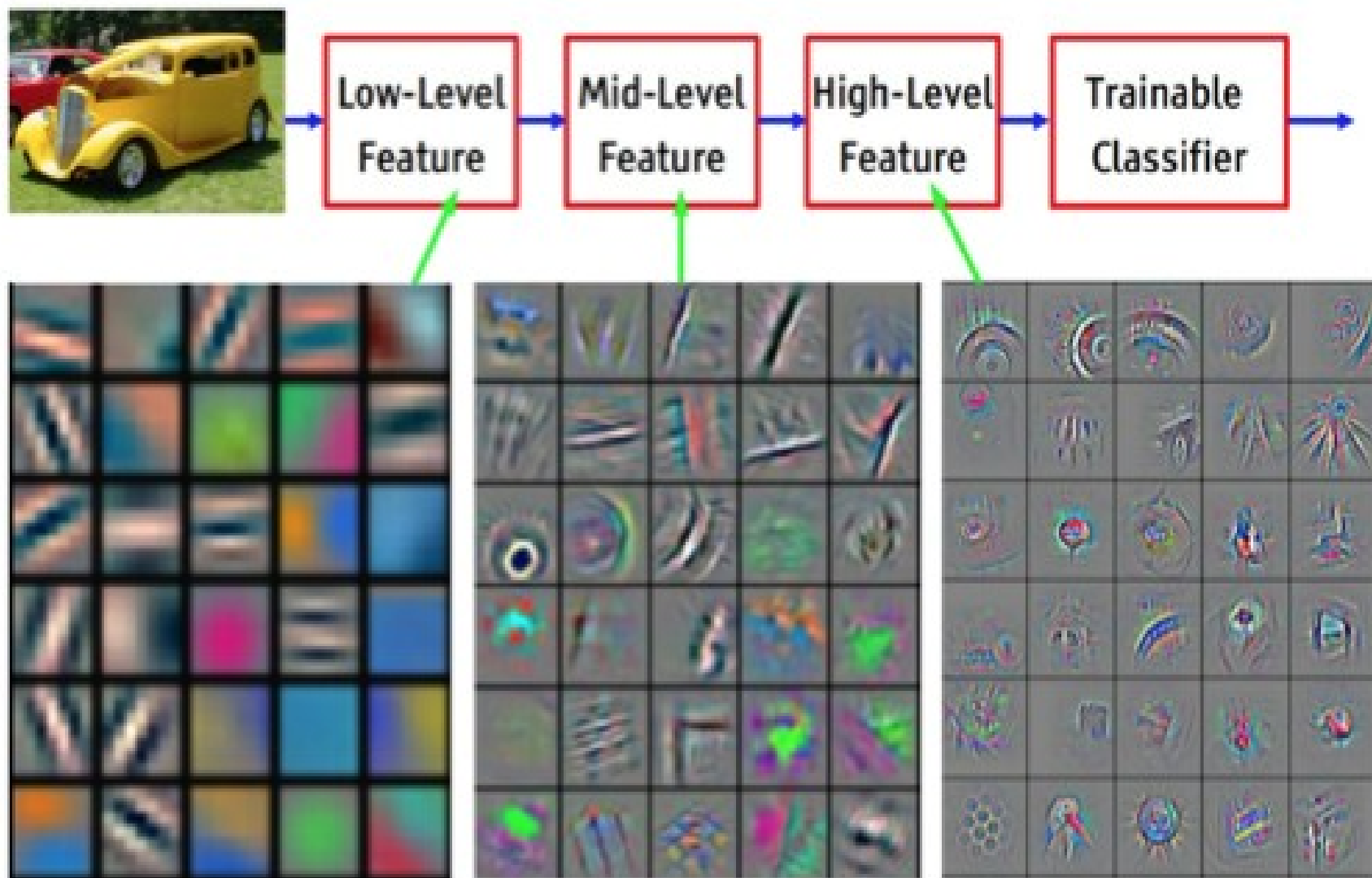
Discrimination of Basic Categories



Scene Recognition using Texture



Convolutional Neural Networks



Feature visualization of convolutional net trained on ImageNet from [Zeiler & Fergus 2013]